

SEMESTER 3

ROBOTICS AND AUTOMATION

SEMESTER S3

MATHEMATICS FOR ELECTRICAL SCIENCE AND PHYSICAL SCIENCE – 3

(Common to B & C Groups)

Course Code	GYMAT301	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hr.30 Min.
Prerequisites (if any)	Basic knowledge in complex numbers.	Course Type	Theory

Course Objectives:

1. To introduce the concept and applications of Fourier transforms in various engineering fields.
2. To introduce the basic theory of functions of a complex variable, including residue integration and conformal transformation, and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fourier Integral, From Fourier series to Fourier Integral, Fourier Cosine and Sine integrals, Fourier Cosine and Sine Transform, Linearity, Transforms of Derivatives, Fourier Transform and its inverse, Linearity, Transforms of Derivative. (Text 1: Relevant topics from sections 11.7, 11.8, 11.9)	9
2	Complex Function, Limit, Continuity, Derivative, Analytic functions, Cauchy-Riemann Equations (without proof), Laplace's Equations, Harmonic functions, Finding harmonic conjugate, Conformal mapping, Mappings of $w = z^2$, $w = e^z$, $w = \frac{1}{z}$, $w = \sin z$. (Text 1: Relevant topics from sections 13.3, 13.4, 17.1, 17.2, 17.4)	9

3	Complex Integration: Line integrals in the complex plane (Definition & Basic properties), First evaluation method, Second evaluation method, Cauchy's integral theorem (without proof) on simply connected domain, Independence of path, Cauchy integral theorem on multiply connected domain (without proof), Cauchy Integral formula (without proof). (Text 1: Relevant topics from sections 14.1, 14.2, 14.3)	9
4	Taylor series and Maclaurin series, Laurent series (without proof), Singularities and Zeros – Isolated Singularity, Poles, Essential Singularities, Removable singularities, Zeros of Analytic functions – Poles and Zeros, Formulas for Residues, Residue theorem (without proof), Residue Integration- Integral of Rational Functions of $\cos\theta$ and $\sin\theta$. (Text 1: Relevant topics from sections 15.4, 16.1, 16.2, 16.3, 16.4)	9

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Determine the Fourier transforms of functions and apply them to solve problems arising in engineering.	K3
CO2	Understand the analyticity of complex functions and apply it in conformal mapping.	K3
CO3	Compute complex integrals using Cauchy's integral theorem and Cauchy's integral formula.	K3
CO4	Understand the series expansion of complex function about a singularity and apply residue theorem to compute real integrals.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	2	-	-	-	-	-	-	-	2
CO2	3	3	-	2	-	-	-	-	-	-	-	2
CO3	3	3	-	2	-	-	-	-	-	-	-	2
CO4	3	3	-	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	10 th edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Complex Analysis	Dennis G. Zill, Patrick D. Shanahan	Jones & Bartlett	3 rd edition, 2015
2	Higher Engineering Mathematics	B. V. Ramana	McGraw-Hill Education	39 th edition, 2023
3	Higher Engineering Mathematics	B.S. Grewal	Khanna Publishers	44 th edition, 2018
4	Fast Fourier Transform - Algorithms and Applications	K.R. Rao, Do Nyeon Kim, Jae Jeong Hwang	Springer	1 st edition, 2011

SEMESTER S3
SOLID MECHANICS

Course Code	PCRUT302	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBEST213	Course Type	Theory

Course Objectives:

1. To help the students to understand the concept of stress and strain in different types of structure/machine under various loading conditions
2. To provide students with an understanding of simple and compound stresses due to forces, stresses and deflection in beams due to bending, torsion in circular, and non-circular section, strain energy, different theories of failure, stress in thin cylinder thick cylinder and spheres due to external and internal pressure.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction- stress, strain and its types, stress strain diagram (ductile & brittle materials), material properties, Relationship between elastic constants (E, G & ν). Calculation of stress, strain and change in length in axially loaded members with single and composite materials, Effects of thermal loading – thermal stress and thermal strain	11
2	Method of sections, Stress vectors on Cartesian coordinate planes passing through a point, stress at a point in the form of a matrix. Equality of cross shear, Cauchy's equation (no derivation). Principal planes and principal stress, meaning of stress invariants, maximum shear stress. Mohr's circle for 2D cases Displacement, gradient of displacement, Cartesian strain matrix, strain-displacement relations (small-strain only), Simple problems to find strain matrix.	11

3	Torsional deformation of circular shafts, assumptions for shafts subjected to torsion within elastic deformation range, torsion formula (no derivation). Torsional rigidity, Polar moment of inertia, basic design of transmission shafts. Types of beams, loading and supports. Shear force and bending moment diagrams for cantilever and simply supported beams (point load & UDL).	11
4	Fundamentals of buckling and stability, critical load, Buckling of columns with pinned ends, Euler's buckling theory for long columns. Critical stress, slenderness ratio, Rankine's formula for short columns. Theories of Failure, Rankine's theory for maximum normal stress, Guest's theory for maximum shear stress, Saint-Venant's theory for maximum normal strain, Hencky-von Mises theory for maximum distortion energy, Haigh's theory for maximum strain energy	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the strength of materials using stress-strain relationships for structural and thermal loading	K3
CO2	Determine the stresses, strains and displacements of structures by graphical (Mohr's circle) approaches	K3
CO3	Explain basic design of circular shafts and thin walled structures subjected to torsional loading and analyse beams subjected to bending moments under different idealised loading conditions	K2
CO4	Analyse column buckling and appreciate the theories of failures and its relevance in engineering design	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3										3
CO3	3	2	3									3
CO4	3	2										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Mechanics of Solids	L. S. Srinath	McGraw Hill Education 4	
2	Strength of Materials,	R. K Bansal	Laxmi Publications	2010
3	Strength of Materials	Rattan	McGraw Hills	2011
4				

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Mechanics of Solids	Popov E	PHI	2002
2	Mechanics of materials in S.I. Units	R .C. Hibbeler,	Pearson Higher Education	2018
3	Design of Machine Elements	V. B Bhandari	McGraw Hill Education	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
2	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
3	https://onlinecourses.nptel.ac.in/noc22_ce46/preview
4	https://onlinecourses.nptel.ac.in/noc22_ce46/preview

SEMESTER S3

MATERIALS AND MANUFACTURING PROCESSES

Course Code	PCRUT303	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understanding the fundamental principles of materials science, developing the ability to analyse and characterize materials, learning about the processing and manufacturing of advanced materials, and gaining knowledge of the applications of materials in engineering and industry.
2. Understanding various stages of manufacturing, including design, production, and assembly

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Crystallography:- Crystal, space lattice, unit cell- BCC,FCC, HCP structures, Coordination number and radius ratio, Polymorphism and allotropy. Phase diagrams: - solid solutions, Hume Rothery's rule - equilibrium diagram of common types of binary systems: Cu-Ni, Pb-Sn. Detailed discussion on Iron-Carbon equilibrium diagram with microstructure and properties. Heat treatments: - annealing, normalizing, hardening, spheroidizing. Tempering. Surface hardening methods: - no change in surface composition methods: - Flame, induction, laser and electron beam hardening processes- change in surface composition methods: carburizing and Nitriding; applications.	11

2	Applications of ferrous and non ferrous alloys steel - low, medium, high carbon steels, stainless steels, tool steels . Cast iron- gray, white, ductile cast irons. copper and its alloys, aluminium and its alloys, magnesium and alloys, Composites:- –classification - fiber phase: - characteristics, classifications - matrix phase:- functions, characteristics of PMC, MMC, and CMC, applications of composites .ceramics, polymers- thermo plastics and thermosetting, Nano materials, bio materials, Optical fibers.	11
3	Primary processes – Casting – sand casting, moulds, types of moulding sand,types of patterns, pattern materials – Forging – forging methods, precision forging, coining, heading, piercing, die design, die materials and lubrication, forging machines – Forming – Extrusion process, hot extrusion, cold	11
4	Extrusion, impact extrusion, wire drawing process – Rolling – principles of rolling, types of rolling mill, hot and cold rolling, rolling mills, ring rolling, thread rolling, applications – rolling of tubes, wheels, axles and I-beams. Importance of material selection for manufacturing processes. Selection of primary forming process and suitable materials for various products. Defects and Design considerations in casting, forging and rolling.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 = 24 marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse the Structure of materials at different levels, basic concepts of crystalline materials like unit cell, FCC, BCC.HCP	K3
CO2	Explain the concept of phase diagram & understand the basic terminologies associated with metallurgy.	K2
CO3	Explain features, classification and applications of ferrous materials and nonferrous materials like Aluminium, Copper, Magnesium, composite, Polymers etc.	K2
CO4	Examine the basics of the primary manufacturing processes and apply the knowledge in designing parts for robotic applications	K3
CO5	Describe the various joining processes and choose the appropriate mechanical and adhesive joining process for the parts.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3						2				2	3
CO2	3						3				2	3
CO3	3						3				2	3
CO4	3	2										3
CO5	3	2										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Material Science & Engineering-Adapted by Balasubramaniam	William D Callister	Wiley	2014
2	Material Science and Engineering,	Raghavan V,	Prentice Hall,	2004
3	Manufacturing Engineering and Technology,	Kalpakkian and Schmid,	Pearson	2013.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Materials Science for Engineers	James F. Shackelford	Pearson	2009
2	Fundamentals of Modern Manufacturing, ,	Mikell P. Groover	John Wiley & Sons, Inc,New Jersey	2010.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/113/102/113102080/
2	https://archive.nptel.ac.in/courses/113/102/113102080/
3	https://archive.nptel.ac.in/courses/112/107/112107219/
4	https://archive.nptel.ac.in/courses/112/107/112107219/

SEMESTER S3

ELECTRONIC CIRCUITS AND LOGIC DESIGN

Course Code	PBRUT304	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. To introduce the fundamentals and applications of op- -amps.
2. To grasp the basics of digital electronics, including logic gates and Boolean algebra
3. To design and implement combinational and sequential circuits to solve specific digital logic problems.
4. To utilize breadboards or FPGAs or simulation software to implement circuits.
5. To work in a group to design and implement a digital electronic system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Operational Amplifiers (Op-Amps) Op-amp Characteristics and ideal op-amp Op-amp configurations: inverting, non-inverting, differential amplifiers Op-amp parameters: gain, bandwidth, input/output impedance. Effect of feedback on op-amp performance Op-amp applications: amplifiers, comparators, integrators, differentiators, V-I and I-V converters, Comparator – Regenerative comparator, zero crossing detector, Window detector, Sample and hold circuit, Rectifiers	11

2	Active Filters and Signal Conditioning Low-pass, high-pass, band-pass, and band-stop filters using op-amps-Active filter design considerations Signal conditioning circuits for sensor interfacing using op-amps A-D and D-A converters using Op-amps D/A converter -specifications -weighted resistor type,R-2R Ladder type. A/D Converters – specifications-Flash type, Successive Approximation type, Dual Slope type	11
3	Fundamentals of Digital Electronics Review of Boolean Algebra, Logic Gates- BCD, Gray code, Excess 3 code De-Morgan's Theorems, Principle of Duality, Minimization of Boolean expressions, Sum of Products (SOP), Product of Sum (POS), Canonical forms, Karnaugh map Minimization. Simplification of logic functions through K–maps and Implementation using logic gates (up to 4 variables). Combinational Logic Circuits Combinatorial Logic Systems - Comparators, Multiplexers, Demultiplexers, Encoder,Decoder. Half and Full Adders, Subtractors, Serial and Parallel Adders, BCD Adder.	11
4	Sequential Logic Circuits Flip-flops and latches: SR, D, JK, and T flip-flops Design of sequential circuits-Counters-Ring counter and Johnsons counter; shift registers -SIPO, SISO, PISO, PIPO. Asynchronous Sequential Circuits- Analysis of state machines using D flip-flops and JK flip flops; Design of state machines - state table, state assignment transition/excitation table	11

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 2 marks (8x2 =16marks)	Each question carries 6 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design linear and nonlinear circuits using Op-amps	K3
CO2	Choose A/D converters and D/A Converters for specific applications	K2
CO3	Explain the fundamental principles of digital electronics, including logic gates, Boolean algebra, and combinational circuits.	K2
CO4	Design and analyse combinational and sequential circuits to solve specific digital logic problems.	K3
CO5	Implement digital/opamp circuits/projects using various hardware components and tools, such as breadboards, FPGAs, and simulation software	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2									3
CO2	3	3	2									3
CO3	3	3	1									
CO4	3	3	2									
CO5	3	3	3	2	3			3	3	3	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op-Amp and Linear Integrated Circuits	Ramakant Gayakwad	Pearson Education India	4 th Edition
2	Linear Integrated Circuits	DRoyChoudhury and Shail B Jain	New Age International Publishers	4 th Edition
3	Digital Fundamentals	Thomas Floyd	Pearson Education India	11 th Edition
4	Fundamentals of Digital Circuits	A.Anand Kumar	Prentice Hall India Private Limited	4 th Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Integrated Circuits	Salivahanan s and Bhaskaran	Tata Mc Graw Hill	2 nd edition
2	Design with operational amplifiers and Analog Integrated circuits	Franco S	Tata Mc Graw Hill	4th edition
3	Digital Circuits and Systems	D V Hall	Tata Mc Graw Hill	
4	Digital Design	Morris Mano	Pearson India	4 th Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	lec15-Introduction to Operational Amplifiers (youtube.com)
2	lec19 - Operational Amplifier Configurations (youtube.com)
3	lec20 - Op-Amp, operational amplifier, Non-inverting amplifier (youtube.com)
4	lec24 - Introduction to Passive and Active Filters and op-amp as Low Pass Filter (youtube.com)

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project

Creativity in solutions and approaches

SEMESTER S3

INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Course Code	GNEST305	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate a solid understanding of advanced linear algebra concepts, machine learning algorithms and statistical analysis techniques relevant to engineering applications, principles and algorithms.
2. Apply theoretical concepts to solve practical engineering problems, analyze data to extract meaningful insights, and implement appropriate mathematical and computational techniques for AI and data science applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to AI and Machine Learning: Basics of Machine Learning - types of Machine Learning systems-challenges in ML- Supervised learning model example- regression models- Classification model example- Logistic regression-unsupervised model example- K-means clustering. Artificial Neural Network- Perceptron- Universal Approximation Theorem (statement only)- Multi-Layer Perceptron- Deep Neural Network- demonstration of regression and classification problems using MLP.(Text-2)	11
2	Mathematical Foundations of AI and Data science: Role of linear algebra in Data representation and analysis – Matrix decomposition- Singular Value Decomposition (SVD)- Spectral decomposition- Dimensionality reduction technique-Principal Component Analysis (PCA). (Text-1)	11

3	Applied Probability and Statistics for AI and Data Science: Basics of probability-random variables and statistical measures - rules in probability- Bayes theorem and its applications- statistical estimation-Maximum Likelihood Estimator (MLE) - statistical summaries- Correlation analysis- linear correlation (direct problems only)- regression analysis- linear regression (using least square method) (Text book 4)	11
4	Basics of Data Science: Benefits of data science-use of statistics and Machine Learning in Data Science- data science process - applications of Machine Learning in Data Science- modelling process- demonstration of ML applications in data science- Big Data and Data Science. (For visualization the software tools like Tableau, PowerBI, R or Python can be used. For Machine Learning implementation, Python, MATLAB or R can be used.)(Text book-5)	11

Course Assessment Method
(CIE: 40 marks , ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	Each question carries 9 marks. Two questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concept of machine learning algorithms including neural networks and supervised/unsupervised learning techniques for engineering applications.	K3
CO2	Apply advanced mathematical concepts such as matrix operations, singular values, and principal component analysis to analyze and solve engineering problems.	K3
CO3	Analyze and interpret data using statistical methods including descriptive statistics, correlation, and regression analysis to derive meaningful insights and make informed decisions.	K3
CO4	Integrate statistical approaches and machine learning techniques to ensure practically feasible solutions in engineering contexts.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3								
CO2	3	3	3	3								
CO3	3	3	3	3								
CO4	3	3	3	3								

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Linear Algebra	Gilbert Strang	Wellesley-Cambridge Press	6 th edition, 2023
2	Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Géron	O'Reilly Media, Inc.	2 nd edition, 2022
3	Mathematics for machine learning	Deisenroth, Marc Peter, A. Aldo Faisal, and Cheng Soon Ong	Cambridge University Press	1 st edition. 2020
4	Fundamentals of mathematical statistics	Gupta, S. C., and V. K. Kapoor	Sultan Chand & Sons	9 th edition, 2020
5	Introducing data science: big data, machine learning, and more, using Python tools	Cielen, Davy, and Arno Meysman	Simon and Schuster	1 st edition, 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data science: concepts and practice	Kotu, Vijay, and Bala Deshpande	Morgan Kaufmann	2 nd edition, 2018
2	Probability and Statistics for Data Science	Carlos Fernandez-Granda	Center for Data Science in NYU	1 st edition, 2017
3	Foundations of Data Science	Avrim Blum, John Hopcroft, and Ravi Kannan	Cambridge University Press	1 st edition, 2020
4	Statistics For Data Science	James D. Miller	Packt Publishing	1 st edition, 2019
5	Probability and Statistics - The Science of Uncertainty	Michael J. Evans and Jeffrey S. Rosenthal	University of Toronto	1 st edition, 2009
6	An Introduction to the Science of Statistics: From Theory to Implementation	Joseph C. Watkins	chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.math.arizo	Preliminary Edition.

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106198/
2	https://archive.nptel.ac.in/courses/106/106/106106198/ https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/resources/lecture-29-singular-value-decomposition/
3	https://ocw.mit.edu/courses/18-650-statistics-for-applications-fall-2016/resources/lecture-19-video/
4	https://archive.nptel.ac.in/courses/106/106/106106198/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender	6

	spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education , employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy and women/transgender empowerment initiatives .	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to	6

	mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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Course Assessment Method

(CIE: 50 marks , ESE: 50)

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5

	reflections)	3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks			50	

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.

- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S3

MACHINE DRAWING AND SOLID

MODELLING LABORATORY

Course Code	PCRUL307	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Engineering Graphics	Course Type	Lab

Course Objectives:

1. To equip students with the knowledge and skills required for proficient manual and computer-aided engineering drawing, including understanding BIS specifications, creating 2D assembly drawings, and using AutoCAD for drafting.
2. To provide students with an understanding of additive manufacturing technologies, focusing on Fused Deposition Modeling (FDM), and to offer hands-on experience in applying these techniques to practical engineering problems.

Expt. No.	Experiments
1	Introduction Principles of drawing Free hand sketching, manual drawing, CAD drawing etc. Code of practice for Engineering Drawing: -BIS specifications – lines, types of lines, dimensioning, sectional views, Welding symbols, riveted joints, keys, fasteners –bolts, nuts, screws, keys.
2	Fasteners: Sketching of conventional representation of 1. welded joints 2. Bolts and Nuts
3	Assembly drawings(2D): Stuffing box and Screw jack
4	Introduction to drafting software AutoCAD, basic commands, keyboard shortcuts. Coordinate and unit setting, Drawing, Editing, Measuring, Dimensioning, Plotting Commands, Layering Concepts, Matching, Detailing, Detailed drawings.
5	Assembly drawings(2D)with Bill of materials: Lathe Tailstock and Universal joint

6	Assembly drawings(2D)with Bill of materials: Connecting rod and Plummer block
7	Introduction to solid modeling software softwares , Basic commands and development of simple 3D drawings, Different file formats for 3D modelling(IGES, STL etc.)
8	Creating assembly models (3D)of Socket and spigot joint
9	Creating assembly models (3D) of Knuckle Joint
10	Creating assembly models (3D) of Rigid flange couplings
11	Introduction to FEA softwares: - FEA softwares like ANSYS, Abaqus etc, Structural analysis only

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate and apply the fundamental principles of engineering drawing, including BIS specifications, line types, dimensioning, sectional views, and welding symbols	K3
CO2	Use free-hand sketching and manual drawing skills, and gain competence in CAD drawing, specifically using AutoCAD, including basic commands, keyboard shortcuts, coordinate settings, and layering concepts.	K3
CO3	Create and interpret detailed 2D assembly drawings, including the preparation of Bill of Materials for components like stuffing boxes, screw jacks, lathe tailstocks, universal joints, connecting rods, and Plummer blocks.	K3
CO4	Demonstrate basic commands and techniques for developing simple 3D drawings using solid modelling software, and create 3D assembly models for mechanical joints and couplings, such as socket and spigot joints, knuckle joints, and rigid flange couplings	K3
CO5	Analyse the stresses and strains induced in rigid bodies using FEA method.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		1	1		1		2		
CO2	3	2	3		3				1	2		
CO3	3	3	3		2					2		
CO4	3		3		3				2	2		
CO5	3		2	2	3		1					2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Drawing	P I Varghese and K C John	VIP Publishers	
2	Machine Drawing	N. D. Bhatt and V.M. Panchal,	Charotar Publishing House	

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1.Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2.Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S3

ELECTRONIC CIRCUITS AND LOGIC DESIGN LABORATORY

Course Code	PCRUL308	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To design and test various circuits using Op-amp.
2. To design and test various combinational and sequential circuits

Expt. No.	Experiments
1	Design and set up of inverting and non-inverting amplifier, RC Integrator and Differentiator circuits using Op-Amp
2	Design and set up active Low pass and high pass filters using Op-Amp
3	Design and set up RC phase shift Oscillator using Op-amp
4	Design and set up Comparator and Schmitt trigger circuit using Op-amp
5	Design and set up Instrumentation Amplifier using Op-amp
6	Set up R-2R D/A convertor and Successive approximation A/D convertor using Op-amp
7	Realization of SOP, and POS functions using K map reduction
8	Realize Half adder and Full adder circuit using NAND gates
9	Realization of Combinational Circuits using Multiplexers
10	Study of counter ICs and Shift register ICs
11	Ring Counter and Johnson counter
12	Introduction to Verilog, Basic gates using Verilog

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.

Endorsement by External Examiner: The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Design, test and analyse circuits using Op-amps	K3
CO2	Design and test various combinational and sequential circuits	K3
CO3	Program basic gates using Verilog	K3
CO4	Collaborate in a team and document the observations and conclusion	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2							2	
CO2	3	3	3	2							2	
CO3	3	3	3	2	3						2	
CO4								3	3	3	3	

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Op-Amp and Linear Integrated Circuits	Ramakant Gayakwad	Pearson Education India	4 th Edition
2	Linear Integrated Circuits	DRoyChoudhury and Shail B Jain	New Age International Publishers	4 th Edition
3	Digital Fundamentals	Thomas Floyd	Pearson Education India	11 th Edition
4	Fundamentals of Digital Circuits	A.Anand Kumar	Prentice Hall India Private Limited	4 th Edition
5	Digital Principles and Design	Digital Principles and Design	Tata Mc Graw Hill	

Continuous Assessment (25 Marks)**1. Preparation and Pre-Lab Work (7 Marks)**

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 4

ROBOTICS AND AUTOMATION

SEMESTER S4
MATHEMATICS FOR ELECTRICAL SCIENCE – 4
(B Group)

Course Code	GBMAT401	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	Basic calculus	Course Type	Theory

Course Objectives:

1. To familiarize students with the foundations of probabilistic and statistical analysis mostly used in varied applications in engineering and science.
2. To expose the students to the basics of random processes essential for their subsequent study of analog and digital communication

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Random variables, Discrete random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Binomial distribution, Poisson distribution, Poisson distribution as a limit of the binomial distribution, Joint pmf of two discrete random variables, Marginal pmf, Independent random variables, Expected value of a function of two discrete variables. [Text 1: Relevant topics from sections 3.1 to 3.4, 3.6, 5.1, 5.2]	9
2	Continuous random variables and their probability distributions, Cumulative distribution function, Expectation, Mean and variance, Uniform, Normal and Exponential distributions, Joint pdf of two Continuous random variables, Marginal pdf, Independent random variables, Expectation value of a function of two continuous variables. [Text 1: Relevant topics from sections 3.1, 4.1, 4.2, 4.3, 4.4, 5.1, 5.2]	9

3	Confidence Intervals, Confidence Level, Confidence Intervals and One-side confidence intervals for a Population Mean for large and small samples (normal distribution and t-distribution), Hypotheses and Test Procedures, Type I and Type II error, z Tests for Hypotheses about a Population Mean (for large sample), t Test for Hypotheses about a Population Mean (for small sample), Tests concerning a population proportion for large and small samples. [Text 1: Relevant topics from 7.1, 7.2, 7.3, 8.1, 8.2, 8.3, 8.4]	9
4	Random process concept, classification of process, Methods of Description of Random process, Special classes, Average Values of Random Process, Stationarity- SSS, WSS, Autocorrelation functions and its properties, Ergodicity, Mean-Ergodic Process, Mean-Ergodic Theorem, Correlation Ergodic Process, Distribution Ergodic Process. [Text 2: Relevant topics from Chapter 6]	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the concept, properties and important models of discrete random variables and to apply in suitable random phenomena.	K3
CO2	Understand the concept, properties and important models of continuous random variables and to apply in suitable random phenomena.	K3
CO3	Estimate population parameters, assess their certainty with confidence intervals, and test hypotheses about population means and proportions using z-tests and the one-sample <i>t</i> -test.	K3
CO4	Analyze random processes by classifying them, describing their properties, utilizing autocorrelation functions, and understanding their applications in areas like signal processing and communication systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	-	-	-	-	-	-	-	2
CO2	3	3	2	2	-	-	-	-	-	-	-	2
CO3	3	3	2	2	-	-	-	-	-	-	-	2
CO4	3	3	2	2	-	-	-	-	-	-	-	2

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability and Statistics for Engineering and the Sciences	Devore J. L	Cengage Learning	9 th edition, 2016
2	Probability, Statistics and Random Processes	T Veerarajan	The McGraw-Hill	3 rd edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Probability, Random Variables and Stochastic Processes,	Papoulis, A. & Pillai, S.U.	McGraw Hill.	4 th edition, 2002
2	Introduction to Probability and Statistics for Engineers and Scientists	Ross, S. M.	Academic Press	6 th edition, 2020
3	Probability and Random Processes	Palaniammal, S.	PHI Learning Private Limited	3 rd edition, 2015
4	Introduction to Probability	David F. Anderson, Timo, Benedek	Cambridge	1 st edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105085/
2	https://archive.nptel.ac.in/courses/117/105/117105085/
4	https://archive.nptel.ac.in/courses/117/105/117105085/

SEMESTER S4

ROBOTIC MANIPULATORS AND CELL DESIGN

Course Code	PCRUT402	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT205	Course Type	Theory

Course Objectives:

1. Students will be able to apply mathematical modelling to kinematics, dynamics, control and trajectory planning of robotic manipulators
2. Students will be able to apply various concepts to design automated work cells that incorporate robotic manipulators.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Components and Structure of Robots: Symbolic Representation, Degrees of Freedom and Workspace, Classification of Robots - Power Source, Application Area, Method of Control, Geometry, Common Kinematic Arrangements (Structure and Workspace) - Articulated Configuration (RRR), Spherical Configuration (RRP), SCARA Configuration (RRP), Cylindrical Configuration (RPP), Cartesian configuration (PPP), Parallel Manipulator, Spherical wrist End Effectors: Mechanical, Magnetic and Vacuum Grippers, Active and Passive Grippers Kinematics of Robots: Rotations - Fundamental and Composite Rotations, Homogeneous coordinates, Translations and Rotations, Composite homogeneous transformations, Screw transformations, Inverse of Homogeneous Transformation Matrices, Other representations of Rotations: Definitions of RPY, Euler Angles and Quaternions	11

2	Forward Kinematics: Definition, Kinematic Chains, Denavit Hartenberg Representation, Assigning the coordinate frames, Algorithm based on DH Convention for Forward Kinematics, Forward Kinematics Examples - Two Link Planar Elbow Manipulator, 3 link cylindrical robot, Spherical Wrist, SCARA Manipulator only Inverse Kinematics: Definition, The General Inverse Kinematics Problem, General properties of solutions to the inverse kinematics problem, Theory of Kinematic Decoupling, Inverse Kinematics Example -SCARA Manipulator only	11
3	Velocity Kinematics: Derivation of the Jacobian – Angular Velocity, Linear Velocity, combining linear and angular Jacobians, Velocity Jacobian Examples given forward kinematic matrices - 2 link planar manipulator, SCARA manipulator only, Singularities – Definition and importance of identifying singularities. Dynamic Analysis: Force-mass-acceleration and torque-inertia-angular-acceleration relationships, Lagrangian Mechanics – Overview, Dynamic Analysis Examples – 1 DOF spring cart system, 2 DOF cart pendulum system, 2 DOF Polar Robot Arm only Robot Control: The control problem, Single axis PID control - Control Law, Block Diagram, Derivation of Closed Loop Transfer Function, advantages and disadvantages	11
4	Trajectory Planning: Definitions of Path planning and Trajectory Planning, Joint space versus Cartesian Space trajectory planning, Cubic Polynomial based trajectory planning, Point to Point and Continuous Path Planning Robot selection considerations for an application: Number of axes, work volume, capacity and speed, stroke and reach, Repeatability, Precision and Accuracy, Operating environment Robotic Work Cell Design: Robot Cell Layouts - Robot-centred Work cell, In-Line Robot Cell, Mobile Robot Cells, Multiple Robots and Machine Interference, Other Considerations in Work cell Design, Work cell Control - Sequence control, Operator interface, Safety monitoring, Interlocks, Error Detection and Recovery, The Work cell Controller Economic Analysis for Robotics: Basic Data Required - Type of Robot Installation, Cost Data Required for the Analysis, Methods of Economic	11

	Analysis - Payback Method, Equivalent Uniform Annual Cost Method, Return on Investment (ROI) Method (problems not required), Subsequent use of the Robot	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe various robotic manipulator configurations and end effectors	K2
CO2	Formulate forward and inverse kinematics for robotic manipulators	K3
CO3	Analyse robotic manipulator configurations using velocity kinematics and dynamic analysis	K4
CO4	Plan trajectories and control schemes for robotic manipulators	K3
CO5	Design robotic work cells for various applications	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3		3							3
CO2	3	3	3		3							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3
CO5	3	3	3		3	2					3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Dynamics and Control	Mark W. Spong, Seth Hutchinson, and M. Vidyasagar	Wiley	2nd edition, 2020
2	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall of India	1996
3	Introduction to Robotics - Analysis, Control, Applications	Saeed Benjamin Niku	John Wiley & Sons	2nd edition, 2010
4	Industrial Robotics – Technology, Programming and Applications	Mikell P. Groover, Mitchel Weiss, Roger N. Nagel, Nicholas G. Odrey, Ashish Dutta	Tata McGraw Hill	2nd edition, 2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics	S K Saha	McGraw-Hill	2008
2	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	2003
3	Introduction to Robotics – Mechanics and Control	John J. Craig	Pearson	4 th edition, 2017
4	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University Press	2006

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/107/106/107106090/
3	https://archive.nptel.ac.in/courses/112/107/112107289/
4	https://archive.nptel.ac.in/courses/112/105/112105249/

SEMESTER S4

MANUFACTURING AUTOMATION

Course Code	PCRUT403	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the principles and types of manufacturing automation.
2. Learn about key technologies and systems used in manufacturing automation.
3. Gain insights into the benefits and challenges of implementing automation in manufacturing.
4. Explore the future trends and advancements in manufacturing automation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Manufacturing Automation: Definition and importance, History and evolution of automation in manufacturing, Benefits of automation, Types of automation: Fixed, programmable, and flexible automation, Reasons for automating, Automation strategies	9
2	Automated Flow Lines and Assembly Systems: Detroit Type Automation and Flow Lines: Automated flow lines: Types and their selection, Methods of work part transport: Continuous, intermittent, and asynchronous, Transfer mechanisms, Controlling automated flow lines, Flow line performance analysis: Average production time, production rate, and line efficiency, Partial automation: Reasons, advantages, and drawbacks, Component quality control and effects of machine jamming, Automated Assembly Systems: Historical developments of the assembly	12

	process, Choice of assembly methods: Cost, production rate, labour availability, and market life, Advantages of automatic assembly, Design for automated assembly, Transfer systems: In-line and rotary.	
3	Feeders and Robotics in Manufacturing: Types of Feeders: Vibratory bowl feeders: Mechanics and analysis, Effect of frequency, track acceleration, and vibration angle, Non-vibratory feeders: Types (e.g., reciprocating tube hopper, center board hopper, external gate hopper, rotary disk feeder), Analysis and design of part orienting devices, feed tracks, and part placing mechanisms. In-Bowl and Out-of-Bowl Toolings: Typical orienting systems: Wiper blade, pressure break, slot in the track, Analysis of part orienting systems, Examples of out-of-bowl toolings, Integration of feeders with automated systems	11
4	Safety and Implementation in Manufacturing Automation: Safety and Standards in Manufacturing Automation: Safety protocols and risk assessment, International standards (ISO, ANSI), Regulatory compliance, Implementation and Integration of Automation Systems: Planning and design of automated systems, Integration with existing manufacturing processes, Economic and Social Impacts of Automation: Cost-benefit analysis, Impact on labour and employment, Ethical considerations.	10

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the Principles and Types of Automation	K2
CO2	Analyze Automated Flow Lines and Assembly Systems	K4
CO3	Design and Integrate Feeder Systems	K6
CO4	Apply Robotics in Manufacturing	K3
CO5	Evaluate Safety Standards and Economic Impacts	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1	1	1	1	1	1	1	1	1	1
CO2	3	3	3	3	2	2	2	2	2	1	2	2
CO3	3	3	3	3	3	2	2	2	2	3	2	3
CO4	3	3	3	3	3	2	2	2	2	3	2	3
CO5	3	2	3	2	2	3	3	3	2	3	3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell P. Groover	Prentice-Hall of India Pvt. Ltd, New Delhi, 1989.	Fourth edition
2	Assembly Automation and Product Design	G. Boothroyd	Taylor & Francis	Second edition, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics in CIM Systems	James A. Rehg, Henry W. Kraebber	Pearson	Fifth edition, 2002
2	Robotics: Control, Sensing, Vision, and Intelligence	K.S. Fu, R.C. Gonzalez, and C.S.G. Lee	McGraw-Hill	First edition, 2008

SEMESTER S4

MICROCONTROLLERS AND EMBEDDED SYSTEMS

Course Code	PBRUT404	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBRUT304	Course Type	Theory

Course Objectives:

1. Students will understand the basic concepts and architecture of microcontrollers and embedded systems.
2. Students will acquire skills in interfacing microcontrollers with sensors and actuators.
3. Students will apply theoretical knowledge through hands on projects, designing and implementing embedded systems to solve real world problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	8051 microcontrollers. Review-Basics of Computer Architecture: Basic block diagram of a computer, System bus and components, RISC and CISC architectures, Von Neumann and Harvard architecture, 8051 microcontrollers: Difference between microprocessor and microcontroller architecture, 8051 architecture, Register organization - special function registers, Memory organization 8051 microcontrollers: Instruction set- Arithmetic, Data transfer, Control transfer, Bit- oriented and Data exchange instructions (familiarization only, not for exams), Addressing modes	11
2	8051 microcontroller Programming and Peripherals. Timers/Counters- TMOD and TCON registers, Serial Communication-SBUF, SCON, PCON registers and their functions (details of different modes of operation not included), Interrupts in 8051- Timer flag interrupt, Serial port interrupt, external interrupts and interrupt priority	11

	Interfacing of peripherals – LED, LCD, ADC, DAC, LM35 temperature sensor, 7 segment LED (embedded C programming)	
3	Embedded Systems Overview and introduction to OS Introduction to Embedded Systems: Definition, Features, applications of embedded systems in Robotics (examples only) Embedded System Architecture: HW - Processor, Controller, SoC, Memory, Peripherals; SW - Application, Middleware, OS, Device Drivers, Tool chain- Assembler, Interpreter, Compiler, Linker, Loader, Debugger Embedded system design process: Requirement Analysis, Specification Development, HW & SW Co-Design and Development, Module Integration, Testing. (2 hours) Introduction to OS: Embedded Operating system basic concepts: Functional layers in a computer system, Kernel Functions (Overview only), Tasks/ Processes: definition, five states of a process/task, RTOS – definition, comparison of General Purpose OS and Real Time OS (3 hours)	11
4	Embedded System Board Study, communication protocols and IoT. Embedded System Board Study (Arduino Uno): Arduino Uno Board: Board Study (Board level Block schematic) - Chip (Features only - architecture not needed), GPIO, Memory, familiarization of Arduino IDE (2 hours) Programming: Sample code for LED and Switch, DC motor, Stepper motor control (2 hours) Internet of Things (IoT): An Introduction to Internet of Things (overview only): Basic components of IoT- Building blocks of IoT, IoT Application framework, The Internet Gateway, Edge IT, IoT Service Providers, IoT Infrastructure, IoT Development service, Comparison of M2M and IoT technology. (2 hours) Role of cloud technology in IoT computing: Definition of cloud computing, types of cloud services (examples for various types only), Need of cloud computing in IoT applications (1 hour) Communication Protocols: UART, I2C, SPI and USB (2 hours)	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x2 =16 marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Interpret the internal architecture of 8051 microcontroller	K2
CO2	Implement the interfacing of 8051 microcontroller with peripheral devices using ALP/Embedded C	K3
CO3	Describe the architecture, design concepts of embedded systems and concepts of embedded operating system	K2
CO4	Explain the fundamentals of Arduino uno board and concepts of IoT	K2
CO5	Implement embedded systems using Arduino and IoT technologies.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2								2	2
CO2	3	3	3			2					2	2
CO3	3	2	2								2	2
CO4	3	2	2								2	2
CO5	3	3	3		3	3	3	3	3	3	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The 8051 Microcontroller and Embedded Systems: Using Assembly and C	Muhammad Ali Mazidi	Pearson	2nd Edition, 2007
2	Embedded Systems: An Integrated Approach	Lyla B Das	Pearson	1e, 2012
3	Beginning Arduino	Michael McRoberts	Apress	1e, 2011
4	The 8051 Microcontroller	Kenneth Ayala	Cengage Learning	3e, 2012
5	Internet of Things	Jeeva Jose	Khanna Publishing House	1e, 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Embedded Systems Architecture, programming and Design	Raj Kamal	Tata McGraw-Hill,	3e, 2013
2	Embedded Systems Architecture- A Comprehensive Guide for Engineers and Programmers	Tammy Noergaard	Newnes – Elsevier	2e, 2012
3	Arduino: The complete guide to Arduino for beginners, including projects, tips, tricks and programming	James Arthur	Ingram	1e, 2019
4	Introduction to IoT	Sudip Misra, Anandarup Mukherjee, Arijit Roy	Cambridge University Press	1e, 2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Microprocessors And Microcontrollers by Prof. Santanu Chattopadhyay IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_ee12/preview
2	Microprocessors And Microcontrollers by Prof. Santanu Chattopadhyay IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_ee12/preview
3	NPTEL Course - Embedded Systems Design, Prof. Anupam Basu, IIT Kharagpur Embedded Systems Design - Course (nptel.ac.in)
4	Arduino by Prof Kannan Moudgalya - Principal Investigator of Spoken Tutorial Project , Indian Institute of Technology Bombay https://onlinecourses.swayam2.ac.in/aic20_sp04/preview NPTEL Course- Introduction To Industry 4.0 And Industrial Internet Of Things, By Prof. Sudip Misra , IIT Kharagpur. Introduction to Industry 4.0 and Industrial Internet of Things - Course (nptel.ac.in) https://www.coursera.org/learn/introduction-to-internet-of-things/ , IIT Bombay

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation / Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

Project Assessment and Evaluation criteria (30 Marks)

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S4
ROBOT SAFETY AND STANDARDS

Course Code	PERUT411	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand basic safety guidelines for robotic applications
2. Identify key elements of the robot safety standard
3. Apply provisions of the standard in daily operations

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ROBOT SAFETY: Safety Basics and Features, Need for Safety and the Role of Engineers Robot Safety: Problems and Hazards, Roles of Robot Manufacturers and Users in Robot Safety, Safety Considerations in Robot Design, Installation and Operations, Robot Safeguard Approaches, Robot-Related Safety Problems Causing Weak Points in Planning, Design, and Operation, Workplace Safety	9
2	ROBOT SYSTEM RELIABILITY: Reliability Basics and Configurations, Classifications of Robot Failures and their Causes and Corrective Measures, Robot Effectiveness Dictating Factors, Robot-Related Reliability Measures, Robot Reliability Analysis, Reliability Analysis of Hydraulic and Electric Robots.	9
3	ROBOETHICS: Roboethics and Levels of Robomortality, Ethics Fundamental Elements and Theories, Top-Down and Bottom Up Roboethics Approach, Ethics in Human-Robot Symbiosis, Robot Rights, Socialized Roboethics, Ethical Issues of Socialized Robots, Case Studies, Additional Roboethics Issues	9
4	ROBOT STANDARDS: The Concept of Standard and Standardization, Characteristics and Benefits of Standardization, Standardization Bodies,	9

	Standard-Setting, Robot Standards: Electrical Interfaces on Robots for Industrial Environments- End Effector, Manipulating Industrial Robots — Mechanical Interfaces, Safety Requirements for Robotics in Industrial Environment, Safety Design for Industrial Robot Systems, Connectors for Electronic Equipment. Safety of Machinery, Performance Criteria and Related Test Methods for Service Robots	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic safety guidelines for robotic applications	K2
CO2	Understand robot system reliability	K2
CO3	Understand roboethics	K2
CO4	Apply provisions of the standard in daily operations	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1			2		2				2
CO2	2		1			2		2				2
CO3	2		1			2		2				2
CO4	2		1			2		2				2
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot System Reliability and Safety: A Modern Approach	B. S. Dhillon	CRC Press, Taylor & Francis Group	2015
2	Roboethics: A Navigating Overview	Spyros G, Taffetas	1st Edition, Springer	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human-Robot Interaction: Safety, Standardization, and Bench marking	Paolo Barattini, Federico Vicentini	CRC Press	2019
2	Robot Ethics: The Ethical and Social Implications of Robotics	Abney, Keith, Bekey, George A, Lin, Patrick	The MIT Press	2012
3	ISO/TC 299, Robotics	ISO	ISO	2018
4	Robot Framework Test Automation	SumitBisht	PACKT Publishing	2013

SEMESTER S4

CNC MACHINES

Course Code	PERUT412	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide knowledge about computer numerical control system.
2. To provide knowledge on write manual part program and program using APT

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Computer Numerical Control (CNC) - Numerical control - Functions of a machine tool - Concept of numerical control - Historical Development - Definition - Advantages of CNC machine tools - Evolution of CNC - Advantages of CNC - Limitations of CNC - Features of CNC - The Machine Control Unit (MCU) for CNC - Classification of CNC Machine Tools - CNC Classifications □ Features of CNC Machining Centers	8
2	Numerical controlled Machines – CNC machines, basic structures of machining and turning centers. Tools, tool holders and tool indexing. Axis configurations and fundamentals of CNC codes. Datum and tool offset settings, Incremental and absolute programming, Canned cycles. Practical programming (simple) G, M codes. APT programming	10
3	CNC Turning & milling – Basics - Fundamentals of CNC lathe. Familiarization of control panel.-Fundamentals of CNC programming.- Part programming techniques - Work piece setting methods.-Tool setting methods.-Programming by using multiple tools.- Project on machine by using different cycles	10
4	Types of CNC machines–pick and place machine-CNC 3D printer– CNC drilling machine– CNC lathe machine–5 axis CNC machine-CNC milling machine-laser cutting CNC machine-grinding CNC machine. Automatic tool changing techniques-its advantages.	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE)

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand about basics of CNC system and CNC machine tools	K2
CO2	Understand the operations in a CNC machine and optimally choose the parameters and settings for a specific machining requirement.	K2
CO3	Decide the datum and tool offset parameters for the required machining operation and to manually program the CNC machine.	K2
CO4	Understand about programs and methods in CNC turning and milling.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	2		2		3							3
CO3	3	3										3
CO4	3	2										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Control of Manufacturing Systems	Yoram Koren	McGraw Hill Education	1st edition, 2017
2	Computer Numerical Control Simplified	Arthur Gill, Peter Smid , S Karr	Industrial Press Inc.	2000
3	Computer Numerical Control Machines and Computer Aided Manufacture	P. Radhakrishnan	New Age International Pvt Ltd	1st edition, 2018
4	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell Groover	Pearson	4 th edition, 2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CAD/CAM: Computer-Aided Design and Manufacturing	M. Groover	Pearson Education	1st edition, 2003
2	Secrets of 5-axis Machining	Karlo Apro	Industrial Press Inc	2008
3	CNC Fundamentals and Programming	Agarwal	Charotar Publishing House Pvt. Ltd	2 nd edition, 2014

Video Links (NPTEL, SWAYAM...)	
Sl. No.	Link ID
1	https://plw.indianrailways.gov.in/uploads/File/dmw/CNC%20Machine%20Programming%20Course.pdf

SEMESTER S4

LEAN MANUFACTURING

Course Code	PERUT413	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NIL	Course Type	Theory

Course Objectives:

1. Understand Lean Manufacturing Fundamentals
2. Apply Lean Tools and Techniques
3. Enhance Quality and Process Improvement
4. Promote Lean Culture and Employee Engagement

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Lean Manufacturing: The Mass Production System vs. Lean Production System, Origins and Necessity of Lean Manufacturing, Lean Revolution: Toyota Production System (TPS), Systems Thinking in Lean, Basic Principles of Lean Production, Customer Focus in Lean, Overview of Waste (MUDA)	9
2	Tools and techniques for Lean Manufacturing: 5S System (Sort, Set in Order, Shine, Standardize, Sustain), Total Productive Maintenance (TPM), Standardized Work and its Elements, Introduction to Just In Time (JIT), Kanban System and Pull Production, Value Stream Mapping (VSM)	9
3	Quality improvement: Jidoka (Automation with a Human Touch), Poka-Yoke (Mistake Proofing) Systems, Six Sigma: Concepts and Methodology, Introduction to Quality Function Deployment (QFD), Failure Modes and Effects Analysis (FMEA), Total Quality Management (TQM) Tools and Techniques	9
4	Employee involvement and continuous improvement: Worker Involvement Techniques (Quality Circles, Suggestions Programs), Kaizen Philosophy and Practices, Hoshin Planning (Systematic Planning Methodology), Lean Culture and its Importance, Overview of Concurrent Engineering, Basics of Design of Experiments (DoE)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental principles and historical development of Lean Manufacturing, emphasizing waste reduction and systems thinking.	K2
CO2	Apply primary Lean tools such as 5S, TPM, standardized work, JIT, and Kanban to improve operational efficiency.	K3
CO3	Implement quality improvement techniques like Jidoka, Poka-Yoke, Six Sigma, and TQM to enhance product quality and reliability.	K4
CO4	Foster employee involvement through techniques like Kaizen and quality circles, and implement systematic planning methodologies to sustain Lean practices.	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	1	1							2
CO2	3	1	1	1	3							2
CO3	3	1	3	1	3							2
CO4	3	3	1	1	1						3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Lean Production Simplified	Pascal Dennis	Tata McGraw Hill	Third edition, 2015
2	The Toyota Way: 14 Management Principles from the World's Greatest Manufacturer	Jeffrey K. Liker	Tata McGraw Hill	First edition, 2004
3	Lean Thinking: Banish Waste and Create Wealth in Your Corporation	James P. Womack and Daniel T. Jones	Simon & Schuster India	Second edition, 2002
4	The Machine That Changed the World: The Story of Lean Production	James P. Womack, Daniel T. Jones, and Daniel Roos	Free Press India	Revised edition, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Lean Six Sigma: Combining Six Sigma Quality with Lean Production Speed	Michael L. George	McGraw Hill	First edition, 2002
2	Toyota Production System: Beyond Large-Scale Production	Taiichi Ohno	Taylor & Francis, Inc.	Reprint Edition, 2019
3	Learning to See: Value Stream Mapping to Add Value and Eliminate MUDA	Mike Rother and John Shook	Lean Enterprise Institute	First edition, 1999

SEMESTER S4

NETWORK THEORY

Course Code	PERUT 414	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)		Course Type	Theory

Course Objectives:

1. To make the students capable of analysing any linear time invariant electrical network.
2. To study time domain, phasor and Laplace transform methods of linear circuit analysis.
3. To study the transient response of networks subject to test signals.
4. To develop understanding of the concept of two port networks

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Mesh and Node Analysis: Introduction to circuit variables and circuit elements, Review of Kirchhoff's Laws. Independent and dependent Sources, Source transformations. wye-delta transformation and delta-wye transformation. Mesh and node analysis of network containing independent and dependent sources. Super mesh and Super node analysis of network containing independent and dependent sources.	9
2	Network Theorems (applied to both dc circuits having dependent source and independent source) Thevenin's theorem, Norton's theorem, Superposition theorem, Maximum power transfer theorem Reciprocity theorem,	9
3	Application of Laplace Transforms: Review of Laplace Transforms Initial value theorem & Final value theorem (Proof not necessary) Transformation of basic signals and circuits into s-	9

	domain. Transient analysis of RL, RC, and RLC networks with impulse, step, pulse, exponential and sinusoidal inputs . Analysis of networks with transformed impedance and dependent sources	
4	Two port networks Impedance, Admittance, Transmission and Hybrid parameters of two port network. Interrelationship among parameter sets. Series and parallel connections of two port networks. Reciprocal and Symmetrical two port network.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyse electrical circuits using the laws of circuit analysis	K4
CO2	Solve complex electrical circuits using network theorems.	K3
CO3	Apply Laplace transform to study the transient behaviour of RLC network for given inputs	K3
CO4	Analyse the two-port network using network parameters.	K4
CO5	Derive the relationship between different parameters of a two-port network.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										3
CO2	3	3										3
CO3	3	3										3
CO4	3	3										3
CO5	3	3										3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Network Analysis and Synthesis	Ravish R.Singh	Mc Graw Hill Education (India) Private Limited	2/e 2019
2	Network Analysis	Valkenburg V	Pearson	3/e 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Circuits and Networks- Analysis and Synthesis	Sudhakar A, Shyammohan S. P	Mc Graw Hill	5/e 2017
2	Networks and Systems	Choudhary R	New Age International	2/e 2013
3	Electric Circuit-Schaum's Outline series	Edminister	Mc Graw Hill	5/e 2009
4	Network Analysis and Applications	William D Stanley	Pearson	4/e 2006
5	Electric Circuits and Networks	K.S Suresh Kumar	Pearson	1/e 2008
6	Network Analysis and Synthesis	Franklin F Kuo	Wiley India	2/e 2012

SEMESTER S4

MACHINE TOOLS AND ADVANCED MANUFACTURING PROCESSES

Course Code	PERUT 416	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT303	Course Type	Theory

Course Objectives:

1. To provide comprehensive knowledge of the wide range of machine tools, parts and operations
2. To understand the mechanics of machining
3. To provide comprehensive knowledge on advanced machining process and effect of its parameters on the machinability.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	General purpose machine tools – types and classification of machine tools – types and classification of lathe – methods of holding work and tool –lathe accessories and attachments –lathe operations - Shaping, Planing and Slotting machines – Work holding devices-types of operations - Drilling and boring Machines – Drill bit nomenclature – boring tools and reamers	9
2	Machinability- Cutting fluids, Milling and milling machine, Milling tool nomenclature Indexing head Different indexing methods -Grinding, honing and lapping – types of grinding machines-operations: Grinding mechanisms – Grinding wheels, Broaching machines –different machines – cutter for broaching – broaching processes.	9
3	Gear cutting –methods in gear production – form cutters –gear generating machines – gear hobbing machines – gear broaching –gear finishing, Micromachining: Diamond turn mechanism, Advanced finishing processes: - Abrasive Flow Machining, Magnetic Abrasive Finishing.- Magnetorheological Finishing, Magnetorheological Abrasive Flow Finishing	9

4	Non Traditional machining processes: - EDM, USM, ECM, LBM, EBM, PAM, IBM, AJM, AWJM, High velocity forming of metals - Sheet metal forming - explosive forming - Electro hydraulic forming - Electro Magnetic Forming.	9
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Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze various machining tools and understand various parts & operations performed by each machine tools.	K4
CO2	Analyze the tool nomenclature of different cutting tools used in each machining processes.	K4
CO3	Understand the limitations of various machining process with regard to shape formation and surface texture.	K2
CO4	To be conversant with the advanced machining process and to appreciate the effect of process parameters on the surface integrity aspects during the advanced machining process.	K2
CO5	To categorize the various nontraditional material removal process based on energy sources and mechanism employed.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	3									2	1
CO2	2	3									2	1
CO3	2	1		2	2							2
CO4	3				2							2
CO5	2				2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Workshop Technology,	Chapman W. A. J.	Viva books (P) Ltd	
2	Production Technology	HMT	Tata McGraw-Hill	
3	Introduction to Micromachining	V.K. Jain	Narosa Publishers	
4	Developments in high speed metal forming	Davies K. and Austin E.R	The machinery publishing Co.	1970

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Machine Tools	Chernov	MIR Publication	
2	Elements of workshop technology	HajraChoudary	Media Publishers.	
3	Advanced Machining Process	V.K. Jain		

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_me04/preview
2	https://onlinecourses.nptel.ac.in/noc21_me04/preview
3	https://onlinecourses.nptel.ac.in/noc22_me119/preview
4	https://onlinecourses.nptel.ac.in/noc22_me119/preview

SEMESTER S4

ELECTRICAL CIRCUITS AND MACHINES

Course Code	PERUT417	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST104	Course Type	Theory

Course Objectives:

1. Equip students with the ability to analyze DC and AC circuits using various circuit theorems and parameters
2. Develop an understanding of resonance in series and parallel circuits, including the concepts of Q Factor, half power frequency, resonant frequency, and bandwidth, and enhance skills in analyzing circuits in the s-domain
3. Provide comprehensive knowledge of the operating principles, construction, types, equations, characteristics, and applications of transformers, DC machines, induction machines, and synchronous machines

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Circuit theorems: DC and AC circuit analysis with dependent and independent sources applying Superposition principle, Source transformation, Thevenin's, Norton's and Maximum Power Transfer theorems - Reciprocity theorem. Two port networks: Driving point and transfer functions – Z, Y, h and T parameters - Conditions for symmetry & reciprocity	9
2	Resonance: Introduction – series resonance – parallel resonance – Definition: Q Factor- Half power frequency resonant frequency – Bandwidth. Transformed circuits in s-domain: Transform impedance/admittance of R, L and C - Mesh analysis and node analysis of transformed circuits in s-domain. Transfer Function representation – Poles and zeros.	9

3	Transformers: Operating principle, classification, construction, emf equation, phasor diagrams, equivalent circuit model, losses & efficiency. D.C. Machines: Operating principle, generator & motor action, construction, types of excitations, emf & torque equations, power stages & efficiency, characteristics & application of d.c generators and d.c motors	9
4	Induction Machines: Three-phase induction motors. Principle of operation, construction, types, emf equation of an AC Machine, torque developed in an induction motor, equivalent circuit model, torque-speed characteristics, Single phase induction motors, Types of Single-Phase Induction motor, starting, application Synchronous Machines: Construction, types & operating principle of synchronous generator, voltage regulation-EMF method, Power Angle characteristics, Synchronous Motor, principle, starting, hunting, damper windings	9

Course Assessment Method

(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply circuit theorems to solve complex DC and AC electric networks and develop the representation of two-port networks.	K2
CO2	Understand the concept of resonance in series and parallel circuits	K2
CO3	Understand the construction, operation, various tests, efficiency, voltage regulation of transformers and DC machines	K2
CO4	Understand the operation, efficiency and speed regulation and applications of induction machines and synchronous machines.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3	2		2	2						3
CO4	3	3	3		2	2						3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory and Problems in Electric circuits	Joseph A. Edminister and Mahmood Nahvi	McGraw Hill	7th Edition, 2017.
2	Network Analysis and Synthesis	Ravish R. Singh	McGraw-Hill	2nd Edition 2019
3	Electric Machinery	Fitzgerald, Kingslay, Umans	McGraw-Hill	7th Edition 2020
4	Electric Machines	Nagrath and Kothari	McGraw-Hill Education	5th Edition 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Circuit Analysis	Hayt and Kemmerly	McGraw Hill	9th Edition, 2020
2	Network Analysis	Van Valkenberg	Prentice Hall India Learning Pvt. Ltd.	Revised 3 Edition, 2019.
3	Electric Machinery and Transformer	Guru, Hiziroglu	Oxford University press	3rd Edition 2001

SEMESTER S4

COMPUTER ORGANISATION AND ARCHITECTURE

Course Code	PERUT418	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Course aims to provide a comprehensive understanding of computer architecture and organization, covering fundamental concepts such as bus structures, instruction cycles, and control methods.
2. Course focuses on practical aspects like assembly language, machine language interpretation, MIPS addressing modes, pipelining, and memory systems, equipping students with the knowledge to efficiently execute and manage programs.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Functional units- basic operational concepts, bus structures-bus operations-Arbitration, instruction Cycle, instruction sequencing and execution-Fundamental concepts-execution of a complete instruction-data path – Single bus and Multiple bus organization- hardwired control- microprogrammed control	9
2	Architecture: Assembly language, Instructions, Operands, Registers, Memory and Constants. Machine Language: R-Type, I-Type, J-Type Instructions, Interpreting machine language code. MIPS Addressing Modes- Register only, Immediate, Base, PC-relative, Pseudo-direct Steps for executing a program – compilation, Assembling, Linking, Loading	9
3	Pipelining: A basic MIPS implementation-Building a Datapath-Control implementation Scheme-Pipelining-pipelined data path and control-handling data hazards and control hazards-exceptions	9

4	Memory System: Basic Concepts-Semiconductor RAM -ROM-Memory Characteristics & Hierarchy-Cache memories-improving Cache Performance-Virtual memory-input output organization - programmed I/O, Interrupt driven I/O, DMA	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the role of functional units and various architectural features	K2
CO2	Summarize data representation and execution procedure of an ALU	K2
CO3	Identify the factors that degrade the pipeline performance and its counter measures	K2
CO4	Explain the performance of memory	K2
CO5	Understand the different ways of communication with I/O devices and interfaces	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Design and Computer Architecture	David Money Harris & Sarah L. Harris	Morgan Kaufmann – Elsevier	2009
2	Computer Organisation and Design	David A. Patterson and John L. Hennessey	Morgan Kaufmann	Fourth Edition
3	Computer Architecture and Organization: From 8085 to core2Duo and Beyond	Subrata Ghoshal	Pearson	2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Organization	Carl Hamacher	Mc Graw Hill	Fifth Edition
2	Computer Architecture and Organisation	John P Hayes	Mc Graw Hill	
3	Computer Organisation and Architecture	William Stallings	Pearson Education	5/e
4	Computer System Architecture	Mano M M	Prentice Hall of India	3/e

SEMESTER S4

ADDITIVE MANUFACTURING

Course Code	PERUT415	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT303	Course Type	Theory

Course Objectives:

1. To provide comprehensive knowledge of the wide range of additive manufacturing processes, capabilities and materials
2. To understand the software tools and techniques used for additive manufacturing.
3. To create physical objects that facilitates product development/prototyping requirements

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Additive manufacturing: Importance of Additive Manufacturing- Basic principle of additive manufacturing- Procedure of product development in additive manufacturing. Classification of additive manufacturing processes, Materials used in additive manufacturing- Benefits & Challenges in Additive Manufacturing.	9
2	Basic Concept — Digitization techniques — Model Reconstruction — Data Processing for Additive Manufacturing Technology: CAD model preparation — Part Orientation and support generation — Model Slicing —Tool path Generation- Introduction to slicing softwares: Cura, 3D Printing-STL Format, STL File Problems, consequence of building validand invalid tessellated models, STL file Repairs: Generic Solution, other Translators, Newly Proposed Formats.	9

3	Principle, process parameters, advantages and applications of: Fused Deposition Modelling (FDM), Selective Laser Sintering (SLS), Stereo Lithography (SLA). Laminated Object Manufacturing (LOM), Electron Beam Melting (EBM), Laser Engineering Net Shaping (LENS), LIGA process, Selection Laser Melting (SLM), 4D printing, hybrid layered manufacturing, segmented object manufacturing.	9
4	Direct processes: - Rapid Prototyping, Rapid Tooling, Rapid Manufacturing; Indirect Processes: - Indirect Prototyping, Indirect Tooling, Indirect Manufacturing. Applications and case studies of Additive Manufacturing: Biomedical- Manufacturing- Aerospace- Automotive- Food- Electronics.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate appropriate level of understanding on principles of additive manufacturing processes.	K2
CO2	Explain slicing operations in additive manufacturing	K2
CO3	Understand liquid and solid based additive manufacturing system	K2
CO4	understand the principles, process parameters, advantages, and applications of advanced manufacturing technologies including FDM, SLS, SLA, LOM, EBM, LENS, LIGA, SLM, 4D printing, hybrid layered manufacturing, and segmented object manufacturing.	K2
CO5	Distinguish between direct and indirect additive manufacturing processes, including rapid prototyping, tooling, and manufacturing, and understand their applications and case studies in biomedical, manufacturing, aerospace, automotive, food, and electronics industries.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3											3
CO3	3											3
CO4	3											3
CO5	3											3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Additive Manufacturing Methodologies: Rapid Prototyping to Direct Digital Manufacturing	Gibson, I, Rosen, D W., and Stucker,B	Springer	2010
2	Rapid prototyping: Principles and applications	Chua C.K., Leong K.F., and Lim C.S	World Scientific Publishers	2010
3	3D Printing and Additive Manufacturing: Principles and Applications: Fourth Edition of Rapid Prototyping	Chee Kai Chua, Kah Fai Leong	World Scientific Publishers	2014
4	Rapid prototyping	Gebhardt A	Hanser Gardener Publication	2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Rapid Prototyping and Engineering applications: A tool box for prototype development	Liou L.W. and Liou F.W	CRC Press	2007
2	Rapid Prototyping: Theory and practice	Kamrani A.K. and Nasr E.A.	Springer	2006
3	Laser Metal Deposition Process of Metals, Alloys, and CompositeMaterials, Engineering Materials and Processes	Mahamood R.M	Springer	2018
4	Stereolithography and other RP&M Technologies: from Rapid Prototypingto Rapid Tooling	Paul F Jacobs	Society of Manufacturing Engineers and the Rapid Prototyplng Association, New York	1996

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/103/112103306/
2	https://archive.nptel.ac.in/courses/112/103/112103306/
3	https://archive.nptel.ac.in/courses/112/103/112103306/
4	https://archive.nptel.ac.in/courses/112/103/112103306/

SEMESTER S3/S4

ECONOMICS FOR ENGINEERS

Course Code	UCHUT346	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of fundamental economic principles essential for effective decision-making in engineering contexts.
2. To enable students to apply economic analysis to production decisions, cost management, and market strategies in engineering practice.
3. To equip students with the ability to evaluate macroeconomic scenarios, financial methods, and investment decisions relevant to engineering projects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basic economic problems – Production Possibility Curve – Utility – Law of diminishing marginal utility –Demand: Factors determining demand – Law of Demand – Demand curve- Price elasticity of demand- measurement of price elasticity and its applications – Supply: factors determining supply - Law of supply – Supply curve- Equilibrium price determination- Changes in demand and supply and its effects on equilibrium price and quantity Production: Production function - Law of variable proportion –Returns to scale- Cobb-Douglas Production Function	6
2	Cost: Cost concepts – Private cost and social cost – Sunk cost – Opportunity cost -Explicit and implicit cost –Short run cost curves –Long run average cost curve -Revenue concepts – Break-even point Market: Perfect Competition – Monopoly - Monopolistic Competition (features and equilibrium of a firm) - Oligopoly – Features – Kinked demand model	6

3	National income: Concepts (GDP, GNP and NNP)– Final goods and Intermediate goods - Methods of Estimation –output method – expenditure method-- Difficulties in the measurement of national income. Inflation: Causes and Effects – Measures to Control Inflation - Monetary and Fiscal policies – Repo and reverse repo rate	6
4	Value Analysis and value Engineering: Cost Value, Exchange Value, Use Value, Esteem Value - Aims, Advantages and Application areas of Value Engineering - Value Engineering Procedure Capital Budgeting: Time value of money - Net Present Value Method - Benefit Cost Ratio – Internal Rate of Return – Payback – Accounting Rate of Return.	6

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/Case Study/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
10	15	12.5	12.5	50

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• Minimum 1 and Maximum 2 Questions from each module. • Total of 6 Questions, each carrying 3 marks (6x3 =18 marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 2 sub divisions. Each question carries 8 marks. (4x8 = 32 marks)	50

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamentals of various economic issues using laws and learn the concepts of demand, supply, elasticity and production function.	K2
CO2	Develop decision making capability by applying concepts relating to costs and revenue, and acquire knowledge regarding the functioning of firms in different market situations.	K3
CO3	Outline the macroeconomic principles of monetary and fiscal systems and national income.	K2
CO4	Make use of the possibilities of value analysis and engineering, and take investment decisions through capital budgeting techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	1	-	-	-	-	1	-
CO2	-	-	-	-	-	1	1	-	-	-	1	-
CO3	-	-	-	-	1	-	-	-	-	-	2	-
CO4	-	-	-	-	1	1	-	-	-	-	2	-

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Managerial Economics	Geetika, Piyali Ghosh and Chodhury	Tata McGraw Hill,	2015
2	Engineering Economy	H. G. Thuesen, W. J. Fabrycky	PHI	1966
3	Engineering Economics	R. Paneerselvam	PHI	2012
4	Financial Management	I M Pandey	Vikas Publishing House	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Economy	Leland Blank P.E, Anthony Tarquin P. E.	Mc Graw Hill	7 TH Edition
2	Indian Financial System	Khan M. Y.	Tata McGraw Hill	2011
3	Engineering Economics and analysis	Donald G. Newman, Jerome P. Lavelle	Engg. Press, Texas	2002
4	Contemporary Engineering Economics	Chan S. Park	Prentice Hall of India Ltd	2001
5	Financial Management: Theory and Practice	Prasanna Chandra	Mc Graw Hill	2007

MODEL QUESTION PAPER				
APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY				
THIRD SEMESTER B. TECH DEGREE EXAMINATION, MONTH AND YEAR				
Course Code: UCHUT346				
Course Name: Economics for Engineers				
Max. Marks: 50			Duration: 2 hours 30 minutes	
	PART A			
		Answer all questions. Each question carries 3 marks	CO	Marks
1		What are the central problems of an economy?	CO1	(3)
2		Point out any three applications of price elasticity of demand.	CO1	(3)
3		What is the social cost of production?	CO2	(3)
4		What is repo rate?	CO3	(3)
5		What is esteem value?	CO4	(3)
6		Write a short note on time value of money.	CO4	(3)
PART B				
Answer any one full question from each module. Each question carries 8 marks				
Module 1				
9	a)	Suppose a country is producing at a point inside the production possibility curve. Draw a PPC and examine this situation.	CO1	(5)
	b)	State the law of demand. Point out any two exceptions of this law.	CO1	(3)
10	a)	A consumer purchases 10 units of a commodity when its price is Rs.100. Later when its price falls to Rs.90, he purchases 8 units only. Estimate price elasticity. What type of a commodity is this?	CO1	(5)
	b)	State the law of variable proportion.	CO1	(3)

Module 2				
11	a)	What is oligopoly? Why price is rigid under oligopoly?	CO2	(5)
	b)	The cost function of a firm is given as $TC=1000+10Q-6Q^2+Q^3$. Calculate fixed cost, variable cost and marginal cost when output is 10 units.	CO2	(3)
12	a)	Suppose a firm is earning super normal profit under monopolistic market condition. Explain this situation by drawing a diagram.	CO2	(5)
	b)	Suppose a firm sells its product at a price of Rs.10 per unit and its average variable cost is Rs.6. If the firm spend Ra.10000 as rent and pay Rs. 6000 as interest every month, estimate its break-even level of output.	CO2	(3)
Module 3				
13	a)	What is inflation? How does inflation affect investment and production.	CO3	(5)
	b)	How will you obtain NNP _{fc} from GDP _{mp} .	CO3	(3)
14	a)	From the data given below (In Rs. Crores) estimate GDP _{mp} and national income. Private final consumption expenditure = 1000, Government expenditure = 500, Invest expenditure = 700, Net exports = 300, Depreciation = 200, NFIA=(-200) and Net indirect tax = 100	CO3	(5)
	b)	What is bank rate? Examine the bank rate policy of the government during inflation.	CO3	(3)
Module 4				
15	a)	Examine the procedures of value engineering.	CO4	(5)
	b)	Examine the application areas of value engineering	CO4	(3)

16	a)	1. Suppose the initial investment of a project is Rs. 3000 (Crores) and the cost of capital or the opportunity cost of capital is 10 percent. Calculate NPV of the project based on the cash flows given below. Year 1 2 3 4 5 Cash flow 1000 900 800 700 600 (In Crores)	CO4	(5)
	b)	Point out any three merits of NPV method.	CO4	(3)

SEMESTER S3/S4

ENGINEERING ETHICS AND SUSTAINABLE DEVELOPMENT

Course Code	UCHUT347	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip with the knowledge and skills to make ethical decisions and implement gender-sensitive practices in their professional lives.
2. Develop a holistic and comprehensive interdisciplinary approach to understanding engineering ethics principles from a perspective of environment protection and sustainable development.
3. Develop the ability to find strategies for implementing sustainable engineering solutions.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of ethics - Personal vs. professional ethics, Civic Virtue, Respect for others, Profession and Professionalism, Ingenuity, diligence and responsibility, Integrity in design, development, and research domains, Plagiarism, a balanced outlook on law - challenges - case studies, Technology and digital revolution-Data, information, and knowledge, Cybertrust and cybersecurity, Data collection & management, High technologies: connecting people and places-accessibility and social impacts, Managing conflict, Collective bargaining, Confidentiality, Role of confidentiality in moral integrity, Codes of Ethics. Basic concepts in Gender Studies - sex, gender, sexuality, gender spectrum: beyond the binary, gender identity, gender expression, gender stereotypes, Gender disparity and discrimination in education, employment and everyday life, History of women in Science & Technology, Gendered technologies & innovations, Ethical values and practices in connection with gender - equity, diversity & gender justice, Gender policy	6

	and women/transgender empowerment initiatives.	
2	Introduction to Environmental Ethics: Definition, importance and historical development of environmental ethics, key philosophical theories (anthropocentrism, biocentrism, ecocentrism). Sustainable Engineering Principles: Definition and scope, triple bottom line (economic, social and environmental sustainability), life cycle analysis and sustainability metrics. Ecosystems and Biodiversity: Basics of ecosystems and their functions, Importance of biodiversity and its conservation, Human impact on ecosystems and biodiversity loss, An overview of various ecosystems in Kerala/India, and its significance. Landscape and Urban Ecology: Principles of landscape ecology, Urbanization and its environmental impact, Sustainable urban planning and green infrastructure.	6
3	Hydrology and Water Management: Basics of hydrology and water cycle, Water scarcity and pollution issues, Sustainable water management practices, Environmental flow, disruptions and disasters. Zero Waste Concepts and Practices: Definition of zero waste and its principles, Strategies for waste reduction, reuse, reduce and recycling, Case studies of successful zero waste initiatives. Circular Economy and Degrowth: Introduction to the circular economy model, Differences between linear and circular economies, degrowth principles, Strategies for implementing circular economy practices and degrowth principles in engineering. Mobility and Sustainable Transportation: Impacts of transportation on the environment and climate, Basic tenets of a Sustainable Transportation design, Sustainable urban mobility solutions, Integrated mobility systems, E-Mobility, Existing and upcoming models of sustainable mobility solutions.	6
4	Renewable Energy and Sustainable Technologies: Overview of renewable energy sources (solar, wind, hydro, biomass), Sustainable technologies in energy production and consumption, Challenges and opportunities in renewable energy adoption. Climate Change and Engineering Solutions: Basics of climate change science, Impact of climate change on natural and human systems, Kerala/India and the Climate crisis, Engineering solutions to mitigate, adapt and build resilience to climate change. Environmental Policies and Regulations: Overview of key environmental policies and regulations (national and international), Role of engineers in policy implementation and compliance, Ethical considerations in environmental policy-making. Case Studies and Future Directions: Analysis of real-	6

	world case studies, Emerging trends and future directions in environmental ethics and sustainability, Discussion on the role of engineers in promoting a sustainable future.	
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**Course Assessment Method
(CIE: 50 marks , ESE: 50)**

Continuous Internal Evaluation Marks (CIE):

Continuous internal evaluation will be based on individual and group activities undertaken throughout the course and the portfolio created documenting their work and learning. The portfolio will include reflections, project reports, case studies, and all other relevant materials.

- The students should be grouped into groups of size 4 to 6 at the beginning of the semester. These groups can be the same ones they have formed in the previous semester.
- Activities are to be distributed between 2 class hours and 3 Self-study hours.
- The portfolio and reflective journal should be carried forward and displayed during the 7th Semester Seminar course as a part of the experience sharing regarding the skills developed through various courses.

Sl. No.	Item	Particulars	Group/ Individual (G/I)	Marks
1	Reflective Journal	Weekly entries reflecting on what was learned, personal insights, and how it can be applied to local contexts.	I	5
2	Micro project (Detailed documentation of the project, including methodologies, findings, and reflections)	1 a) Perform an Engineering Ethics Case Study analysis and prepare a report 1 b) Conduct a literature survey on 'Code of Ethics for Engineers' and prepare a sample code of ethics	G	8
		2. Listen to a TED talk on a Gender-related topic, do a literature survey on that topic and make a report citing the relevant papers with a specific analysis of the Kerala context	G	5
		3. Undertake a project study based on the concepts of sustainable development* - Module II, Module III & Module IV	G	12
3	Activities	2. One activity* each from Module II, Module III & Module IV	G	15
4	Final Presentation	A comprehensive presentation summarising the key takeaways from the course, personal reflections, and proposed future actions based on the learnings.	G	5
Total Marks				50

*Can be taken from the given sample activities/projects

Evaluation Criteria:

- **Depth of Analysis:** Quality and depth of reflections and analysis in project reports and case studies.
- **Application of Concepts:** Ability to apply course concepts to real-world problems and local contexts.
- **Creativity:** Innovative approaches and creative solutions proposed in projects and reflections.
- **Presentation Skills:** Clarity, coherence, and professionalism in the final presentation.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop the ability to apply the principles of engineering ethics in their professional life.	K3
CO2	Develop the ability to exercise gender-sensitive practices in their professional lives	K4
CO3	Develop the ability to explore contemporary environmental issues and sustainable practices.	K5
CO4	Develop the ability to analyse the role of engineers in promoting sustainability and climate resilience.	K4
CO5	Develop interest and skills in addressing pertinent environmental and climate-related challenges through a sustainable engineering approach.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1						3	2	3	3	2		2
CO2		1				3	2	3	3	2		2
CO3						3	3	2	3	2		2
CO4		1				3	3	2	3	2		2
CO5						3	3	2	3	2		2

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Ethics in Engineering Practice and Research	Caroline Whitbeck	Cambridge University Press & Assessment	2nd edition & August 2011
2	Virtue Ethics and Professional Roles	Justin Oakley	Cambridge University Press & Assessment	November 2006
3	Sustainability Science	Bert J. M. de Vries	Cambridge University Press & Assessment	2nd edition & December 2023
4	Sustainable Engineering Principles and Practice	Bhavik R. Bakshi,	Cambridge University Press & Assessment	2019
5	Engineering Ethics	M Govindarajan, S Natarajan and V S Senthil Kumar	PHI Learning Private Ltd, New Delhi	2012
6	Professional ethics and human values	RS Naagarazan	New age international (P) limited New Delhi	2006.
7	Ethics in Engineering	Mike W Martin and Roland Schinzinger,	Tata McGraw Hill Publishing Company Pvt Ltd, New Delhi	4" edition, 2014

Suggested Activities/Projects:

Module-II

- Write a reflection on a local environmental issue (e.g., plastic waste in Kerala backwaters or oceans) from different ethical perspectives (anthropocentric, biocentric, ecocentric).
- Write a life cycle analysis report of a common product used in Kerala (e.g., a coconut, bamboo or rubber-based product) and present findings on its sustainability.
- Create a sustainability report for a local business, assessing its environmental, social, and economic impacts
- Presentation on biodiversity in a nearby area (e.g., a local park, a wetland, mangroves, college campus etc) and propose conservation strategies to protect it.
- Develop a conservation plan for an endangered species found in Kerala.
- Analyze the green spaces in a local urban area and propose a plan to enhance urban ecology using native plants and sustainable design.
- Create a model of a sustainable urban landscape for a chosen locality in Kerala.

Module-III

- Study a local water body (e.g., a river or lake) for signs of pollution or natural flow disruption and suggest sustainable management and restoration practices.
- Analyse the effectiveness of water management in the college campus and propose improvements - calculate the water footprint, how to reduce the footprint, how to increase supply through rainwater harvesting, and how to decrease the supply-demand ratio
- Implement a zero waste initiative on the college campus for one week and document the challenges and outcomes.
- Develop a waste audit report for the campus. Suggest a plan for a zero-waste approach.
- Create a circular economy model for a common product used in Kerala (e.g., coconut oil, cloth etc).
- Design a product or service based on circular economy and degrowth principles and present a business plan.
- Develop a plan to improve pedestrian and cycling infrastructure in a chosen locality in Kerala

Module-IV

- Evaluate the potential for installing solar panels on the college campus including cost-benefit analysis and feasibility study.
- Analyse the energy consumption patterns of the college campus and propose sustainable alternatives to reduce consumption - What gadgets are being used? How can we reduce demand using energy-saving gadgets?
- Analyse a local infrastructure project for its climate resilience and suggest improvements.
- Analyse a specific environmental regulation in India (e.g., Coastal Regulation Zone) and its impact on local communities and ecosystems.
- Research and present a case study of a successful sustainable engineering project in Kerala/India (e.g., sustainable building design, water management project, infrastructure project).

Research and present a case study of an unsustainable engineering project in Kerala/India highlighting design and implementation faults and possible corrections/alternatives (e.g., a housing complex with water logging, a water management project causing frequent floods, infrastructure project that affects surrounding landscapes or ecosystems).

SEMESTER S4

MANUFACTURING AND PROTOTYPING LABORATORY

Course Code	PCRUL407	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To demonstrate the parts of various machine tools and impart hands on experience on lathe, milling and grinding machines
2. To develop fundamental knowledge on tool materials, cutting fluids and tool wear Mechanisms
3. To study process parameters and practice on arc and gas welding technologies.

Expt. No.	Experiments
1	Exercises on Centre Lathe: - Facing, plain turning, step turning, taper turning and form turning, groove cutting, knurling and chamfering, multi-start thread, square thread and internal thread, Inspection of the work using measuring instruments
2	Exercises on Milling Machine: - Face milling, end milling – spur and helical gear cutting milling of key ways, T slot cutting, Inspection of the work using measuring instrument
3	Exercises on Grinding machine: - Exercise on surface grinding, cylindrical grinding and tool grinding, Inspection of the work using measuring instruments
4	Exercises on Welding: - Exercises on arc and gas welding - butt welding and lap welding of M.S. sheets.
5	Preparation of program and Exercise on CNC lathe including simulation: -turning, step turning, taper turning, thread cutting, circular interpolation etc.
6	Use of CAM software for generation of CNC code from CAD data.
7	Exercises on Rapid Prototyping: - 3D printing - simple box, helix shape, U bracket

Course Assessment
Method (CIE: 50 marks,
ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine different machine tools with understanding of work holders and operating principles to produce different part features to the desired quality.	K3
CO2	Identify the various operations and the required machines and attachments for prototyping the robotic components.	K3
CO3	Demonstrate CNC machines and programming them for machining robotic components.	K3
CO4	Use CAD/CAM for generating CNC code for production machines to realise parts.	K3
CO5	Identify the properties of 3D printed parts and use the rapid prototyping effectively to make prototypes	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3								1		3	3
CO2	3										3	3
CO3	3				2						3	3
CO4	3	2	2		3						3	3
CO5	3	2	2		2						3	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Production Technology	HMT	Tata McGraw Hill.	2017
2	Workshop Technology Part I	W. A. J. Chapman,	Routledge	5th edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	CNC Manufacturing Technology: Lab Workbook	Rick Calverley	Goodheart-Willcox Pub;	Workbook edition (2020)
2	The 3D Printing Handbook: Technologies, design and applications	Ben Redwood, Filemon Schöffner	3D Hubs	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
Expt 5	https://archive.nptel.ac.in/courses/112/105/112105211/
Expt 7	https://archive.nptel.ac.in/courses/112/104/112104230/

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S4

MICROCONTROLLERS AND IOT LABORATORY

Course Code	PCRUL408	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To demonstrate the basics of Assembly Language Programming and Embedded 'C' programming of a microcontroller.
2. To understand the microcontroller interfacing concepts for programming and design of a simple embedded system.
3. To describe the development of a small low-cost IoT enabled embedded system using Arduino/Raspberry Pi open platform.
4. Improve the team building, communication and management skills of the students.

Expt. No.	Experiments
The programs shall be written in assembly language. The interfacing modules may be developed using Embedded C (use KEIL simulator). Any 10 experiments with part C compulsory	
Part A: 8051 Assembly language Programming	
1	All Arithmetic Operations- addition, subtraction (32 bit), multiplication and division (8 bit).
2	A block of Data Transfer from one memory location to another memory location. (internal/external memory).
3	Sorting of an array of numbers stored in the memory location (ascending/descending order)
4	Time delay generation using timers and relay interface with 8051/AT89S51 microcontroller.
5	DAC interface with 8051/AT89S51 for different wave form generation
6	Display (LED/Seven segments/LCD) and keyboard interface with 8051/AT89S51.
Part B: Arduino Programming using Arduino IDE	
7	Toggle the ON/OFF status of an LED with a Switch using Arduino UNO.
8	Interfacing of Ultrasonic sensor using Arduino UNO.

9	Interfacing Stepper Motor using Arduino UNO.
10	ADC operation of Arduino using Potentiometer and Display with Arduino UNO.
11	Sending sensor data from Arduino to cloud using WIFI module ESP8266 and Things Speak account.
Part C: Project	
12	Develop and test the mini IOT projects using Arduino with ESP8266/RaspberryPi.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate simple programs for 8051 using assembly language programming.	K2
CO2	Apply the concepts of Interfacing to connect external devices with the microcontroller.	K3
CO3	Design an Embedded System based on the required application.	K3
CO4	Build and test the mini IOT projects successfully using Arduino/RaspberryPi.	K3
CO5	Relate open-source platforms for designing IoT devices.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2						2			
CO2	3	2	2						2			
CO3	3	3	3	2					2	1	2	3
CO4	3	3	3	2	3	1			2	1	2	3
CO5	3	3	2		3				2	1	2	3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"The 8051 Microcontroller and Embedded Systems: Using Assembly and C"	Muhammad Ali Mazidi	Pearson	2nd Edition, 2007
2	"Beginning Arduino"	Michael McRoberts	Apress	1e, 2011
3	"The 8051 Microcontroller"	Kenneth Ayala	Cengage Learning	3e, 2012
4	"Embedded Systems Architecture, programming and Design"	Raj Kamal	Tata McGraw-Hill	3e, 2013
5	"Internet of Things"	Jeeva Jose	Khanna Book Publishing	1/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things	RMD Sundaram Shriram K Vasudevan, & Abhishek S Nagarajan	Wiley	2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/WUYAjsxwU4

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 5

ROBOTICS AND AUTOMATION

SEMESTER S5

INDUSTRIAL AUTOMATION

Course Code	PCRUT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the industrial automation methodologies
2. To understand the elements of industrial automation
3. To identify the sensors and actuators used in automation
4. To understand the control in automation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automation methodologies: Concept of Mechanization and Automation, Types of Automation, Automated flow lines, Fundamentals of Transfer Lines. Trends in manufacturing – GT and Cellular Manufacturing, Flexible manufacturing systems – features of FMS, Computer integrated manufacturing – Need for AI and expert systems in CIM, Automated assembly system – flexible assembly automation,	11
2	Elements of CNC systems: servomotor and servo system design trends, stepper motors and controls, adaptive control, Automated tool changers and pallet changers Material Handling and Identification Technologies: Overview of Material Handling Systems, Principles and Design Consideration, Material Transport Systems, Storage Systems, Automated Storage and Retrieval Systems, Overview of Automatic part Identification Methods, Automatic Guided Vehicles. Sensor systems for automated inspection, Automatic gauging and size control systems, thickness measurement, Machine vision systems.	11

3	Sensors and actuators for automation, Classification of position, proximity and motion sensors, electromechanical switches, LVDT, RVDT. Electrical, Hydraulic, and pneumatic actuators and their comparison (Concept only), Examples - use of Electrical, Hydraulic, and Pneumatic actuators in industrial automation. Pneumatic/Hydraulic Automation: control valves – direction, pressure and flow, sequential control of single /multiple actuator systems, cascade and Karnaugh Veitch map methods, step-counter systems. Electro pneumatic/electrohydraulic automation: Symbols: Basic electrical elements – relay, solenoid, timers, pneumatic – electrical converters, Design of circuits and hands on models on material handling systems.	11
4	Automation Control: Sequence control and programmable controllers – logic control and sequencing elements, Programmable Logic Controller – Architecture, PLC ladder diagram, Relay logic, Analog and digital I/Os, Timers, Counters, Program control instructions. Case studies on PLC ladder programming. Motion controllers-VFD, External relays and contactors. SCADA – Building blocks of SCADA systems, RTU and HMI subsystem, Architecture of Distributed Control System (DCS), Interfacing PLC to SCADA/DCS using communication link (RS232, RS485), Protocols (Modbus ASCII/RTU) and OPC.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of automation methodologies and trends in manufacturing automation.	K2
CO2	Explain the design aspects of modern CNC machines, different automated inspection methods and operation of different types of material handling devices.	K2
CO3	Design different pneumatic and hydraulic circuits based on their applications.	K3
CO4	Develop the basic concepts of PLC programming.	K3
CO5	Illustrate SCADA systems and Distributed Control systems for industrial automation.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2	-	-	-	-	-	-	3
CO2	3	2	2	1	2	1	-	-	-	-	-	2
CO3	3	3	2	2	2	1	-	-	-	-	-	2
CO4	3	3	2	2	2	1	-	-	-	-	-	2
CO5	3	3	2	2	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems and Computer Integrated Manufacturing	Mikell P Groover	Pearson	4th Edition, 2014
2	Programmable Logic Controllers	Frank D. Petruzella	McGraw Hill	4th Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Control of Manufacturing Systems	Yoram Koren	Tata McGraw-Hill	1st Edition, 2005
2	Standard Handbook of Industrial Automation	Douglas M Considine & Glenn D Considine	Springer US	1986
3	Pneumatic Control for Industrial Automation	Peter Rohner& Gordon Smith	John Wiley and Sons	1987
4	Pneumatics, Concepts, Design and Applications	Jagadeesha T	University Press	2015

SEMESTER S5

KINEMATICS AND DESIGN OF MACHINE ELEMENTS

Course Code	PCRUT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	ENGINEERING MECHANICS	Course Type	Theory

Course Objectives:

1. Understand the fundamentals of the theory of kinematics of machines.
2. Enable students to attain the basic knowledge required to understand, analyse, design and select machine elements required in transmission systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of mechanisms: Machines and mechanisms, lower and higher pairs, kinematic chains, kinematic inversions of four bar and single slider crank chains. mobility /degrees of freedom (DoF), Kutzbach's formula Velocity Analysis: Definition of the velocity analysis problem-graphical method of velocity analysis-four bar and slider crank mechanisms only	11
2	Acceleration analysis: Definition of the acceleration analysis problem, angular acceleration of a rigid link and relative acceleration of points, graphical method of acceleration analysis-four bar and slider crank mechanisms only. Corioli's acceleration- Concept only (no derivation) Static force analysis: Free body diagrams, static force analysis of four bar and slider crank mechanisms only,	11
3	Introduction to Design- Definition, steps in design process, preferred numbers, standards and codes in design. Shock and impact loads, fatigue loading, endurance limit stress, factors affecting endurance limit, factor of safety Gears- classification, Gear nomenclature, Tooth profiles, Materials of gears,	11

	virtual or formative number of teeth, gear tooth failures, Lewis equation, Buckingham's equation for dynamic load, wear load, endurance strength of tooth, surface durability, heat dissipation – lubrication of gears – Merits and demerits of each type of gears. Design of spur gear	
4	Rolling contact bearing- Types, Selection of a bearing type, bearing life, static and dynamic load capacity, axial and radial loads, equivalent bearing load, selection of bearing life. Design of Ball Bearings Basic idea of flat and V belt, materials for belts, Selection of V-belt drives, Advantages and limitations of V-belt drive	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate the principles of kinematic pairs, chains and their classification, DOF, inversions and planar mechanisms	K3
CO2	Determine the acceleration of a point in planar mechanisms.	K3
CO3	Analyse static forces in planar mechanisms	K3
CO4	Review concepts of statics and strength of materials	K3
CO5	Provide basic design methods for gear drives, belt drives and bearings	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2									2	3
CO2	3	3	3								2	3
CO3	3	3	3								2	3
CO4	2	1									2	3
CO5	3	2	2								2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Theory of Machines	Rattan S S	Tata McGraw-Hill	2009
2	Kinematics and Dynamics of Machinery	Norton R L	McGraw Hill Education Pvt. Ltd	13 th edition, 2017
3	Mechanical Engineering Design,	J. E. Shigley,	McGraw Hill,	2003
4	Design of Machine elements	V.B.Bhandari	McGraw Hill	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Kinematics, Dynamics and Design of Machinery	Kenneth J. Waldron, Gary L. Kinzel, Sunil K. Agrawal,	Wiley	3rd Edition, 2016
2	Theory of Machines and Mechanisms	John J. Uicker. Jr, Gordon R. Pennock, Joseph E. Shigley,	Oxford HED	4th Edition, 2014
3	Machine Design- An integrated Approach	Robert L.Norton	Pearson Education, New Delhi	5 th edition, 2014

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105268/
2	https://archive.nptel.ac.in/courses/112/105/112105268/
3	https://archive.nptel.ac.in/courses/112/105/112105124/
4	https://archive.nptel.ac.in/courses/112/105/112105124/

SEMESTER S5

ELECTRIC DRIVES AND CONTROL

Course Code	PCRUT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide students with a comprehensive understanding of electric drives, their components, and their applications, particularly in robotics.
2. To introduce students to various special electrical machines, their construction, working principles, and their roles in modern automation systems.
3. To impart knowledge about power electronics components such as SCRs, controlled rectifiers, DC-DC converters, and inverters, including their operation, control, and protection.
4. To explore the control and operation of DC and AC motor drives, including closed-loop control techniques and advanced methods for speed and torque control

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Electric Drives: –Block diagram. Advantages, Uses in Robotic Applications. Introduction to Special Electrical Machines- DC servo motor, AC servo motors, stepper motor- variable reluctance - permanent magnet- hybrid, BLDC, PMSM- Construction, working and principle of operation	9
2	SCR: Structure, working, V-I and switching characteristics- Turn on methods of SCR- Gate triggering circuits, R and RC triggering circuits, line synchronised triggering – natural and forced commutation (concept only). Protection of SCR. Requirements of isolation and synchronization	9

	in gate drive circuits- Opto and pulse transformer-based isolation. Controlled Rectifiers –Single phase fully controlled bridge rectifier with R, RL and RLE loads (continuous & discontinuous conduction) – output voltage equation. Three-phase fully controlled converter with RLE load	
3	DC-DC converters – Step down and step up choppers – control methods- two-quadrant & four-quadrant chopper. DC Motor drives- Solid state speed control of DC motors- Armature control and field control, Single phase fully controlled converter drives (Rectifier and inverter mode). Chopper-controlled DC drives- Regenerative braking control- Four quadrant chopper drives. Closed loop speed and torque control. Inverters — 1-phase full-bridge voltage source inverters inverter with R & RL loads- 3-phase bridge inverter with R load – 120° & 180° conduction mode. Voltage control in inverters– Pulse Width Modulation – single pulse width, multiple pulse width & sine PWM-elimination of harmonics- Variable Voltage Variable Frequency Drive (Block diagram only).	9
4	Control of Servomotors-Components of typical servo system-with DC and brushless DC servo motor, Feedback system -Sizing of servomotors Position control of Stepper motor- Drive circuit – modes of excitation- open loop and closed loop control of Stepper Motor- applications Permanent Magnet Motor drives- Speed control of BLDC motor- converter circuits, modes of operation - applications. Speed control of PMSM-Self-control- Sensorless Control Digital Control of Electric Drives: Digital Controllers; Microcontroller and DSP-Based Control	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the construction, working principles, and applications of special electrical machines such as DC servo motors, AC servo motors, stepper motors, BLDC, and PMSM.	K2
CO2	Analyze the operation, characteristics, and control methods of power electronic components including SCRs, controlled rectifiers, DC-DC converters, and inverters.	K4
CO3	Apply knowledge of DC and AC motor drives to design and implement speed and torque control systems using various control methods.	K3
CO4	Design and simulate drive circuits for different types of motors, including closed-loop control systems, using digital controllers such as microcontrollers and DSPs.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	-	-	-	-	-	-	-	-	-	-
CO2	3	3	2	2	1	-	-	-	-	-	-	-
CO3	3	2	3	2	2	-	-	-	-	-	-	-
CO4	3	2	3	3	2	1	-	-	-	-	-	-

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power electronics converters applications and design	Ned Mohan, Tore m Undeland, William P Robbins,	John Wiley and Sons.	Third Edition, 2022
2	Power semiconductor control drives	Dubey G. K.	Prentice Hall, Englewood Cliffs, New Jersey,	Third Edition
3	Special Electrical Machines	E. G. Janardhanan	PHI Learning Private Limited.	2014
4	Electrical machines	Nagarath.I.J& Kothari .D.P	Tata McGraw-Hill.	1998

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electric Drives: Concepts and Applications	Vedam Subrahmanyam	McGraw-Hill Education	2 nd Edition 2001
2	Fundamentals of Electric Drives	Gopal K. Dubey	Narosa Publishing House	2 nd Edition, 2002
3	Electric Motor Drives: Modeling, Analysis, and Control	R.Krishnan	Prentice Hall	2st Edition 2001
4	Control of Electric Machine Drive Systems	Seung-Ki Sul	Wiley-IEEE Press	1 st Edition, 2011

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Fundamentals of Electric Drives by Prof. Shyama Prasad Das, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc22_ee94 NPTEL :: Electrical Engineering - NOC: Fundamentals of Electric Drives
2	Power Electronics by Prof. L. Umanand, IISc Bangalore https://onlinecourses.nptel.ac.in/noc21_ee01/preview
3	Fundamentals of Electric Drives by Prof. Shyama Prasad Das, IIT Kanpur: https://onlinecourses.nptel.ac.in/noc22_ee94 NPTEL :: Electrical Engineering - NOC: Fundamentals of Electric Drives
4	Control of Electrical Drives by Prof. K. R. Rajagopal, IIT Kharagpur

SEMESTER S5
MOBILE ROBOTICS

Course Code	PBRUT504	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT402	Course Type	Theory

Course Objectives:

1. To give students the ability to apply mathematical modelling to kinematics, localization and path planning of mobile robots
2. To give students the ability to apply various strategies to system level control of mobile robots

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Locomotion: Introduction, Key issues for locomotion, Legged Mobile Robots - Leg configurations and stability, Wheeled Mobile Robots - Wheeled locomotion: the design space, Wheel design, Wheel geometry, Stability, Maneuverability, Controllability, Aerial robots – advantages of UAVs over conventional manned aircraft, Types of UAVs - Fixed-wing UAV or airplane – characteristics, Rotary-wing UAV - Multi-rotor, Tricopter, Quadcopter or quadrotor, Hexacopter, Y3/Y6, Octocopter Mobile Robot Kinematics: Introduction, Kinematic Models and Constraints - Representing robot position, Forward kinematic models, Wheel kinematic constraints for Fixed standard wheel, Steered standard wheel and Castor wheel only, Robot kinematic constraints Example of Robot kinematic models and constraints: Differential-drive robot only Mobile Robot Maneuverability - Degree of mobility, Degree of steerability, Robot maneuverability, Mobile Robot Workspace - Degrees of freedom, Holonomic robots	11

2	Mobile Robots Fundamental problems: Path Planning, Localization, Sensing, Mapping, Simultaneous Localization and Mapping Mobile Robot Localization: Introduction, The Challenge of Localization - Noise and Aliasing, Localization-Based Navigation versus Programmed Solutions, Belief Representation - Single-hypothesis belief, Multiple-hypothesis belief, Differences between Markov and Kalman Localization (detailed explanation not required) Representation and Reasoning about Space: Representing Space - Spatial Decomposition, Geometric Representations, Topological Representations Representing the Robot: Configuration Space, Simplification of C-space	11
3	Path Planning for Mobile Robots: Free-space path versus Semi-free-space path, Non-homotopic paths versus Homotopic paths Constructing a discrete search space - Visibility graph, Tangent graph, Retraction Methods Searching a Discrete State Space - Graph search - Depth-first Search, Breadth-first Search, Dijkstra's Algorithm, Best first search – A and A* Algorithm Searching a Continuous State Space – Potential Fields, Vector field histogram, Bug 1, 2 and Tangent Bug algorithms Probabilistic path planning – RPP, PRM, RRT	11
4	System Control: Horizontal Decomposition, Vertical Decomposition, Hybrid Control Architectures, Middleware, Alternative Control Formalisms, The Human Robot Interface Off-Board Communication: Tethered, Untethered – types of untethered communication	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Derive kinematic constraints and models for mobile robots	K2
CO2	Describe the techniques used in mobile robot localization	K2
CO3	Apply various algorithms to path planning for mobile robots	K3
CO4	Apply various techniques for mobile robotic system control	K3
CO5	Implement a mobile robot that incorporates functionalities suitable for an application	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3
CO5	3	3	3	3	3	-	-	-	3	3	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Autonomous Mobile Robots	R Siegwart, IR Nourbakhsh, D Scaramuzza	The MIT Press	2nd edition, 2011
2	Computational Principles of Mobile Robotics	Gregory Dudek, Michael Jenkin	Cambridge University Press	2nd edition, 2010
3	Chapman & Hall_CRC Artificial Intelligence and Robotics Series - A First Course in Aerial Robots and Drones-	Yasmina Bestaoui Sebbane	CRC Press	1st edition, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier Science Publishing Co Inc	1st edition, 2013
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	1st edition, 2019
3	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier Science Publishing Co Inc	1st edition, 2013

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106298/
2	https://archive.nptel.ac.in/courses/112/106/112106298/
3	https://archive.nptel.ac.in/courses/112/106/112106298/
4	https://archive.nptel.ac.in/courses/112/107/112107289/

PBL Course Elements

L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity

Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5

ARTIFICIAL INTELLIGENCE FOR ROBOTICS

Course Code	PERUT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the Fundamentals of Robotics and AI:
2. Utilize supervised and unsupervised learning techniques for robotic perception and action.
3. Develop and evaluate novel AI algorithms for specific robotic applications through projects and case studies.
4. Analyze recent advancements and research trends in AI and robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Artificial intelligence: Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview, Intelligent Agents-Agents and Environments, Good behaviour- The concept of rationality, nature of Environments, structure of Agents, Expert systems - basic overview, Application areas of AI - vision and speech processing, robotics.	9
2	Learning - Forms of learning: Supervised Learning Algorithms- Decision Trees, Logistic Regression, Support vector Machines, Unsupervised Learning Algorithms: K-means clustering k-Nearest Neighbors, Principle Component Analysis, Reinforcement based learning	9
3	Deep Feedforward Networks: Perceptrons – Single layer & Multi-layer, Learning XOR, Convolutional Neural Networks -basic outline and functions of each layers only, Sequence Modeling-Need for sequence models, basic RNN architecture and types	9

4	Robotics - Robotic perception, Localization and mapping, Machine learning in robot Perception, Application domains. Mobile Robot Navigation-Path planning, Path Planning and obstacle avoidance-A* algorithm	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the role of AI in solving problems in different domains and its evolution.	K3
CO2	Explain the different learning techniques used in Machine learning	K2
CO3	Identify the need for multilayer neural network for solving complex tasks	K2
CO4	Identify suitable neural network architectures for different types of data and tasks.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								2
CO2	3	3	3	2	2							2
CO3	3	3	3	2	2							2
CO4	3	3	3	2	2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016
2	Artificial Intelligence - A Modern Approach	Stuart J. Russell and Peter Norvig,	Pearson	Third Edition, 2016
3	Pattern Recognition and Machine Learning,	Bishop, C. ,M.,	Springer	2006.
4	A survey of deep learning techniques for autonomous driving.	Grigorescu, Sorin, et al		Journal of Field Robotics 37.3 (2020): 362-386.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to AI Robotics,	Robin R. Murphy	The MIT Press	Second Edition: 2019
2	Artificial Intelligence and Machine Learning,	Chandra S.S.V, Anand Hareendran S	PHI	2021
3	Models, Learning and Inference	Simon J. D. Prince	Cambridge University Press	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2	https://archive.nptel.ac.in/courses/106/106/106106139/
3	https://archive.nptel.ac.in/courses/106/106/106106184/
4	https://archive.nptel.ac.in/courses/112/106/112106298/

SEMESTER S5

DESIGN FOR MANUFACTURING AND ASSEMBLY

Course Code	PERUT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT303, PCRUL307	Course Type	Theory

Course Objectives:

1. To understand how design choices impact manufacturing costs and processes.
2. To gain skills in tolerance analysis and understanding datum systems for precise manufacturing.
3. To learn to minimize part count, standardize designs, and create assembly-friendly products.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Design Impact and Material Considerations - Design Impact on Cost: Understanding how design choices affect manufacturing costs. Design for 'X' – DFM Approach: Overview of Design for Manufacturing principles and the DFM Framework. Material and Process Evaluation: Selecting appropriate materials and processes. General DFM Guidelines: Key guidelines for designing manufacturable products. Machining and Casting Guidelines and Examples: Specific considerations for machining and casting processes. Minimize Finishing Requirements: Strategies to reduce post-processing needs. Facilitating Machining Design Features: Datum features, functional and manufacturing aspects. Component Design for Machining: Considerations and redesign examples. Form Design of Castings and Weldments: Design principles for castings and weldments.	9
2	Tolerance Analysis and Selective Assembly - Tolerance Analysis: Process capability and metrics, cost aspects of tolerances. Feature and Geometric Tolerances: Understanding and applying tolerances in design. Relationship	9

	between Tolerance Grades and Machining Processes: Attainable tolerance grades. Cumulative Effect of Tolerances: Sure fit law, normal law, and truncated normal law. Interchangeable Part Manufacture and Selective Assembly: Concepts and examples. Control of Axial Play: Techniques for controlling play, including secondary machining operations and laminated shims.	
3	Datum Systems and True Position Theory - Datum Systems: Degrees of freedom, grouped datum systems, and their applications. Computational Accuracy: Translational and rotational accuracy calculations. Geometric Analysis: Applications of geometric analysis in design. True Position Theory: Understanding true position theory, including co-ordinate and conventional methods of feature location. Virtual Size Concept: Concepts of floating and fixed fasteners, projected tolerance zones. Assembly with Gasket and Zero True Position Tolerance: Practical applications and examples. Functional Gauges: Use and design of functional gauges. Paper Layout Gauging: Techniques and examples.	9
4	Principles of Design for Assembly - Minimize Part Count: Strategies to reduce the number of parts in a design. Standardization and Minimize Part Variety: Principles of standardization and variety reduction. Design Guidelines for Manual Assembly: Best practices for designing for manual assembly. DFA Analysis: Conducting DFA analysis and calculating DFA index. Design for Automated Assembly: Principles and guidelines for automated assembly. Introduction to DFMA Software: Overview of software tools used for DFMA, including practical applications and case studies.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the impact of design decisions on manufacturing costs	K2
CO2	Apply geometric tolerance principles to design features for machining and casting.	K3
CO3	Analyze the effectiveness of different tolerance grades in machining processes.	K4
CO4	Evaluate the design for assembly (DFA) of a product using DFA analysis tools.	K5
CO5	Create a comprehensive report on datum systems and their application in ensuring part accuracy.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1				2						
CO2	3	2	3				1	1				
CO3	3	2		2					2			
CO4	3	2			3					3		
CO5	3	2									2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Design for Manufacture and Assembly	Geoffrey Boothroyd, Peter Dewhurst, Winston A. Knight	CRC Press	3 rd Edition; Year: 2010
2	Manufacturing Processes for Engineering Materials	Serope Kalpakjian, Steven R. Schmid	Pearson	6 th Edition; Year: 2016
3	Engineering Design: A Systematic Approach	Gerhard Pahl, Wolfgang Beitz, Jörg Feldhusen, Karl-Heinrich Grote	Springer	3 rd Edition; Year: 2007
4	Mechanical Tolerance Stackup and Analysis	Bryan R. Fischer	CRC Press	2 nd Edition; Year: 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Modern Manufacturing: Materials, Processes, and Systems	Mikell P. Groover	Wiley	7 th Edition; Year: 2020
2	The Principles of Product Development Flow: Second Generation Lean Product Development	Donald G. Reinertsen	Celeritas Publishing	1 st Edition; Year: 2009
3	Design for Manufacturing and Assembly: Concepts, Designs and Methods	O. Molloy, S. Tilley, E.A. Warman	Springer	1 st Edition; Year: 1998
4	Product Design for Manufacture and Assembly	Boothroyd G., Dewhurst P., Knight W.	Marcel Dekker, New York	4 th Edition Year: 2012
5	Geometric Dimensioning and Tolerancing for Mechanical Design: A Self-Teaching Guide to ANSI Y14.5M-1982 Standards	Gene R. Cogorno	McGraw-Hill Education	1 st Edition; Year: 1990

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/107/103/107103012/
2	https://archive.nptel.ac.in/courses/107/103/107103012/
3	https://archive.nptel.ac.in/courses/107/103/107103012/
4	https://archive.nptel.ac.in/courses/107/103/107103012/

SEMESTER S5

MAINTENANCE AND SAFETY ENGINEERING

Course Code	PERUT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart knowledge in maintenance
2. To know about the fundamentals of maintenance and to implement it
3. To study about safety engineering practices
4. To analyze the hazards in protection
5. To know about the safety in machine operation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ORGANISATION AND MANAGEMENT OF MAINTENANCE FUNCTION: Scope of Responsibilities, Organization, Manpower Requirements, Selection and Training – Maintenance vs. Reliability Engineering – Operating Policies of Effective Maintenance – Reliability Based Maintenance – TPM – Concepts, Metrics, Benefits, Implementation, Common Barriers, Success Factors and Enablers	9
2	MAINTENANCE BEST PRACTICES: Key Performance Indicators – Fundamental Requirements of Preventive Maintenance, Sustaining Maintenance: Preventive Maintenance, Periodic Replacement, Corrective Maintenance, Overhauls and Upgrades, Breakdown Maintenance – Computerized Planning and Scheduling – Maintenance Repair and Operations – Storeroom Excellence	9

3	MAINTENANCE ENGINEER'S TOOLBOX: Life Cycle Cost Analysis – Fault-Tree Analysis – Cause and Effect Analysis – Sequence of the Events Analysis – The Why-Why / Root Cause Analysis, Statistical Analysis Tools – Failure Mode Criticality and Risk Assessment	9
4	INDUSTRIAL HAZARDS AND SAFETY: Hazards: Types – Physical, Chemical, Biological, Ergonomical, Electrical - Fire: Chemistry of Fire, Detection and Suppression Systems – Safety Acts: Factories Act 1948, Environment Act 1986, OSHA, HASAWA, ISO Related to Safety	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze and define the scope of responsibilities in maintenance management, including organizational structures and manpower requirements	K1
CO2	demonstrate an understanding of different maintenance strategies and their applicability in industrial settings	K1, K2
CO3	students with a comprehensive set of tools, techniques, and skills essential for managing and optimizing maintenance activities effectively in industrial environments	K2
CO4	students will be equipped with the knowledge and practical skills necessary to effectively manage industrial hazards and ensure a safe working environment in any industrial setting	K1, K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	-	-	1	-	1	1	1	1	-	-	1
CO2	1	1	1	1	1	2	2	2	2	1	-	2
CO3	1	-	2	-	-	-	-	-	2	-	-	2
CO4	2	2	2	1	2	2	2	2	2	2	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Maintenance Engineering Handbook	Keith Mobley	McGraw Hill Education	2014
2	Implementing TPM	Robinson C J, Ginder A P	Productivity Press	1995

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Maintainability, Maintenance and Reliability for Engineers	Dhillon B S	CRC Press	2006
2	Industrial Accident Prevention	Heinrich H W	National Safety Council	1998
3	Personal Protective Equipment	National Safety Council	National Safety Council	1998
4	Accident Prevention Manual for Industrial Operations	National Safety Council	National Safety Council	1995

SEMESTER S5
SIGNALS AND SYSTEMS

Course Code	PERUT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To study and analyse continuous and discrete-time signals and systems, focusing on their properties, representations, and necessary analytical methods.
2. To understand signal transformation between domains and to develop filtering methods based on DFT

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Elementary Signals, Classification and representation of continuous time and discrete time signals, Signal operations. Continuous time and discrete time systems – Classification. Representation of systems: Differential equation representation of continuous time systems. Difference equation representation of discrete systems.	9
2	Sampling of continuous time signals, Sampling theorem for low pass signals, aliasing. Z transform, ROC, Inverse transform, properties, Unilateral transform. Frequency domain representation of discrete time signals, Discrete time Fourier series and its properties. Discrete time Fourier transform (DTFT) and its properties.	9
3	The Discrete Fourier Transform-IDFT-Properties of DFT-Circular convolution – Linear Filtering methods based on the DFT- linear convolution using circular convolution, overlap save and overlap add methods.	9
4	Computation of DFT: Radix-2 Decimation in Time and Decimation in Frequency FFT -Algorithms - IDFT computation using Radix-2 FFT	9

	Algorithms. FIR and IIR Filters. - FIR Filter Structures: Direct Form, Cascade Form IIR Filter Structures: Direct Form, Transposed Form, Cascade Form and Parallel Form (Design is not required).	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Outline the different types of signals and systems	K2
CO2	Obtain frequency domain representation of discrete time signals	K3
CO3	Develop filtering methods based on DFT	K3
CO4	Calculation of DFT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1	2									3
CO3	3	2	2									3
CO4	3	2	2									3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals and Systems	Alan V. Oppenheim and Alan Willsky,	Pearson Education	2/e, 2009
2	Discrete Time Signal Processing	Oppenheim A.V, Schafer R.W and Buck J.R	Prentice Hall	3/e, 2007
3	Signals & Systems,	Simon Haykin,	John Wiley,	2/e, 2003
4	Digital Signal Processing	Proakis J. G. and Manolakis D. G.	Pearson Education	4/e, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Signals and Systems	Anand Kumar	PHI	3/e, 2013.
2	Signals and Systems	Mahmood Nahvi	Mc Graw Hill (India)	3/e, 2015
3	Digital Signal Processing	Nagoor Kani	Mc Graw Hill Education, New Delhi	2/e, 2013
4	Understanding Digital Signal Processing	Lyons, Richard.G	Pearson Education, India	3/e, 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/104/108104100/
2	https://archive.nptel.ac.in/courses/108/105/108105055/
3	https://onlinecourses.nptel.ac.in/noc21_ee20/preview
4	https://auece.digimat.in/nptel/courses/video/108106168/L47.html

SEMESTER S5

INDUSTRIAL ENGINEERING

Course Code	PERUT526	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Introduce students to the fundamental concepts of industrial engineering, including its history, development, role, and applications.
2. Equip students with the knowledge and skills to design, analyze, and improve production systems, including plant layout, line balancing, and product design.
3. Develop students' understanding of quality control principles and techniques, including statistical process control, sampling inspection, and reliability engineering

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Concept of Industrial Engineering -History & development- Role of industrial engineer-Applications of industrial engineering. Production-production function-production system-analysis-Productivity-factors affecting productivity-increasing productivity of resources. Plant location, Layout & Line balancing - Concept-factors governing plant location-location economics-Design of layout of factories-Product, process and combination layout-systematic layout planning. Design of assembly lines, Line balancing methods. Computer applications in layout designs-CORELAP, ALDEP, CRAFT	9
2	Product Design and Analysis – New product development – concept & steps- process planning and design-Value Engineering-Standardization-Simplification- Make or Buy Decisions – Ergonomics in product design – Concurrent engineering Forecasting - Simple moving average method-weighted moving average	9

	method-double moving average method- simple exponential smoothing – adjusted exponential smoothing- linear regression-semi average method- Delphi method	
3	Inspection & Quality Control – Definition- Purpose- types- SQC- Probability concepts- Estimation- Significance Testing- ANOVA- Sampling Inspection – Sampling Plans –Control charts & their applications- Life testing – Reliability and Reliability Prediction – Monte Carlo Simulation- Zero defect concept – Quality Circle	9
4	Decision Making – Definition – Importance – types of Decisions – Theories – Techniques- Decision making process-Scientific approach- quantitative methods in decision making – Decision under risk – Decision under Uncertainty – Decision Tree	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a systematic plant layout	K3
CO2	Develop a comprehensive plan for the design and development of a new product, considering factors such as functionality, manufacturability, cost, ergonomics, and safety.	K3
CO3	Apply a variety of forecasting techniques to analyze time series data and generate accurate forecasts.	K3
CO4	Design and implement a comprehensive quality control program that utilizes inspection techniques, statistical methods, and quality management philosophies to ensure product or service consistency and continuous improvement.	K2
CO5	Confidently navigate the decision-making process by applying various theories, techniques, and quantitative methods to analyze situations, assess risks and uncertainties, and make well-informed choices that lead to desired outcomes.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	1								1	1
CO2	3	2									1	1
CO3	3	2	1								1	1
CO4	3	2	1								1	1
CO5	3	2	1								1	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial engineering & management	O.P.Khanna	Dhanpat rai publications	2020
2	Operations Research	R. Paneerselvam	PHI Publication	Second. 2016
3	Production & Operations Mangement	R. Paneerselvam	PHI Publication	Third, 2012
4	Statistical Quality Control	Eugene.L.Grant	McGraw Hill	Seventh, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial engineering & management	O.P.Khanna	Dhanpat rai publications	2020
2	Operations Research	R. Paneerselvam	PHI Publication	Second. 2016
3	Production & Operations Mnagement	R. Paneerselvam	PHI Publication	Third, 2012
4	Statistical Quality Control	Eugene.L.Grant	McGraw Hill	Seventh, 2017

SEMESTER S5

MEASUREMENTS AND INDUSTRIAL INSTRUMENTATION

Course Code	PERUT527	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To familiarise with the measurement system and mechanism for measuring parameters
2. To familiarise with the measurement interfaces, transmitters and Instrumentation diagrams

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to measurements systems - Methods of Measurement- Functional Elements of measurement system (Measurement System Architecture)- Examples of measurement systems-Standards Used in Measurements- Electrical Standards- Time and Frequency - Physical Standards -Basic SI Units-Calibration- process of calibration-methods- Expression of uncertainty- Accuracy and Precision -indications of precision, significant figures- Errors in Measurements - Static error, static correction, Propagation of errors, Systematic and Random errors in measurement - Statistical Evaluation of measured data: Histogram, Mean, Measure of dispersion from the mean, Range, Deviation, Average deviation, Standard Deviation, Variance Design of measurement systems-Example	9
2	Force measurement -Introduction to force measurement-Spring Balance - Load cells – Elastic load cells and Strain gauge load cell -Piezoelectric force transducers. Torque measurement – Introduction to torque measurement-Strain gauge torque meter- Inductive torque meter-optical torque measurement.	9

	Displacement Measurement (Linear and Angular): Optical Encoders - Synchro/Resolver Sensors -Time of Flight Ultrasonic Sensors- Laser Interferometers Speed measurement – Introduction to Speed measurement -Mechanical tachometer (Revolution counter)– electrical tachometer (Eddy current tachometer)-optical tachometer (Stroboscopic tachometer) Acceleration-Strain gauge-based accelerometers Quartz accelerometers MEMS accelerometers Thickness Measurement- Micrometers- Laser scanners Position, Location and Altitude Measurement-MEMS devices -Global Positioning System (GPS) Range measurement – Laser range sensor Slant angle measurement – tilt sensor	
3	Optical Variables Measurement - Photometry and Radiometry- Photomultipliers -photodiodes Charge coupled devices (CCD cameras) Radiation Measurement -Basic principle - Geiger Muller counter – Scintillation counter Thermal Variables Measurement-Temperature measurement- Thermistors- Thermopile-Infrared sensors - Pyroelectric sensors- Sensor ICs - LM34, LM35, Thermal Imaging -Calorimetry Pressure measurement -Bourdon tubes-Fiber optic pressure sensor- Differential pressure transmitter for pressure measurement-Dead weight tester Vacuum measurement-Thermal Conductivity gauge, Pirani Gauge Flow Measurement - Electromagnetic Flow meters -Ultrasonic Flow meters- Thermal Mass Flow meters- Flow transmitter Humidity measurements-Definitions and units-psychrometer, dew point hygrometer, electrical resistive hygrometer Qualitative and Quantitative methods of chemical measurement - Beer-Lambert's law Spectrometry – NMR – Chromatography	9
4	Smart transmitters: Transmitters-types of signals- pneumatic instrument communication signals (3-15 psig,)- electronic instrument communication signals (4-20 ma), signal transducers_ I to P converter, P to I converter-features of smart and intelligent transmitters- smart transmitter with HART communicator. Digital interfaces in measurement systems: IEEE -488_ instrumentation bus (GPIB)- objectives of industrial communications - serial data	9

	communications links- RS-232C and D interfaces- RS-422, RS-423 and RS-485 interfaces - Communication protocols- field bus-profibus-modbus-foundation fieldbus. Instrumentation diagrams: instruments' labeling- ISA standards-Data Sheets- P&ID diagrams.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To design and evaluate measurement systems, understand and apply measurement standards and calibration processes, assess measurement accuracy and precision, analyze errors, and statistically evaluate measured data to ensure reliability and accuracy in various engineering and scientific applications.	K2
CO2	To understand and apply various principles and techniques for measuring physical quantities such as force, torque, displacement, speed, acceleration, thickness, position, location, altitude, range, and slant angle using advanced instrumentation and sensor technologies.	K3
CO3	To comprehend and apply a wide range of measurement techniques and instrumentation for optical variables, radiation, thermal variables, pressure, vacuum, flow, humidity, and chemical analysis to enhance precision and accuracy in various scientific and engineering applications.	K3
CO4	To understand and implement smart transmitter technologies, digital interfaces in measurement systems, and effectively interpret instrumentation diagrams, ensuring proficiency in modern industrial communication protocols and adherence to ISA standards for accurate and efficient process control.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3				2						2
CO2	3	3	2			2						2
CO3	3	3	2			2						2
CO4	3	3	2			2						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Doebelin's Measurement Systems	Ernest.O.Doebelin and Dhanesh.N.Manik	McGraw Hill Education	6th Edition, 2011
2	Process Measurement and Analysis	B.G.Liptak	CRC Press	4th Edition, 2003
3	Principles of Industrial Instrumentation	Patranabis D	Tata McGraw Hill	3rd Edition, 2010.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Instrumentation Reference Book.	B.E.Noltingk	Butterworth Heinemann	2ndEdition, 1995
2	Process / Industrial Instruments& Controls Handbook	Douglas M. Considine	McGraw Hill, Singapore	5th Edition,199 9.
3	Applied Instrumentation in Process Industries – A survey	Andrew W.G	Gulf Publishing Company, Houston	Vol I & Vol II, 2001
4	Industrial Flow measurement	Spitzer D. W	ISA press	3rd Edition, 2005

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/108/105/108105064/#
2	https://archive.nptel.ac.in/courses/108/105/108105064/#
3	
4	https://archive.nptel.ac.in/courses/108/105/108105064/#

SEMESTER S5

ALGORITHMS AND DATA STRUCTURES

Course Code	PERUT 528	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GBEST204 Programming in C	Course Type	Theory

Course Objectives:

1. Understand the principles of algorithm analysis, including time and space complexity.
2. Analyse the complexity of algorithms using asymptotic notations.
3. Implement linear and non-linear data structures.
4. Illustrate various techniques for searching and sorting.
5. Design algorithms using fundamental design techniques: greedy, divide and conquer, and dynamic programming.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Algorithms Performance Analysis- Space Complexity, Time Complexity, Asymptotic Notation-Big Oh, Big Omega, Big Theta, Complexity Calculation of Simple Algorithms Introduction to Data Structures Definition, Classification of Data Structures: Primitive and Non-primitive, Linear and Non-linear, Operations on Data structures	9
2	Linear data structures Array: Representation of arrays, Applications of arrays Stack: Definitions & Concepts, Operations on Stacks, Applications of Stacks-Infix to postfix conversion, Postfix Evaluation Queue: Definitions & Concepts, Operations on Queue, Types of queue - Circular Queue, Priority Queue, Applications of Queue	9

	Linked lists: - singly linked list, doubly linked list, Circular linked list, operations on linked, Application of linked list-memory management-first fit, best-fit, worst-fit	
3	Non-linear data structures Tree-Terminologies, Representation, Types, Traversals Graph-Terminologies, Representation, Traversal-DFS, BFS, Minimum spanning tree-Kruskal's, Prim's algorithm, Shortest path-Dijkstra's algorithm	9
4	Searching and Sorting Algorithms Linear search, Binary search, Bubble sort, Selection sort, Insertion sort, Quick sort, Merge sort. Hashing -Hash table, Hashing functions Algorithm Design: Greedy algorithm, Divide and conquer, Dynamic programming, Backtracking, Branch and bound,	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Examine a given algorithm and express its time and space complexities in asymptotic notations	K3
CO2	Make use of appropriate data structures to solve real world computational problem in an efficient way.	K3
CO3	Discover appropriate sorting and searching algorithms that can be used in specific circumstances.	K4
CO4	Apply appropriate hash function to enable efficient access of data in the given set.	K3
CO5	Design efficient algorithms using various algorithm design techniques	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1		1				-	-	2
CO2	2	2	2	1		1				-	-	1
CO3	2	2	3	1		1				-	-	1
CO4	2	2	3	1		1				-	-	1
CO5	3	3	3	1		1				-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Data Structures in C	Ellis Horowitz, Sartaj Sahni and Susan Anderson-Freed,	Universities Press	2008
2	Classic Data Structures	Samanta D.	Prentice Hall India,	2/e, 2009
3	Data Structures: A Pseudocode Approach with C	Richard F. Gilberg, Behrouz A. Forouzan,	Cengage Learning	2/e, 2005
4	Data Structures and Algorithms	Aho A. V., J. E. Hopcroft and J. D. Ullman	Pearson Publication	2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Data Structures	Peter Brass	Cambridge University Press	2008
2	Theory and Problems of Data Structures	Lipschuts S	Schaum's Series	1986
3	Algorithms + Data Structures = Program	Wirth N	Prentice Hall	2004
4	A Structured Approach to Programming.	Hugges J. K. and J. I. Michtm	PHI	1987

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/
2	https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/
3	https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/
4	https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/ https://nptel.ac.in/courses/106/105/106105193/

SEMESTER S5

ROBOT OPERATING SYSTEM (ROS)

Course Code	PERUT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	Robotics Basics, Basic knowledge of Linux command line tools, Basic programming in Python	Course Type	Theory

Course Objectives:

1. Introduce the ROS architecture and its components
2. Introduce the application of ROS

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ROS Fundamentals: ROS Graph, ROS master, roscore, Catkin: Workspaces and ROS Packages, roslaunch, Namespaces, and Remapping Using roslaunch, tf: Coordinate Transforms - Poses, Positions, and Orientations, Publishing to a Topic, Subscribing to a Topic, Latched Topic, Defining Message Types, Mixing Publishers and Subscribers	9
2	Services and Actions: Defining a Service, implementing a Service, Using a Service, Defining an Action, Implementing a Basic Action Server, Using an Action, Implementing a More Sophisticated Action Server, Using the More Sophisticated Action.	9
3	Robots and Simulators: Subsystems - Actuation: Mobile Platform, Actuation: Manipulator Arm, Sensors, Computation Complete Robots – PR2, Fetch, Robonaut 2, TurtleBot, Simulators – Stage, Gazebo. Other Simulators	9
4	Building Maps of the World and Navigating About the World: Maps in ROS-representation and formats, SLAM (Simultaneous	9

	Localization and Mapping), rosbag and its functionalities, Recording and playing back data, Techniques for building maps from sensor data, GMapping and Cartographer for mapping, Map Server in ROS, Loading and serving maps, localization techniques (e.g., AMCL), Implementing localization in ROS.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Two Course Project applying the principles of nonlinear control system analysis:(10 marks each)

Project phases: Proposal, Implementation, Testing, Final Report, Presentations and Viva Voce:

Faculty may direct students for the selection of topics for projects from real time examples.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply ROS fundamentals and topics	K3
CO2	Explain ROS services and actions	K3
CO3	Describe robots used in ROS	K2
CO4	Evaluate the applications of simulators in ROS	K5
CO5	Apply ROS to build maps of the World and navigate	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	1	3	-	-	1	2	2	1	1
CO2	3	1	2	1	3	-	-	1	2	2	1	1
CO3	3	1	2	1	3	-	-	1	2	2	1	1
CO4	3	1	2	1	3	-	-	1	2	2	1	1
CO5	3	1	2	1	3	-	-	1	2	2	1	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey, and William D. Smart	O'Reilly	2015
2	A Gentle Introduction to ROS	Jason M. O'Kane		2016
3	Robot Operating Systems (ROS) for Absolute Beginners	Lentin Joseph	Apress	Second Edition 2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Operating System Cookbook	Kumar Bipin	Packt Publishing	2018
2	A Systematic Approach to learning Robot Programming with ROS”	Wyatt Newman	CRC press	2017

SEMESTER S5

AUTOMATION LABORATORY

Course Code	PCRUL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Lab

Course Objectives:

1. To impart practical knowledge on various industrial automation technologies.
2. To understand the functioning and programming of Programmable Logic Controllers (PLCs).
3. To develop skills in electro-pneumatic and electro-hydraulic systems.
4. To simulate and control industrial processes using SCADA.
5. To familiarize with Automation Studio software for designing and simulating automation systems

Expt. No.	Experiments
1	Introduction to PLC ○ Understanding PLC hardware and software. ○ Basic programming in Ladder Logic.
2	PLC Programming - I ○ Development of basic automation tasks using PLC. ○ Simple start-stop circuits.
3	PLC Programming - II ○ Advanced PLC programming concepts. ○ Timer and counter functions.
4	Basic Pneumatic Systems ○ Introduction to pneumatic components.

	○ Designing and implementing basic pneumatic circuits.
5	Electro-Pneumatic Systems ○ Integration of electrical control with pneumatic systems. ○ Troubleshooting electro-pneumatic circuits.
6	Basic Hydraulic Systems ○ Introduction to hydraulic components and systems. ○ Designing basic hydraulic circuits.
7	Electro-Hydraulic Systems ○ Integrating electrical control with hydraulic systems. ○ Practical implementation and troubleshooting.
8	Introduction to SCADA Systems ○ Basics of SCADA. ○ Configuring SCADA for monitoring and control applications.
9	SCADA Simulation ○ Simulating industrial processes using SCADA. ○ Developing human-machine interfaces (HMIs).
10	Automation Studio Software ○ Designing and simulating automation systems using Automation Studio. ○ Final project integrating PLC, pneumatic, hydraulic, and SCADA systems.
11	Comprehensive Project 1 • Experiment: Combine PLC and SCADA systems into a single automated process.
12	Comprehensive Project 2 • Experiment: Combine PLC, electro-pneumatic, and electro-hydraulic systems into a single automated process.

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate the ability to program and troubleshoot PLCs.	K3
CO2	Apply knowledge to design and implement electro-pneumatic systems.	K3
CO3	Apply knowledge to design and implement electro-hydraulic systems.	K3
CO4	Develop and simulate SCADA-based projects.	K4
CO5	Utilize Automation Studio software for creating automation solutions.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	3	2			2	2	2	2
CO2	3	3	3	2	3	2			2	2	2	2
CO3	3	3	3	2	3	2			2	2	2	2
CO4	3	3	3	2	3	2			2	2	2	2
CO5	3	3	3	2	3	2			2	2	2	2

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers: Principles and Applications	John W. Webb and Ronald A. Reis	Pearson	2014
2	Industrial Automation and Control	Stamatios Manesis and George Nikolakopoulos	Wiley	2018
3	Programmable Logic Controllers	W. Bolton	Elsevier	2015
4	Automation, Production Systems, and Computer-Integrated Manufacturing by	Mikell P. Groover	Pearson Education	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Programmable Logic Controllers	Gary Dunning	Cengage Learning	2012
2	Hydraulic and Pneumatic Control Systems: Design and Troubleshooting	John Watton	Cambridge University Press	2009
3	SCADA Systems: Application and Architecture	Stuart G McCrady	CRC Press	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
EXP 1-3	https://onlinecourses.swayam2.ac.in/ntr24_ed63/preview
EXP 1-3,8,9	https://onlinecourses.swayam2.ac.in/nou24_ee02/preview

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER S5

ROBOT OPERATING SYSTEM (ROS) LAB

Course Code	PCRUL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	ROS Installed Ubuntu System, Basic knowledge of Linux command line tools, Basic programming in Python— 2.7/3.5/3.6, Robotics Basics	Course Type	Lab

Course Objectives:

1. The lab introduces the student to a Robot Operating System and familiarisation of few ROS tools.

ROS Essentials

1. Installing and Configuring Your ROS Environment—ROS2/ROS noetic/Compatible versions
2. Familiarisation with ROS (Master, nodes, topics, messages, services, parameters and actions)
3. Familiarisation with ROS Tools – Gazebo , Moveit , Rviz
4. Creating Workspace and Package in ROS

Expt. No.	Experiments
1	Develop a Simple Publisher and Subscriber, Simple Service and Client, Recording and playing back data, Reading messages from a bagfile
2	Working with Turtlesim- • Understanding ROS communication: Topics and services • Basic control and movement of Turtlesim Writing simple ROS nodes and scripts
3	Visualization with Rviz- • Introduction to RViz and its capabilities • Visualizing robot models and sensor data

	Creating and configuring RViz displays
4	Introduction to tf – • Broadcast the state of a robot to tf • Get access to frame transformations Adding a frame
5	Building a Visual Robot Model with URDF from Scratch, Building a Movable Robot Model with URDF, Adding Physical and Collision Properties to a URDF Model, Create a 3DOF robotic arm from scratch.
6	Simulation with Gazebo • Introduction to Gazebo and its integration with ROS • Setting up simulation environments Simulating sensors and actuators in Gazebo
7	Motion Planning with MoveIt- • Introduction to MoveIt and its features • Configuring MoveIt for robotic arms Performing motion planning and execution
8	Navigation with ROS • Overview of ROS navigation stack • Localization and mapping with SLAM Path planning and obstacle avoidance
9	Familiarise ROS Serial Arduino for hardware interface. (hardware Interface)
10	Obstacle avoidance using a differentially driven mobile robot using ROS ((hardware Interface)

Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the applications of ROS in real world complex scenarios	K2
CO2	Evaluate the usage of turtlesim and Rviz in ROS	K5
CO3	Evaluate the use of Gazebo and MoveIt in ROS	K5
CO4	Analyze the concepts behind navigation in ROS	K4
CO5	Analyze the issues in hardware interfacing with ROS environment	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	3	2			2	2		3
CO2	3	2	2	2	3	2			2	2		3
CO3	3	2	2	2	3	2			2	2		3
CO4	3	2	2	2	3	2			2	2		3
CO5	3	2	2	2	3	2			2	2		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Operating Systems (ROS) for Absolute Beginners	Lentin Joseph	Apress	Second Edition 2018
2	Learning ROS for Robotics Programming	Aaron Martinez, Enrique Fernández	Packt Publishing	2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Operating System Cookbook	Kumar Bipin	Packt Publishing	2018
2	A Systematic Approach to learning Robot Programming with ROS”	Wyatt Newman	CRC press	2017

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students’ preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6
ROBOTICS AND AUTOMATION

SEMESTER S6

ROBOTIC CONTROL SYSTEMS

Course Code	PCRUT601	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Model the systems using transfer function approach as well as state space approach
2. Employ time domain analysis to predict and diagnose transient performance parameters of the system for standard input functions.
3. Compute the time domain and frequency domain specifications of a system
4. Analyse dynamic systems for their stability and performance using root locus and Bode plot
5. Analyse systems using state space approach

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Control System- Definition, Open loop vs closed loop control systems- components of a typical control system- Necessity of a control system in a Robot, Linear time invariant Systems- Transfer function, Necessity of knowing the transfer function, Modelling –Mechanical Translational and Electromechanical systems (Force voltage and Force current analogy) . block diagram representation -Block diagram reduction, signal flow graph, Basic problems using Masons gain formula.	11
2	Time domain analysis of control systems: Transient and steady state responses - time domain specifications - first and second order systems - step responses of first and second order systems. Steady state error analysis - static error coefficient of type 0,1, 2 systems .Concept of stability: Time response for various pole locations - stability of feedback system -Routh's	11

	stability criterion.	
3	Root locus - General rules for constructing Root loci –stability from root loci - effect of addition of poles and zeros. Frequency domain analysis: Frequency domain specifications (derivations not required)- Analysis based on Bode plot -Log magnitude vs. phase plot. Design of P, PI and PID controller using Ziegler-Nichols tuning method.	11
4	State space analysis of systems: Introduction to state concept - state equation of linear continuous time systems, matrix representation of state equations. Phase variable and canonical forms of state representation-controllable, observable, -state transition matrix, relationship between state equations and transfer function. Controllability & observability-Kalman's method. State feed-back design via pole placement technique.	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Develop a mathematical model for linear time invariant systems.	K3
CO2	Compare the responses of first and second order systems for various inputs.	K2
CO3	Determine the stability of linear systems by various time domain techniques.	K2
CO4	Determine the stability of linear systems by frequency domain techniques.	K2
CO5	Obtain the state space representation of continuous time systems	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	1	-							2
CO2	3	3	2	1								2
CO3	3	-	3	2	1							2
CO4	3	-	3	2	1							2
CO5	3	2	1	2	-							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Control Systems Engineering	I. J. Nagrath and M. Gopal	New Age International Pvt Ltd	6
2	Modern Control Engineering	Katsuhiko Ogata	Pearson Education India	5
3	Control Systems	A. Anand Kumar	PHI	2
4	Control Systems Principles and Design	M. Gopal	McGraw Hill Education (India) Pvt. Ltd.	4

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Control Systems Engineering	Nise N. S	Wiley Eastern	6
2	Modern Control Systems	Dorf R. C. and R. H. Bishop	Pearson Education	13
3	Robotics- Fundamental Concepts and Analysis	Ashitava Ghosal	Oxford University press	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_de09/preview
2	https://onlinecourses.nptel.ac.in/noc22_de09/preview
3	https://onlinecourses.nptel.ac.in/noc22_de09/preview
4	https://onlinecourses.nptel.ac.in/noc24_ee90/preview

SEMESTER S6

FIELD AND SERVICE ROBOTICS

Course Code	PCRUT602	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBRUT504	Course Type	Theory

Course Objectives:

1. To give students the ability to use robots to solve problems in the field
2. To give students the ability to apply robotics to various services

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Field Robots - Definition Robotics in Agriculture and Forestry: Forestry - Technological Developments, Robot Locomotion in Forestry, Forestry Automation, Machine Perception and SLAM in the Forest Environment, Autonomy, Teleoperation, and Fleet Management, Autonomous Robots for Silviculture and Treatment, Broad Acre Applications in Agriculture - Overview of Automatic Guidance, Sowing, Weeding, Spraying, and Broad-Acre Harvesting, Horticulture - Picking of Fruit and Vegetables, Color Sorting and Produce Grading, Livestock Management - Robot Milking, Sheep Shearing, Livestock Inspection, Robotic Animals, Unmanned Vehicles in Agriculture, Applications of robots in Precision agriculture, Greenhouse farming, Seeding and harvesting, Aerial robots and Ground Robots for Agriculture Environmental Monitoring and Disaster Management Robots: Environmental Monitoring - Environmental Sensing, Sensors, Sensor Networks, Early Environmental Robotics, Energy-Efficient Sampling Platforms, Natural Habitat Mapping, Marine and Atmospheric Plume	9

	Detection and Localization, Data Muling WSNs, Robotics in Disaster Management - Surveying and Mapping, Rescue, Reconstruction Case Studies: Any one general application each of robotics in Agriculture and Forestry, Environmental Monitoring and Disaster Management	
2	Construction and Infrastructure Robots: Automation and robotics in the construction sector and precast concrete industry, timber construction, steel component production, Service robot systems and humanoid construction robots, Guidelines for construction robot development, Strategies for an automation and robot oriented construction process, Construction components for a robot oriented design & construction - Clear product structure, Design focused on simplicity, handling and assembly, Families of components, Component interfaces, Integral product structures, The accuracy problem, Robots in Inspection and Monitoring of Buildings and Infrastructure: Unmanned Aerial Vehicle (UAV), Unmanned Ground Vehicles (UGV), Wall-Climbing Robots, Cable-Crawling Robots, Marine Robots, Hinged Microbots, Legged Robots, Hybrid Robots, Multi-Robot Systems Mining Robots: Scope for Robotics in Mining - Underground Transport, Drilling, Explosive Placement, Secondary Rock Breaking, Metalliferous Roadway Development, Geology-Based Guidance Systems Military and Defence Robots: Applications of robots in military and defence Case Studies: Any one general application each of robotics in Construction and Infrastructure, Mining, Military and Defence	9
3	Marine and Underwater Robots: Types of Underwater Vehicles – Submarines, Submersibles, ADS, ROVs, AUVs, Hybrid Underwater Vehicles, Basics of moving a vehicle through water – Newton’s Laws, Forces on Underwater Vehicles, Navigation – Specifying Position or Location – Landmarks, Dead Reckoning, Standard Coordinate Systems and GPS, Triangulation, Navigational Instruments Case Study: A detailed look at a Work Class ROV – mechanical subsystems, Power source, Control room, Operations, crew, Deck cables, umbilical and tether, LARS, the ROV Space Robots: Orbital Robotic Systems, Surface Robotic Systems, Planetary Rovers – Architecture, Mission Planning, Propulsion and Locomotion, Planetary Navigation Educational and Entertainment Robots: Role of robots in Education,	9

	Case Study: Any one general application each of robotics in Education and Entertainment	
4	Service Robots: Definition, Mapping and Navigation for Service Robots: Map Building and Localization - Localization with Wi-Fi and Other Networked Radio Communications, Ultra Wideband Localization, Radio Frequency Identification Localization, North-Star Active Infrared Localization - Two-dimensional planar code localization, Laser-ranging localization, Camera-Based Localization, Time-of-Flight Camera Localization, Navigation, Path Planning, and Obstacle Avoidance - Structured Light Obstacle Avoidance, Linear and Planar Distance Sensors and Obstacle Avoidance, Optic Flow, Ground-Plane Segmentation, and Feature-Based Obstacle Avoidance, Optimization of Path Planning for Obstacle Avoidance, Optimization of Robot Behavior and Interaction for Obstacle Avoidance Household Service Robots: Introduction, Work Environments for Household Service Robots, Functionalities of Household Service Robots Examples of Service Robotic System Design: Surveillance Robot – Introduction, Block diagram of the system, System Initialization - Passive Acoustic Location, Target Detection Using the Frame Differencing Method, Feature Selection and Segmentation, Video Surveillance - Tracking Using Particle Filter, Target Model Update, Abnormal Behavior Detection, MFCC Feature Extraction, Support Vector Machine Service Robot for Elders - Robot System - System Architecture, Hardware Structure, Human Robot Interaction - Touch and Watch Interaction, Speech Interaction, Motion Interaction Medical and Healthcare Robots: Key Benefits, Surgical Robots - Surgeon extender robots, Auxiliary surgical support robots, Control Paradigms in Surgical Robots, Other Applications of Robots in Healthcare Logistics Robots: Application areas of robots in Logistics, Warehouse robots – Types of Warehouse Robots	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply field robots to sustainable management of agriculture, forestry, environment and disasters	K3
CO2	Apply field robots to effective execution of tasks in construction and infrastructure development, mining and defence	K3
CO3	Make use of robots to solve problems in the fields of underwater and space exploration, education and entertainment	K3
CO4	Select suitable characteristics and techniques for robots in the service sector	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	3	3	-	-	-	-	3
CO2	3	3	3	-	3	3	3	-	-	-	-	3
CO3	3	3	3	-	3	3	3	-	-	-	-	3
CO4	3	3	3	-	3	3	-	-	-	-	-	3
CO5	3	3	3	-	3	3	3	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Field and Service Robotics: Results of the 9th International Conference: 105 (Springer Tracts in Advanced Robotics)	Luis Mejias, Peter Corke , Jonathan Roberts (Editors)	Springer	2014
2	Service Robots	Antonio J. R. Neves (Editor)	Intechopen	2018
3	Robotics for Environmental Monitoring	Matthew Dunbabin and Lino Marques	IEEE Robotics & Automation Magazine	2012
4	Construction Automation and Robotics	Thomas Bock	Intechopen	2016
5	Robots in Inspection and Monitoring of Buildings and Infrastructure: A Systematic Review	Srijeet Halder, Kereshmeh Afsari	MDPI, Applied Sciences	2023
6	Underwater Robotics - Science, Design and Fabrication	Steve W Moore, Harry Bohm, Vickie Jensen	Marine Advanced Technology Edu	2010
7	Dynamics and Control of Autonomous Space Vehicles and Robotics	Ranjan Vepa	Cambridge University Press	2019
8	Household Service Robotics	Yangsheng Xu, Huihuan Qian, Xinyu Wu	Academic Press Inc;	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Field Robotics: A Systems Approach	Robert H. Sturges Jr	Wiley–Blackwell	2015
2	Robotics In Service	Joseph F Eng Elberger	Kluwer Academic Publishers	1989
3	Service Robots and Robotics – Design and Application	Marco Ceccarelli	Engineering Science Reference	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ge31/preview
2	https://onlinecourses.nptel.ac.in/noc24_ge31/preview
3	https://archive.nptel.ac.in/courses/112/105/112105249/
4	https://archive.nptel.ac.in/courses/112/107/112107289/

SEMESTER S6
AERIAL ROBOTICS

Course Code	PERUT631	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to UAV Systems -Introduction, Classes of UAV Systems, Basic Aerodynamics, Stability and Control, Propulsion, Payloads, Launch and Recovery	9
2	Hardware Anatomy of Quadcopter -Power Train – Propellers, Motors- Total Lift - Electronic Speed Controllers – Flight Battery – Radio transmitter and receiver – Flight Controller – Sensor, GPS, Compass, Camera Assembling for Quad copter – Connectors, Mounting of Propellers and Powering up.	9
3	Testing and Maintenance of Quadcopter -Key Flight Safety Rules - Preflight Checklist and Flight Log Information – Flight Instructions - Repair and Maintenance: Crash analysis, Common issues, Voltage testing.	9
4	Design ,Control of Quadcopter and Case Studies -Flight Performance Analysis – Dynamics and Design – Design Challenge – Guidance, Navigation, and Control of Drones, Applications of drones in agriculture, smart cities, delivery drones, health care, defense surveillance	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principle of UAV systems different Aerodynamics, Stability and Control, Propulsion	K2
CO2	Demonstrate the hardware anatomy of quadcopter Power Train ,gps and Propeller functionality	K2
CO3	Classify the use of various Testing and Maintenance of Quadcopter , Crash analysis and Voltage testing.	K2
CO4	Model the Control of Quadcopter Performance Analysis and Challenge.	K3
CO5	Explain the Case Studies Applications of drones in different areas	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2										2
CO2	2	2			2							2
CO3	2	2			1							2
CO4	3	3	2	2	2				2	1		3
CO5	2	2	1		1							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Unmanned Aircraft Systems: UAVs Design, Development and Deployment	Wiley Publications	Reg Austin	2010
2	Build Your Own Quadcopter -Power Up Your Designs with the Parallax Elev-8	Donald Norris	McGraw-Hill Education	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Building a Quadcopter with Arduino	VasilisTzivaras	Packt Publishing	2016
2	Drones and Unmanned Aerial Systems	Zavrsnik	Springer	2015
3	Drone Technology- Future Trends and Practical Applications	Sachi Nandan Mohanty	Scrivener Publishing	2023
4	Building your own Drones	John Baichwal	Que Publishing	2016

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/S-XiFIRVkgQ
2	https://youtu.be/mv81oj5PBwM
3	https://youtu.be/9c769xiEXn0?list=PLFW6lRTa1g83B1HdU2mece6QLeBrtsPL7
4	https://youtu.be/9c769xiEXn0?list=PLFW6lRTa1g83B1HdU2mece6QLeBrtsPL7

SEMESTER S6

ADVANCED INDUSTRIAL AUTOMATION

Course Code	PERUT632	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To provide an in-depth understanding of advanced industrial automation systems and technologies.
2. To develop the ability to design, implement, and manage automation solutions for industrial applications.
3. To develop proficiency in programming and using Programmable Logic Controllers (PLCs).
4. To foster an understanding of the integration of automation with business and manufacturing processes

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Advanced Industrial Automation- Automation Hierarchy: Field, Control, and Supervisory Levels. Sensors and Actuators: Types of Sensors, Signal Conditioning, Types of Actuators. Communication Protocols in Automation Safety in Automation Systems, Data Acquisition and Monitoring. Human-Machine Interface (HMI). Communication Protocols in Automation, Modbus, Profibus, Ethernet/IP.	9
2	Advanced Control Systems- Control System Fundamentals Open-loop vs Closed-Loop Control, PID Control PID Tuning Methods. Distributed Control Systems (DCS), Supervisory Control and Data Acquisition (SCADA) Advanced Control Strategies: Feedforward Control,	9

	Cascade Control, Model Predictive Control (MPC). Case Studies and Applications.	
3	Advanced PLC Programming and Applications: Advanced PLC Programming Techniques, Timers, Counters, and Sequencers. PLC Networking and Communication, SCADA Systems. Data Acquisition and Monitoring, PID Control using PLCs. PLC Troubleshooting and Maintenance, Industrial Applications of PLCs, Case Studies and Real-world Examples.	9
4	Recent Trends in Industrial Automation - Industrial Internet of Things (IIoT), Big Data Analytics in Automation, Cybersecurity in Industrial Automation, Artificial Intelligence and Machine Learning in Automation, Digital Twin Technology, Integration of Control Systems with IT Systems, Case Studies on Smart Factories and Industry 4.0. Future Trends and Innovations in Industrial Automation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the advanced industrial automation systems and technologies.	K2
CO2	Develop the ability to control automation solutions for industrial applications.	K3
CO3	Develop proficiency in Programmable Logic Controllers (PLCs) programming.	K3
CO4	Identify the integration of automation with business and modern tools.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3	2	2					2	2	2
CO2	3	3	3	3	2	2	2		2	2	2	3
CO3	3	2	3	2	3	2	2		2	2	2	3
CO4	3	3	3	3	3	2	2	2	2	2	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Automation and Robotics	A.K. Gupta and S.K. Arora	University Science Press	2nd Edition, 2017
2	Programmable Logic Controllers	Frank D. Petruzella	McGraw-Hill Education	5th Edition, 2016
3	Advanced Industrial Control and Applications	K.S. Fu, R.C. Gonzalez, and C.S.G. Lee	McGraw-Hill Education	1st Edition, 1987
4	Process Control: Modeling, Design, and Simulation	B.W. Bequette	Prentice Hall	1st Edition, 2003

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Industrial Control Technology	Peng Zhang, Bin Wei	Springer	1st Edition, 2021
2	Industrial Control Technology: A Handbook for Engineers and Researchers	Peng Zhang, Bin Wei	CRC Press	1st Edition, 2021

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/ntr24_ed63/preview
2	https://onlinecourses.swayam2.ac.in/nou24_ee02/preview

SEMESTER S6

MECHATRONIC SYSTEM DESIGN

Course Code	PERUT633	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Focus on understanding the interdisciplinary nature of mechatronics and the synergy between various components to create efficient and reliable systems.
2. Enable students to develop and apply advanced modeling, simulation, and control techniques for mechatronic systems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Mechatronics Definition, scope, and applications of mechatronic systems. Interdisciplinary approach: integration of mechanical, electronic, and software components.	9
2	Mechanical Components Introduction to mechanical components: actuators, sensors, and mechanical structures. Selection criteria and compatibility considerations. Electronic Components Overview of electronic components: microcontrollers, sensors, and signal conditioning circuits. Interfacing techniques and circuit design for integration.	9
3	Modeling Techniques Introduction to modeling methodologies: mathematical, physical, and empirical models. Modeling mechanical, electrical, and control subsystems.	9

	Simulation Tools Hands-on experience with simulation software (e.g., MATLAB/Simulink, LabVIEW). Simulating mechatronic systems to analyze performance and behavior.	
4	Control Theory Basics Introduction to control systems: feedback, PID control, and system stability. Design considerations for mechatronic control systems. Implementation and Testing Practical implementation of control algorithms on hardware platforms. Testing and validation of mechatronic systems using simulation and real-world experiments.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the interdisciplinary nature of mechatronics by examining how mechanical, electronic, and software components are integrated to create efficient and reliable systems.	K2
CO2	apply advanced modeling and simulation techniques to predict and analyze the behavior of mechatronic systems, enabling them to make informed design decisions.	K3
CO3	develop integrated mechatronic systems by designing and integrating mechanical, electronic, and software components, demonstrating proficiency in system-level design and integration	K3
CO4	evaluate and assess control system designs for mechatronic applications, demonstrating competence in designing, implementing, and evaluating control strategies to achieve desired system performance	K5

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1			1							1
CO2	3	1			3							1
CO3	3	1		3	3							1
CO4	3	1		3	3							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Mechatronics: Principles and Applications"	Godfrey C. Onwubolu	CRC Press	2015
2	"Mechatronics: A Foundation Course"	Clarence W. de Silva	CRC Press	2020
3	"Mechatronics: Electronic Control Systems in Mechanical and Electrical Engineering"	W. Bolton	Pearson	2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Mechatronics: Integrated Mechanical Electronic Systems"	R. K. Rajput	S. Chand & Company Ltd.	2016
2	"Mechatronics: Principles, Technologies and Applications"	J. K. Choudhury	CRC Press	2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=0RATDMszxoE
2	https://www.youtube.com/watch?v=1_Jfxh5Eoul
3	https://www.youtube.com/watch?v=jsAqyIGr-5o
4	https://www.youtube.com/watch?v=wkfEZmsQqiA

SEMESTER S6

ELECTROMAGNETICS

Course Code	PERUT634	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3-0-0-0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	1.Mathematics course including calculus, vector calculus, and differential equations. 2.GBPHT121Physics for Engineers	Course Type	Theory

Course Objectives:

1. To familiarize the students with the fundamentals of electrostatics, magnetostatics, time-varying fields and electromagnetic waves.
2. To equip students with expertise in analysing transmission line parameters, and utilizing the Smith chart for effective impedance matching and VSWR calculations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Co-ordinate Systems: Rectangular, Cylindrical and Spherical Co- ordinate Systems – Co-ordinate transformation, Vector Calculus: Gradient of a Scalar field, Divergence of a Vector field and Curl of a Vector field- their physical interpretation; Divergence Theorem, Stokes' Theorem	9
2	Electrostatics: Coulomb's Law, Electric field intensity, Electric Flux and Flux Density; Gauss's law and its application to determine the field due to an infinite line charge, infinite sheet charge; Poisson's and Laplace's equations Magnetostatics: Biot-Savart's Law, Magnetic Field intensity, Magnetic flux, Magnetic flux density, Magnetic Scalar and Vector Potential, Ampere's circuital law and simple applications; Continuity equation for current,	9

	Maxwell's equation from fundamental laws.	
3	Wave Propagation: Wave Equations from Maxwell's Equations; Solution of Wave equation Uniform Plane Waves, Propagation of Uniform Plane waves in free space, lossless and lossy dielectric medium, good conductor- phase velocity and group velocity, Intrinsic Impedance, Attenuation constant and Propagation constant in all medium; Skin effect and skin depth, Poynting Vector and Poynting Theorem, Wave polarisation and types of polarisation.	9
4	Transmission Lines: Uniform lossless transmission line - line parameters, Transmission line equations & solutions, Propagation constant, Characteristic impedance, Wavelength, phase velocity and group velocity, Derivation of input impedance of transmission line, Reflection coefficient and VSWR, Transmission line as circuit elements (L and C), Development of Smith chart, Calculation of line impedance and VSWR using smith chart.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Summarize the basic mathematical concepts related to electromagnetic vector fields.	K2
CO2	Apply the basic concepts and laws related to electrostatics and magnetostatics to diverse engineering problems.	K3
CO3	Analyse electromagnetic wave propagation in different media.	K3
CO4	Examine the characteristics of transmission lines using the basic laws of circuit analysis.	K3
CO5	Utilize the Smith chart for analysing and solving transmission line problems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	1	1								2
CO2	3	3	1	1								2
CO3	3	3	1	1								2
CO4	3	3	1	1								2
CO5	3	3	1	1								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Elements of Electromagnetics	Mathew N O Sadiku	Oxford University Press	6/e, 2014
2	Engineering Electromagnetics	Hayt W. H. and J. A. Buck	McGraw-Hill	8/e, 2014
3	Electromagnetics	John D. Kraus	TMH	5/e, 2010

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Electromagnetic waves and Radiating Systems	Jordan and Balmain	PH	2/e,2013
2	Elements of Engineering Electromagnetics	Nannapaneni Narayana Rao	Pearson	6/e, 2006
3	Electromagnetics, Schaum's Outline Series	Joseph A. Edminister	Tata McGraw-Hill	Revised 2nd Edition.
4	Fundamentals of Engineering Electromagnetics	Cheng D K	Addison-Wesley	2/e,1989
5	Electromagnetic Field Theory	Gangadhar K. A. and P. M. Ramanathan	Khanna Publishers	2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/115101005
2	https://nptel.ac.in/courses/115101005
3	https://nptel.ac.in/courses/115101005
4	http://www.nptelvideos.com/course.php?id=582

SEMESTER S6

QUALITY AND RELIABILITY

Course Code	PERUT636	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Demonstrate the approaches and techniques to assess and improve process and/or product quality and reliability.
2. Introduce the principles and techniques of Statistical Quality Control and their practical uses in product and/or process design and monitoring
3. Illustrate the basic concepts and techniques of modern reliability engineering tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Quality value and engineering – quality systems Quality engineering in product design and production process – system design, parameter design, tolerance design, Quality costs – quality improvement	9
2	Statistical Process control - X, R, p, c charts, other types of control charts, process capability, process capability analysis, process capability index. Acceptance sampling by variables and attributes, design of sampling plans, single, double, sequential and continuous sampling plans, design of various sampling plan	9
3	Loss function, tolerance design – N type, L type, S type; determination of tolerance for these types. Online quality control– variable characteristics attribute characteristics, parameter design.	9
4	Quality function deployment – House of quality, QFD matrix, and total quality management concepts. Quality information systems, quality circles, introduction to ISO 9000 standards. Reliability – Evaluation of design by tests - Hazard Models, Linear, Raleigh,	9

	Wei bull. Failure Data Analysis, reliability prediction based on wei bull distribution, Reliability improvement.	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Attain the basic techniques of quality improvement, fundamental knowledge of statistics and probability	K1,K2
CO2	Use control charts to analyze for improving the process quality	K1,K2
CO3	Describe different sampling plans	K1,K2
CO4	Acquire basic knowledge of total quality management and Reliability	K1,K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	-	-	-	-	1	-	-	-	-	-	1
CO2	1	2	-	-	-	1	1	-	-	-	-	1
CO3	1	-	2	-	-	-	-	-	-	-	-	1
CO4	1	-	1	-	1	2	1	1	1	-	2	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Statistical Process Control	Eugene Grant, Richard Leavenworth	McGraw Hill	
2	Quality Engineering in Production Systems	G Taguchi	McGraw Hill	1989
3	Optimization & Variation Reduction in Quality	W.A. Taylor	Tata McGraw Hil	1991

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Jurans Quality Planning and Analysis	Frank. M.Gryna Jr	McGrawHill	
2	Taguchi Techniques for Quality Engineering, (2ndEdition)	Philipposs	McGraw Hill	1996
3	Reliability Engineering, (3rdEdition)	LS Srinath	Affiliated East West Pvt Ltd	1991
4	Reliability Engineering	E.Bala Guruswamy	Tata McGraw Hill	1994

SEMESTER S6
ENERGY SYSTEMS

Course Code	PERUT637	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Introduces various types of renewable energy sources
2. Various means of generating and storing energy and the importance of renewable energy

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Energy Scenario: Indian Energy Scenario, World Energy Scenario, Indian Energy Sector Reforms, Energy and Environment, Energy Security, Energy conservation act Energy Efficient Systems: Reducing pollution and improving efficiency in buildings, Green Building Standards, Types of lamps and their efficiencies	9
2	Renewable Energy Resources: Solar Thermal System-Working Principle-Block diagram, Solar Photovoltaic System- Working Principle-Block diagram, Solar cell efficiency calculation, Wind Energy Systems- Working Principle-Block diagram, wind power equation, Energy from Waves and tides- Working Principle-Block diagram, Ocean Thermal Energy System-Working Principle-Block diagram, Energy from Biomass	9
3	Energy Storage: Importance of Energy Storage- Means of Storing Energy- Principle of operation and performance comparison. Compressed air storage, Fly wheel Energy Storage, Battery Storage-Battery: Specification, Charging/Discharging rate, Primary and secondary cells-Dry cell, lead acid, lithium ion, Lithium air, Nickel Cadmium, Nickel Metal Hydride Fuel Cell: Working Principle, efficiency	9

4	Energy Standards – International Energy Standards-ISO50001, Bureau of Energy Efficiency, star rating Energy Management: Significance and general principles of Energy Management, Energy audit-types and procedure, Energy audit report, Instruments for energy auditing Energy Economics: Traditional Types of Rates - Single-Part Rates - Two-Part Rates - ThreePart Rates – Numerical problems	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Illustrate Indian and global energy scenario	K2
CO2	Elaborate different conventional and non-conventional energy generation schemes and the economics of generation	K2
CO3	Analyse principle of operation and performance comparison of various energy storage schemes	K4
CO4	Identify major Global and Indian standards for Energy Management	K2
CO5	Appraise various aspects of energy economics	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										2
CO2	3	3										2
CO3	3	3										2
CO4	3	3										2
CO5	3	3										2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Energy Storage for Power Systems	A.G.Ter-Gazarian	The Institution of Engineering and Technology (IET) Publication, UK	Second Edition
2	Guide to Energy Management	Barney L. Capehart, Wayne C. Turner and William J. Kennedy	The Fairmont Press Inc., 2012	Seventh Edition
3	Electric Power Systems Planning	S. Pabla	Mac Millan India Ltd	First Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Renewable Energy Sources and Emerging Technology	K.C. Kothari, D.P.Ranjan, Rakeshsingal	PHI	2nd Revised edition
2	Energy Resources: Conventional & Non-Conventional	M.V.R. Koteswara Rao	BS Publications/BSP Books (2017)	First Edition
3	Efficient Lighting Applications & Case Studies	Albert Thumann, Scott Dunning	The Fairmont Press, Inc.	First Edition
4	Energy Economics-Concepts, Issues, Markets and Governance	Subhes C. Bhattacharyya	Springer	First Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=hMtZl4Op5VQ
2	https://archive.nptel.ac.in/courses/121/106/121106014/
3	https://onlinecourses.nptel.ac.in/noc21_mm34/preview
4	https://onlinecourses.nptel.ac.in/noc20_hs68/preview

SEMESTER S6

COMPILER DESIGN AND OPERATING SYSTEMS

Course Code	PERUT638	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PERUT 528	Course Type	Theory

Course Objectives:

1. To understand theory of compiler implementation, compiler parsing techniques, intermediate machine representation and optimization.
2. To impart fundamental understanding of the purpose, functions and the key design issues of an operating system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to compilers-Phases of compiler, The role of Lexical Analyzer, Role of the Parser, Basic parsing approaches - Eliminating left recursion, left factoring. Top-Down Parsing - Recursive Descent parsing, Nonrecursive Predictive Parsing, Bottom-up parsing-Shift Reduce Parsing, Simple LR Parser, More Powerful LR Parsers	9
2	Syntax directed definitions-Inherited and Synthesized Attributes. S-attributed definitions, L-attributed definitions. Intermediate Code Generation-Variants of Syntax Tree. Three address code-Quadruple, Triple. Run Time Environments, Local optimizations, Principal Sources of Optimizations.	9
3	Operating system overview - Functions, Service - System calls, Types-System boot process. Processes: Process states, Process control block, threads, scheduling, -Inter-process communication - shared memory systems, Message passing systems. Process Scheduling: Basic concepts- Scheduling criteria -Scheduling algorithms- First come First Served, Round robin scheduling	9

4	Process synchronization: Race conditions – Critical section problem – Peterson’s solution - Semaphores – Synchronization problems - Producer Consumer. Deadlocks: Necessary conditions, Resource allocation graphs - Deadlock avoidance – Banker’s algorithms. Memory Management: Concept of address spaces, Contiguous memory allocation, fixed and variable partitions. File System: File concept - Attributes, Operations, types	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the phases in compilation process and role of lexical analyser for a given source program	K2
CO2	Develop appropriate parsers for programming languages and syntax directed translation schemes for a given context free grammar.	K3
CO3	Illustrate intermediate code generation and code optimization techniques in compilation.	K2
CO4	Illustrate the relevance, functions, concepts of process management and process scheduling mechanisms employed in Operating Systems.	K2
CO5	Explain process synchronization, deadlocks, memory management, file management in Operating Systems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	2	2							2
CO2	3	3	3	2	2							2
CO3	3	3	3	2								2
CO4	3	3	3	2						2		2
CO5	3	3	3	2						2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Compilers-Principles, Techniques and Tools	Aho A.V, Ravi Sethi and D. Ullman	Addison Wesley, 2006	2nd edition, 2006
2	Operating System Concepts	Abraham Silberschatz, Peter Baer Galvin, Greg Gagne.	Wiley India 2015.	9th Edition, Wiley India 2015.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	System Programming and Operating Systems	D.M. Dhamdhere,	Tata McGraw Hill	3 rd Edition, 1996.
2	Modern Operating Systems.	Andrew S Tanenbaum	Prentice Hall	4th Edition, 2015.
3	Operating systems	William Stallings	Pearson, Global Edition	6th Edition, 2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106105190
2	https://nptel.ac.in/courses/106105190
3	https://archive.nptel.ac.in/courses/106/102/106102132
4	https://archive.nptel.ac.in/courses/106/102/106102132

SEMESTER S6

INDUSTRIAL INTERNET OF THINGS

Course Code	PERUT635	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	UCEST105 Algorithmic Thinking with Python	Course Type	Theory

Course Objectives:

1. To understand the basic concepts of Industrial Internet of Things (IIoT) systems.
2. To become proficient in the design, development and implementation of IIoT systems

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to IoT and IIoT: IOT Definition, Importance of IoT, IoT architecture, Technology Behind IoT - Server End Technology, Sources of Internet of Things, M2M Communication. IoT Application Areas. The Various Industrial Revolutions, Role of Internet of Things (IoT) & Industrial Internet of Things (IIoT) in Industry, Industry 4.0 revolutions, Support System for Industry 4.0.	9
2	Implementation Systems for IIoT: Sensors and actuators for industrial processes, sensor networks: network topologies, types of networks. Study of different sensors and their characteristics: Light sensor, temperature sensor with thermistor, voltage sensor, ADC and DAC, Temperature and Humidity Sensor DHT11, Motion Detection Sensors, Wireless Bluetooth Sensors, Level Sensors. Actuators: Thermal, hydraulic, pneumatic, electromechanical. Communication Protocols: UART Communication Protocol, I2C Protocol device interfacing and decoding of signal, SPI Protocol device interfacing and decoding of signal. Application layer protocols: HTTP, CoAP, MQTT.	9

3	Data Handling in IIoT: Role of microcontrollers in IIoT: Introduction to Raspberry Pi-Installation, programming, Interfaces (serial, SPI, I2C) Cloud Services: Configuration of the cloud platform, Sending data from the IoT nodes to the gateways using different communication options; Transferring data from gateway to the cloud. Google, AWS, Microsoft Azure cloud architect. Case study: Program different Sensors for IoT applications Security and privacy issues in IoT	9
4	Industrial IoT Applications: Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management. Oil, chemical and pharmaceutical industry, applications of UAVs in industries	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Two Course Project applying the principles of nonlinear control system analysis:(10 marks each)

Project phases: Proposal, Implementation, Testing, Final Report, Presentations and Viva Voce:

Faculty may direct students for the selection of topics for projects from real time examples.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. - Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts of IoT and IIoT	K2
CO2	Analyse basic sensors and communication protocols in IIoT	K4
CO3	Implement IoT system using Raspberry Pi	K3
CO4	Explain the basic cloud services in IIoT	K2
CO5	Describe various applications of IIoT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1							2
CO2	3	3	3		1							2
CO3	3	3	3		2							2
CO4	3	3	3		1							2
CO5	3	3	2		1							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Internet of Things : Architecture and Design Principles	Rajkamal	McGraw Hill (India) Private Limited.	2e/2022
2	Industry 4.0: The Industrial Internet of Things	Alasdair Gilchrist	Apress	2e/2019
3	Introduction to Industrial Internet of Things and Industry 4.0	Sudip Misra, Chandana Roy, Anandarup Mukherjee	CRC Press	1e/2021
4	Raspberry Pi Cookbook, Software and Hardware Problems and solutions	Simon Monk	O'Reilly Media, Inc.	4e/2022
5	Sensors, Actuators and Their Interfaces	Nathan Ida	SciTech Publishers	2e/2020
6	Internet of Things: Architectures, Protocols and Standards	Simone Cirani, Gianluigi Ferrari, Marco Picone, Luca Veltri	Wiley	1e/2018

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hands on Industrial Internet of Things	Giacomo Veneri, Antonio Capasso,	Packt Press	2018
2	Industrial Internet of Things (IIoT)	R. Anandan, Suseendran Gopalakrishnan, Souvik Pal, Noor Zaman	Wiley-Scrivener	2022
3	Industrial Internet of Things Technologies and Research Directions	Anand Sharma, Sunil Kumar Jangir, Manish Kumar, Dilip Kumar Choubey, Tarun Shrivastava, S. Balamurugan	CRC Press	1e/2022
4	Learning Internet of Things	Peter Waher	Packt Press	2015

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc20_cs69/preview
2	https://archive.nptel.ac.in/courses/106/105/106105195/
3	https://www.plcdojo.com/courses/learn-iot-using-raspberry-pi-with-mqtt-cloud
4	https://dituniversity.digimat.in/nptel/courses/video/106105195/L54.html

SEMESTER S6

SOFT COMPUTING AND COMPUTER VISION FOR ROBOTS

Course Code	PBRUT604	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Equip students with a solid understanding of the basic principles and significance of soft computing techniques, including fuzzy logic, genetic algorithms, and neural networks, and their differentiation from traditional hard computing methods
2. Introduce students to the concepts of hybrid systems such as neuro-fuzzy and genetic fuzzy systems, and provide a foundational understanding of machine learning techniques, including supervised, unsupervised, and reinforcement learning
3. Enable students to grasp the fundamental concepts of computer vision
4. Enable students to apply different vision algorithms for industrial applications.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Soft Computing : Definition and significance of soft computing; Differences between hard and soft computing, Introduction to Fuzzy Logic: Fuzzy Sets and Membership Functions, Fuzzy Rules and Inference Systems. Introduction to genetic algorithms-Working Principle, Operators: Selection, crossover, and mutation. Introduction to neural networks, Learning rules and various activation functions, Single layer Perceptrons multilayer perceptrons, Backpropagation algorithm, Architecture	11
2	Hybrid systems for Computer vision- Neuro-fuzzy systems, Genetic fuzzy system (Concept only). Introduction to Machine Learning in Robotics- Supervised and unsupervised-learning, Reinforcement learning (Concept	11

	only). Basics of Soft Computing Tools and Platforms: Open-source libraries (TensorFlow, Keras) (simple programs only)	
3	Introduction to Computer Vision: Industrial Machine Vision Basic concepts of image formation: Radiometry and geometric transformations. Geometric Camera Models-Pinhole camera model, Perspective projection and camera parameters. Techniques for 3D reconstruction. Spatial domain and Frequency domain methods for image enhancement. Image Segmentation: Thresholding., Region-based segmentation.	11
4	Pattern Analysis Clustering: K-Means, Mixture of Gaussians, Classification: Discriminant Function, Supervised, Un-supervised, Semi-supervised; Classifiers: Bayes, KNN, ANN models; Dimensionality Reduction: PCA, LDA, ICA Vision algorithms: Vision algorithms for identifying end wires- binary vision algorithm, grey-scale vision algorithm, active ranging using structured light technique in industrial vision	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of fuzzy logic, genetic algorithms, and neural networks.	K2
CO2	Analyze the performance of hybrid systems such as neuro-fuzzy and genetic fuzzy systems in computer vision applications.	K4
CO3	Explain the fundamentals of image processing methods required in a particular computer vision application	K2
CO4	Suggest methods for pattern recognition, image segmentation, and object detection in computer vision.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										
CO2	3	2	3	2	2							
CO3	3	3										
CO4	3	2	3	3	2	1						
CO5	3	2										

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Soft Computing and Intelligent Systems Design: Theory, Tools and Applications	F. O. Karray and C. de Silva		
2	Principles of Soft Computing	S N Sivanandam, Deepa	Wiley Publishers	3rd Edition
3	Neural networks, fuzzy systems and evolutionary algorithms : synthesis and applications	S Rajashekar and G A Vijayalakshmi Pai	PHI	2nd edition, 2017
4	Computer Vision: Principles and Practice	S. K. Parulkar, R. Raman	New Age International Publishers	1st edition 2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Image Processing: Principles and Applications	P. K. Chattopadhyay	Prentice Hall	1st edition 2011
2	Pattern Recognition and Image Analysis	A. K. Jain, B. Dubey	PH Learning	3rd edition 2014
3	Fuzzy Logic with Engineering Applications	Timothy J Ross	Wiley	4th Edition 2016
4	Computer Vision: Algorithms and Applications"	Richard Szeliski	Springer	1st edition 2010
5	Machine Learning: A Probabilistic Perspective	Kevin P Murphy	MIT press	1st edition 2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	Lecture 1 Introduction to soft computing (youtube.com) the complete series
2	
3	Computer Vision and Image Processing – Fundamentals and Applications [Intro Video] (youtube.com) (NPTEL IIT Guwahati)
4	

SEMESTER S6

DESIGN THINKING AND PRODUCT DEVELOPMENT

(Common to Group A & Group B)

Course Code	GXEST605	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	2:0:0:0	ESE Marks	60
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To guide students through the iterative stages of design thinking, including empathizing with users, defining problems, ideating solutions and developing Proof of Concepts (PoC) and technical feasibility studies.
2. To promote the development of critical thinking skills by engaging students in integrative inquiry, where they ask meaningful questions that connect classroom knowledge with real-world applications.
3. To equip students with the ability to involve in product design considering the sustainability, inclusivity, diversity and equity aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of design thinking and product development: Overview of stages of product development lifecycle; Design thinking -Definition-Design thinking for product innovation; Bringing social impact in ideation-Identifying societal needs-understanding multi-faceted issues-community engagement and empathetic design- technological innovation meeting societal needs; Understanding and Bridging the divide using Human Centered Design (HCD); Designing for inclusivity in product development-embracing user diversity - Long term impact - sustainability encompassing environmental,economic and social dimensions; Technology Readiness Level in the Innovation Life-cycle; Performing a self-check on innovative	6

	ideas - Originality of idea- understanding innovation landscape - patentability - understanding the economic landscape - Unique Selling Proposition (USP) - Repeatability and Manufacturability - Sustainability - Leveraging business models for comprehensive analysis	
2	Empathize: Design thinking phases; Role of empathy in design thinking; Methods of empathize phase - Ask 5 Why/ 5 W+H questions; Empathy maps - Things to be done prior to empathy mapping - Activities during and after the session; Understanding empathy tools - Customer Journey Map - Personas. Define: Methods of Define Phase: Storytelling, Critical items diagrams, Define success.	6
3	Ideation : Stages of ideation; Techniques and tools - Divergent thinking tools - Convergent thinking tools - Idea capturing tools; Cross-industry inspiration; Role of research in ideation - Market research - consumer research - leveraging research for informed ideation; Technological trends - navigating the technological landscape - Integrating emerging technologies; Feasibility studies - technical, economic, market, operational, legal, and ethical feasibility; Ideation session- techniques and tips. Proof of Concept (PoC) : Setting objectives; Risk assessment; Technology scouting; Document and process management; Change management; Knowledge Capture; Validating PoC; Story telling in PoC presentation	6
4	Design: Navigating from PoC to detailed design; Developing Specification Requirement Document (SRD)/Software Requirement Specification (SRS); Design for manufacturability; Industrial standards and readability of code; Design to cost; Pre-compliance; Optimized code; Design Failure Mode and Effects Analysis (DFMEA); Forecasting future design changes. Prototyping: Alpha prototypes; Beta prototypes; Transition from design to prototype; Goals and expectations for Alpha and Beta prototypes; Effective strategies for maintaining timeline in prototyping; Testing and refining Alpha prototypes; Transitioning to Beta prototypes. Pilot build: Definition and purpose of a pilot build; setting objectives; Identification and selection of manufacturing partner for pilot build; Testing procedures in pilot build; Scaling from pilot build to full-scale production / implementation.	6

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignments	Internal Examination	Reflective Journal and Portfolio	Total
5	20	10	5	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Empathize to capture the user needs and define the objectives with due consideration of various aspects including inclusivity, diversity and equity	K5
CO2	Ideate using divergent and convergent thinking to arrive at innovative ideas keeping in mind the sustainability, inclusivity, diversity and equity aspects.	K6
CO3	Engage in Human Centric Design of innovative products meeting the specifications	K5
CO4	Develop Proof of Concepts (PoC), prototypes & pilot build of products and test their performance with respect to the Specification Requirement Document.	K4
CO5	Reflect on professional and personal growth through the learnings in the course, identifying areas for further development	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2		2	3	3	3	2	2		3
CO2	3	2	3		2	3	3	3	2	2		3
CO3	3	2	3		2	3	3	2	2	2		3
CO4	3	2	2		3	3	3	2	2	2		3
CO5	3					3	3	2	2	2		3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Product Sense: Engineering your ideas into reality	Dr. K R Suresh Nair	NotionPress.com	2024
2	Change by Design: How Design Thinking Transforms Organizations and Inspires Innovation	Tim Brown	HarperCollins Publishers Ltd.	2009
3	Design Thinking for Strategic Innovation	Idris Mootee	John Wiley & Sons Inc.	2013

Sample Assignments:

1. Evaluate and prepare a report on how the aspects including inclusivity, diversity and equity are taken into consideration during the empathize and define phases of the Miniproject course.
2. Evaluate and prepare a report on how the aspects including sustainability, inclusivity, diversity and equity are taken into consideration during the ideate phase of the Miniproject course.
3. Evaluate and prepare a report on how User-Centric Design (UCD) is used in the design and development of PoC of the product being developed in the Miniproject course.
4. Prepare a plan for the prototype building of the product being developed in the Miniproject course.
5. Report on the activities during the empathize phase including the maps & other materials created during the sessions.
6. Report on the activities during the define phase including the maps & other materials created during the sessions.
7. Report of all the ideas created during the ideation phase of the Miniproject course through the tools including SCAMPER technique, SWOT analysis, Decision matrix analysis, six thinking hats exercise
8. Prepare a full scale production plan for the product being developed in the Miniproject course.
9. Create a Stanford Business Model Canvas related to the Miniproject.

An industrial visit of at least a day for experiential learning and submit a report on the learnings, for example industry standards and procedures.

SEMESTER S6
FUNDAMENTALS OF ROBOTICS

Course Code	OERUT611	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Understand and discuss the Robotic systems on the basis of various parameters
2. Provide insight into the types of sensors and actuators used in robotics
3. Explain the various types of end effectors.
4. Educate on the principles of kinematics and motion planning.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke & reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control. Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.	9
2	Sensors and Actuators Sensor classification- touch, force, proximity, vision sensors. Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors.	9

	Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.	
3	Robotic configurations and end effectors Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot.	9
4	Kinematics and Motion Planning Robot Coordinate Systems- Fundamental and composite rotations, homogeneous co-ordinates and transformations, Kinematic parameters, D-H representation, Direct Kinematics. The Arm equation forward Kinematic analysis of a typical robots up to 3 DOF. Motion Planning- joint space trajectory planning-cubic polynomial, linear trajectory with parabolic blends; Cartesian space planning, Point to point vs continuous path planning.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise with anatomy, specifications and applications of Robots	K2
CO2	Choose the appropriate sensors and actuators for robots	K3
CO3	Choose appropriate Robotic configuration and gripper for a particular application	K3
CO4	Obtain kinematic model of robotic manipulators	K4
CO5	Plan trajectories in joint space and Cartesian space	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										
CO2	2	1										
CO3	2	1										
CO4	3	2	2									
CO5	3	2	2									

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics	S K Saha,	Mc Graw Hill Eduaction	
2	Fundamentals of robotics – Analysis and control	Robert. J. Schilling	Prentice Hall of India	1996.
3	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	2003.
4	Introduction to Robotics (Mechanics and control)	John. J. Craig	Pearson Education Asia	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University press.	
2	Robotics Technology and Flexible Automation	S. R. Deb	Mc Graw Hill Eduaction	Second Edition

SEMESTER S6

INTRODUCTION TO INDUSTRIAL AUTOMATION

Course Code	OERUT612	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. To familiarise students with industrial automation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Automation methodologies: Concept of Mechanization and Automation, Types of Automation, Automated flow lines, Fundamentals of Transfer Lines. Trends in manufacturing – GT and Cellular Manufacturing, Flexible manufacturing systems – features of FMS, Computer integrated manufacturing – Need for AI and expert systems in CIM, Automated assembly system – flexible assembly automation,	9
2	Elements of CNC systems: servomotor and servo system design trends, stepper motors and controls, Automated tool changers and pallet changers. Material Handling and Identification Technologies: Overview of Material Handling Systems, Principles and Design Consideration, Material Transport Systems, Storage Systems, Automated Storage and Retrieval Systems, Overview of Automatic part Identification Methods, Automatic Guided Vehicles. Sensor systems for automated inspection, Automatic gauging and size control systems, thickness measurement, Machine vision systems.	9
3	Sensors and actuators for automation, Classification of position, proximity and motion sensors, electromechanical switches, LVDT, RVDT. Electrical, Hydraulic, and pneumatic actuators and their comparison (Concept only), Examples - use of Electrical, Hydraulic, and Pneumatic actuators in industrial automation.	9 h

	Pneumatic/Hydraulic Automation: control valves – direction, pressure and flow, sequential control of single /multiple actuator systems, cascade and Karnaugh Veitch map methods. Electro pneumatic/electrohydraulic automation: Symbols: Basic electrical elements – relay, solenoid, timers, pneumatic – electrical converters, Design of circuits and hands on models on material handling systems.	
4	Automation Control: Sequence control and programmable controllers – logic control and sequencing elements, Programmable Logic Controller – Architecture, PLC ladder diagram, Relay logic, Analog and digital I/Os, Timers, Counters, Program control instructions. Case studies on PLC ladder programming. SCADA – Building blocks of SCADA systems, RTU and HMI subsystem, Architecture of Distributed Control System (DCS), Interfacing PLC to SCADA/DCS using communication link (RS232, RS485), Protocols (Modbus ASCII/RTU) and OPC.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts of automation methodologies and trends in manufacturing automation.	K2
CO2	Explain the design aspects of modern CNC machines, different automated inspection methods and operation of different types of material handling devices.	K2
CO3	Design different pneumatic and hydraulic circuits based on their applications.	K3
CO4	Develop the basic concepts of PLC programming.	K3
CO5	Illustrate SCADA systems and Distributed Control systems for industrial automation.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	2							3
CO2	3	2	2	1	2	1						2
CO3	3	3	2	2	2	1						2
CO4	3	3	2	2	2	1						2
CO5	3	3	2	2	2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems and Computer Integrated Manufacturing	Mikell P Groover	Pearson	4th Edition, 2014
2	Programmable Logic Controllers	Frank D. Petruzella	McGraw Hill	4th Edition, 2011

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Control of Manufacturing Systems	Yoram Koren	Tata McGraw-Hill	1st Edition, 2005
2	Standard Handbook of Industrial Automation	Douglas M Considine & Glenn D Considine	Springer US	1986
3	Pneumatic Control for Industrial Automation	Peter Rohner& Gordon Smith	John Wiley and Sons	1987
4	Pneumatics, Concepts, Design and Applications	Jagadeesha T	University Press	2015

SEMESTER S6

COMPUTER INTEGRATED MANUFACTURING

Course Code	OERUT613	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to CAD and CAM: Overview of CAD/CAM, Importance in modern manufacturing, Manufacturing Planning and Control, Overview of Manufacturing Planning, Key aspects of Manufacturing Control Introduction to CIM: Concepts and significance of CIM, CIM system components: hardware and software, Production Types and Models, Types of production: batch, mass, and continuous, Manufacturing models and metrics, Automation and Lean Production, Basic elements of an automated system, Levels of automation, Lean Production and Just-In-Time (JIT) principles	9
2	Production Planning, Control, and CAPP: Process Planning and CAPP, Steps in Computer-Aided Process Planning, Benefits and implementation of CAPP, Production Planning Techniques, Aggregate Production Planning, Master Production Schedule Material Requirement Planning (MRP) and Capacity Planning: Introduction to MRP, Techniques for Capacity Planning Control Systems and Shop Floor Control: Overview of manufacturing control systems, Shop floor control techniques Inventory Control and ERP: Basic principles of inventory control, Introduction to Manufacturing Resource Planning-II (MRP-II): Overview of Enterprise Resource Planning (ERP)	9

3	Cellular Manufacturing and Group Technology: Group Technology (GT) and Part Families, Concept of Group Technology Classification and coding of part families: Production Flow Analysis, Techniques for production flow analysis, Benefits of production flow analysis Cellular Manufacturing Concepts: Composite part concept, Designing and layout of machine cells Quantitative Analysis in Cellular Manufacturing: Rank Order Clustering Method, Arranging machines in a GT cell using Hollier Method Machine Cell Design and Layout: Principles of machine cell design, Layout strategies for cellular manufacturing	9
4	Advanced Manufacturing Systems: Flexible Manufacturing Systems (FMS), Types and components of FMS, Applications and benefits, Planning and control of FMS Automated Guided Vehicle Systems (AGVS): Overview of AGVS, Applications and benefits, Vehicle guidance technology Vehicle Management and Safety in AGVS: Techniques for vehicle management, Safety considerations in AGVS Industrial Robotics: Anatomy and classification of robots, Robot control systems and end effectors, Sensors in robotics Robot Accuracy and Applications: Robot accuracy and repeatability, Industrial applications of robots	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Comprehend the fundamentals and importance of CIM, including CAD/CAM, production types, and automation principles.	K2
CO2	Analyze and apply knowledge of production planning, control systems, and CAPP in manufacturing processes.	K4
CO3	Understand and evaluate cellular manufacturing concepts, including Group Technology and machine cell design.	K5
CO4	Demonstrate knowledge of advanced manufacturing systems, such as FMS, AGVS, and industrial robotics, and their applications in modern manufacturing.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2	2	1	1	1	1	1	1	2
CO2	3	3	3	3	3	2	2	1	2	2	2	3
CO3	3	3	3	2	3	2	2	1	2	2	2	3
CO4	3	3	3	3	3	2	2	1	2	2	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automation, Production Systems, and Computer-Integrated Manufacturing	Mikell P. Groover	Pearson Education India	5 th edition, 2018
2	Computer Integrated Manufacturing	James A. Rehg and Henry W. Kraebber	Pearson Education India	3 rd edition, 2004
3	Computer-Aided Manufacturing	Tien-Chien Chang, Richard A. Wysk, and Hsu-Pin Wang	Pearson Education India	4 th edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Flexible Manufacturing Systems in Practice: Design, Analysis, and Simulation	Joseph Talavage and Roger G. Hannam	CRC Press	1 st edition, 1988
2	CAD/CAM: Principles and Application	P.N. Rao	McGraw Hill Education (India)	3 rd edition 2010

SEMESTER S6

INTRODUCTION TO AI IN ROBOTICS

Course Code	OERUT614	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Understand the Fundamentals of Robotics and AI:
2. Utilize supervised and unsupervised learning techniques for robotic perception and action.
3. Develop and evaluate novel AI algorithms for specific robotic applications through projects and case studies.
4. Analyze recent advancements and research trends in AI and robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Artificial intelligence : Introduction, its importance, The Turing test, Foundations of artificial intelligence, A brief historical overview, Intelligent Agents-Agents and Environments, Good behaviour: The concept of rationality, nature of Environments, structure of Agents, Expert systems - basic overview, Application areas of AI - vision and speech processing, robotics.	9
2	Learning - Forms of learning: Supervised Learning Algorithms- Decision Trees, Logistic Regression, Support vector Machines, Unsupervised Learning Algorithms: K-means clustering k-Nearest Neighbors, Principle Component Analysis, Reinforcement based learning .	9
3	Deep Feedforward Networks: Perceptrons – Single layer & Multi-layer, Learning XOR ,Convolutional Neural Networks -basic outline and functions of each layers only, Sequence Modeling-Need for sequence models, basic	9

	RNN architecture and types.	
4	Robotics - Robotic perception, Localization and mapping, Machine learning in robot Perception, Application domains.Mobile Robot Navigation-Path planning ,Path Planning and obstacle avoidance-A* algorithm	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Analyze the role of AI in solving problems in different domains and its evolution.	K3
CO2	Explain the different learning techniques used in Machine learning	K2
CO3	Identify the need for multilayer neural network for solving complex tasks	K3
CO4	Identify suitable neural network architectures for different types of data and tasks.	K3
CO5	Explain the different ways of perception of the environment by a robot and its use in path planning	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2								
CO2	3	3	3	2	2							
CO3	3	3	3	2	2							
CO4	3	3	3	2	2							
CO5	3	3	3	2	2							

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Deep Learning	Ian Goodfellow, Yoshua Bengio, Aaron Courville	MIT Press	2016
2	Artificial Intelligence - A Modern Approach	Stuart J. Russell and Peter Norvig,	Pearson	Third Edition, 2016
3	Pattern Recognition and Machine Learning,	Bishop, C. ,M.,	Springer	2006.
4	A survey of deep learning techniques for autonomous driving.	Grigorescu, Sorin, et al	Journal of Field Robotics	37.3 (2020): 362-386.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to AI Robotics,	Robin R. Murphy	The MIT Press	Second Edition: 2019
2	Artificial Intelligence and Machine Learning,	Chandra S.S.V, Anand Hareendran S	PHI	2021
3	Models, Learning and Inference	Simon J. D. Prince	Cambridge University Press	2012

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc22_cs56/preview
2	https://archive.nptel.ac.in/courses/106/106/106106139/
3	https://archive.nptel.ac.in/courses/106/106/106106184/
4	https://archive.nptel.ac.in/courses/112/106/112106298/

SEMESTER: S6

ROBOTICS AND MACHINE VISION LAB

Course Code	PCRUL607	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:2:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT402	Course Type	Lab

Course Objectives:

1. Develop Kinematic Models and Verify Performance
2. Familiarise Trajectory Planning Techniques
3. Implement Advanced Control Strategies
4. Integrate Sensing and Localization Techniques
5. Implement deep learning algorithms for object detection and tracking

Expt. No.	Experiments
1	Programming of Robotic Manipulators
2	Obtain forward and inverse kinematic models (check end effector and joint positions with theoretical and actual values)
3	Trajectory Planning for Robotic Manipulators(Joint space and Cartesian space trajectory planning)
4	Control of Mobile Robots- Control of mobile robot for moving to a point(x_g, y_g) , following a line ($ax+by+c=0$),moving to a specific target orientation (θ_g) (Closed loop control considering kinematic models)
5	Obstacle avoidance for a Differential Drive Robot using Ultrasonic Sensor
6	Localization of a Mobile Robot using LIDAR
7	Calibration of Sensors
8	Deep Learning based Object Detection and Tracking using Yolo Algorithm
9	Visual Servoing- to enable the robotic arm to autonomously locate and manipulate objects based on visual feedback from an attached camera.
10	Target Tracking from a Moving Vehicle

Course Assessment Method (CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
5	25	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- *Submission of Record: Students shall be allowed for the end semester examination only upon submitting the duly certified record.*
- *Endorsement by External Examiner: The external examiner shall endorse the record*

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Test forward, inverse kinematic modelling and path planning of robotic manipulators	K3
CO2	Test basic control algorithms in mobile robots to move to a point, to follow a line, to follow a path and for obstacle avoidance.	K3
CO3	Familiarise about localisation of mobile robots	K4
CO4	Calibrate sensors used in robots	K4
CO5	Design and develop sensor-based systems in robots	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO- PO Mapping (Mapping of Course Outcomes with Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	2		2			2	2		3
CO2	3	2	2	2		2			2	2		3
CO3	3	2	2	2		2			2	2		3
CO4	3	2	2	2		2			2	2		3
CO5	3	2	2	2		2			2	2		3

1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics	John. J. Craig,	Pearson Education Asia	2002.
2	Introduction to Robotics	S K Saha,	Mc Graw Hill Education	2014
3	Robotics-Fundamental concepts and analysis	Ashitava Ghosal	Oxford University press.	
4	Robotics, Vision and Control: Fundamental Algorithms in MATLAB,	Peter Corke	Springer	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics Technology and Flexible Automation,	S. R. Deb	McGraw Hill Education (India) Private Limited	Second Edition
2	Robotics and Control	R K Mittal and I J Nagrath	Tata McGraw Hill	New Delhi, 2003

Video Links (NPTEL, SWAYAM...)	
Experiment No.	Link ID
1	https://archive.nptel.ac.in/courses/112/106/112106304/
2	https://archive.nptel.ac.in/courses/112/106/112106304/
3	https://onlinecourses.nptel.ac.in/noc21_me108/preview
4	https://archive.nptel.ac.in/courses/112/106/112106298/
5	https://archive.nptel.ac.in/courses/112/106/112106298/
6	https://onlinecourses.nptel.ac.in/noc22_me38/preview

Continuous Assessment (25 Marks)

1. Preparation and Pre-Lab Work (7 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (7 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (6 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

Final Marks Averaging: The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 7

ROBOTICS AND AUTOMATION

SEMESTER S7

INDUSTRIAL NETWORKING

Course Code	PERUT741	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Basic knowledge about networking in industries.
2. Understand the evolution of computer networks using the layered network architecture.
3. Understand the concepts of data communications
4. Be familiar with the Transmission media and Tools

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Modern instrumentation and control systems – OSI model – Protocols – Standards – Common problems and solutions – Grounding/shielding and noise - EIA-232 interface standard – EIA-485 interface standard – Current loop and EIA-485 converters. FIBRE OPTICS: Introduction – Fibre optic cable components and parameters – Basic cable types – Connection fibres – troubleshooting	9
2	MODBUS Overview – Protocol structure – Function codes – Modbus plus protocol – Data Highway – AS interface (AS-i) – Device Net: Physical layer – Topology – Device taps – Profibus PA/DP/FMS: Protocol stack – System operation	9
3	ETHERNET SYSTEMS IEEE/ISO standards – Medium access control – frames – Reducing collisions – Auto negotiation – LAN system components – Structured cabling – Industrial Ethernet – Troubleshooting Ethernet. CAN BUS: Concepts of bus access and arbitration – CAN: Protocol-Errors: Properties – detection – processing – Introduction to CAN 2.0B	9
4	WIRELESS COMMUNICATIONS Radio spectrum – Frequency allocation – Radio modem – Intermodulation – Implementing a radio link – RFID: Basic principles of radio frequency identification – Transponders – Interrogators. APPLICATIONS Automotive communication technologies – Design of automotive X-by-Wire systems (Assignment only)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the concepts of data communications to design computer networks using subnetting and routing concepts	K3
CO2	Compare the various medium access control techniques.	K2
CO3	Compare and contrast the characteristics of the physical layer.	K2
CO4	Analyze the different protocols.	K4
CO5	Compare and contrast the different network components.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	1	1	1							1
CO2	1	1	1	1	1							1
CO3												
CO4												
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Practical Industrial Data Networks: Design, Installation and Troubleshooting	Steve Mackay, Edwin Wright, Deon Reynders and John Park	Newnes (Elsevier)	2004
2	Practical Filebus, DeviceNet and Ethernet for Industry	IDC Technology		2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	The Industrial Communication Technology Handbook	Richard Zurawski	Taylor and Francis	2005
2	Multiplexed Networks for Embedded Systems	Dominique Paret	John Wiley & Sons	2007
3	RFID Design Fundamentals and Applications	Albert Lozano-Nieto	CRC Press	2011

SEMESTER S7

NON-LINEAR CONTROL

Course Code	PERUT742	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCRUT601 Robotic Control Systems	Course Type	Theory

Course Objectives:

1. To understand the behavior of nonlinear system
2. To familiarize the analysis of nonlinear system with various methods

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction: Characteristics of non-linear systems - Methods of analysis – Classifications of nonlinearities – Common physical nonlinearities – Saturation or limiter – Deadzone or Threshold – Types of relays - Types of springs – Backlash in Gears – Inherent nonlinearity. Linearisation of Nonlinear system - Autonomous and non autonomous systems-modelling of simple pendulum- mass spring system.	9
2	Phase plane analysis Phase Plane Analysis: Singular points- Phase portraits – Analytical method for the construction of Phase trajectories – Graphical method f construction of Phase trajectory – Isoclines Method – Analysis and classification of singular points - Local behaviour of nonlinear systems - Limit cycles on phase plane - Stability- poincare- bendixon theorems.	9
3	Describing function analysis Describing Function: Definition - Describing Function Analysis- Describing functions of Typical nonlinearities- Ideal Relay – Relay with deadzone – Simple deadzone – Saturation or Limiter	9

	Closed loop stability using describing function- Stability of limit cycles – Accuracy of describing function analysis - Relative stability from describing from describing function.	
4	Stability of the system- Lyapunov Stability Theory Stability of nonlinear systems- Concept - equilibrium points- stability in small and stability in large- Lyapunov's stability definitions – Localisation and stability in the small - Stability by the method of Lyapunov – First method of Lyapunov- Concept of Lyapunov's stability theorems - Sign definiteness of scalar function Lyapunov stability theorems - Stability of nonlinear system by method of Lyapunov – Krasovskii' theorem- Variable gradient method of generating Lyapunov function	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand nonlinear system behaviour	K2
CO2	Analyse nonlinear system using phase plane method	K3
CO3	Understand and apply describing function method for analysing nonlinear system	K3
CO4	Analyse the stability of nonlinear system using Lyapunov method	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										2
CO2	3	2	2	2								2
CO3	3	2	2	2								2
CO4	3	2	2	2								2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nonlinear systems	H. K. Khalil	Prentice Hall	3rd Edition, 2002.
2	Applied Nonlinear Control	Jean Jacques Slotine, Weiping Li	Prentice Hall Inc.	1991
3	Nonlinear Systems; Analysis, Stability and Control	Shankar Sastry	Springer	1999
4	Advanced Control System	Nagoor Kani	Rba Publication	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Nonlinear systems analysis	M. Vidyasagar	Society of Industrial and Applied Mathematics	2nd Edition, 2002
2	Nonlinear Control Systems, Analysis and Design	H. Marquez	Wiley	2003
3	Nonlinear Control Systems	A. Isidori,	Springer	3rd Edition, 1995
4	Nonlinear Differential Equations and Dynamical Systems	F. Verhulst	Springer	2nd Edition, 1996.

SEMESTER S7

DESIGNING THE MECHANISMS FOR AUTOMATED MACHINES

Course Code	PERUT743	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. This course is designed to build a basic understanding on the various concepts involved in the design of different mechanisms of automated machines.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Automated Systems <i>Overview of Automated Systems:</i> Structure of automatic industrial systems, Linear and circular configurations, Non-industrial robots <i>Manufacturing Process Layouts:</i> Processing layout, Case studies: Chain links, springs, soldering printed circuits, galvanizing steel strips, Approaches to automating manufacturing processes, Timing layout and productivity determination,	9
2	Drive Systems and Dynamic Analysis Mechanical and Electromagnetic Drives: Mechanical drive systems: Falling weight, inclined belt, spring systems, Electromagnetic drives and electric drives: Induction motors, stepper motors Hydraulic and Pneumatic Drives: Hydraulic and pneumatic drives, Brakes and variable moment of inertia Dynamic Analysis: Calculation of selection parameters, Calculation of braking time for different systems	9
3	Kinematics and Control Kinematics and Control Mechanisms: Position function and input-output relations, 4 slot Geneva mechanism, cam mechanisms, Amplifiers and flexible cam systems	9

	Dynamic Accuracy and Damping: Dynamic accuracy and vibrations, Damping: Electrical and mechanical	
4	Transportation, Feeding, and Functional Systems Transportation Devices: Linear and rotational transportation Conveyors, indexing tables, Case studies on transportation devices Feeding and Orientation Devices: Feeding systems: Liquid, powder, wire, rod Orientation: Passive, active, logical, and electromagnetic fields Functional Systems and Manipulators: Automatic assembling and inspection systems, Manipulators: Dynamics, kinematics, vibrations	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify and describe the structure and components of automated industrial systems	K1
CO2	Analyze and evaluate the performance of various drive systems in industrial automation	K4
CO3	Apply principles of kinematics and control mechanisms to design and optimize automated machines.	K3
CO4	Design and implement transportation, feeding, and functional systems for automated manufacturing processes.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1	1	1	1	2	2	1	3
CO2	3	3	2	3	2	1	1	1	2	2	2	3
CO3	3	3	3	2	2	1	1	1	2	2	2	3
CO4	3	3	3	2	3	2	2	1	3	2	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotics Designing the Mechanisms for Automated Machinery	Ben-Zion Sandler	Academic Press	2 nd edition, 1999
2	Industrial Automation: Hands On	Frank Lamb	McGraw-Hill Professional	1 st edition, 2013
3	Mechatronics: Principles and Applications	Godfrey C. Onwubolu	Elsevier India Edition	1 st Edition, 2005
4	Robotics: Control, Sensing, Vision, and Intelligence	K.S. Fu, R.C. Gonzalez, C.S.G. Lee	McGraw Hill Education (India)	1 st Edition, 2008

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mechatronics	Bolton W	Pearson Education	6 th edition, 2015
2	Introduction to Mechatronics and Measurement systems	Davis G.Alciatore and Michael B. Histan	McGraw Hill Education	4 th edition, 2011

SEMESTER S7

ANALOG AND DIGITAL COMMUNICATION

Course Code	PERUT744	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the concept of analog communication.
2. To analyse different digital communication schemes.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Analog Communication Block diagram of a communication system. Need for analog modulation. Amplitude modulation. Equation and spectrum of AM signal. DSB-SC and SSB systems. Block diagram of SSB transmitter and receiver. Frequency and phase modulation. Narrow and wide band FM and their spectra. FM transmitter and receiver.	9
2	Stochastic processes and Source coding Review of probability. Bayes theorem and conditional probability. Stochastic processes, Stationarity. Conditions for WSS and SSS. Autocorrelation and power spectral density. LTI systems with WSS as input. Sampling and Quantization. Pulse code modulation, Transmitter and receiver. Companding. Practical 15 level A and mu-law companders. DPCM transmitter and receiver. Delta modulation.	9
3	G-S Procedure and Effects in the Channel Definition of signal .Basic operation on signals. Gram-Schmitt procedure. Signal space. Baseband transmission through AWGN channel. Inter Symbol Interference. Nyquist criterion for zero ISI. Raised cosine and Square-root raised cosine spectrum, Partial response signalling and	9

	duobinary coding. Equalization. Vector model of AWGN channel. Matched filter and correlation receivers. MAP receiver, Maximum likelihood receiver and probability of error.	
4	Digital Modulation Schemes Digital modulation schemes. Baseband BPSK system and the signal constellation. BPSK transmitter and receiver. Base band QPSK system and Signal constellations. Plots of BER Vs SNR with analysis. QPSK transmitter and receiver. Quadrature amplitude modulation and signal constellation.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the existent analog communication systems.	K1
CO2	Apply the concepts of random processes and waveform coding techniques in digital transmission.	K3
CO3	Apply GS procedure to develop digital receivers	K3
CO4	Apply equalizer design to counteract ISI..	K3
CO5	Apply digital modulation techniques in signal transmission	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3										
CO2	3	3	2	3	3	2						2
CO3	3	3	2	3	3	2						2
CO4	3	3	2	3	3	2						2
CO5	3	3	2	3	3	2						2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Communication Systems	Simon Haykin	Wiley.	Fourth edition ,2006
2	Digital Communications: Fundamentals and Applications	Sklar	Pearson	Second edition ,2001
3	Digital Telephony	John C. Bellamy	Wiley	Third edition ,2000
4	Analog and Digital Communication	Couch	Pearson Education India	Eighth edition ,2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Digital Communication	R. Gallager	Oxford University Press	2008
2	Digital Communication	John G Proakis, Masoud Salehi	McGraw Hill Education	Fifth edition, 2014.
3	Principles of Communication Systems	H.Taub and Schilling	McGraw Hill Education	Fourth edition ,2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105143/
2	https://archive.nptel.ac.in/courses/117/105/117105077/
3	https://archive.nptel.ac.in/courses/108/102/108102120/
4	https://archive.nptel.ac.in/courses/108/102/108102120/

SEMESTER S7

OPERATIONS RESEARCH

Course Code	PERUT746	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics of operations research- OR models- applications, Linear programming: problem formulation-Graphical method-simplex method- Big M method-Two phase method-Duality in linear programming.	9
2	Transportation Problem: Formulation- balanced & unbalanced transportation problems, method for finding initial solution- North west corner method- Least cost entry method-Vogel's approximation method (VAM), test for finding optimality- MODI method. Assignment Problem: Formulation-optimal solution- Hungarian Algorithm- Variants of assignment problems.	9
3	Introduction to queuing theory: terminologies- classification of queuing models- Kendall's notation-Single server models Multi server models. Network analysis: basic terms- network construction-time analysis-Critical Path Method (CPM)-Programme Evaluation Review Technique (PERT).	9
4	Decision theory: steps in decision theory approach-decision making conditions- decisions under risk-decision under uncertainty conditions- decision tree analysis. Game theory: games with saddle points- Dominance property- games without saddle points for 2x2 games- graphical method for $m \times 2$ & $2 \times n$ games.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Formulate and solve linear programming problems.	K2
CO2	Optimize transportation and assignment problems using different optimization methods.	K2
CO3	Evaluate different parameters of queuing models.	K2
CO4	Construct and evaluate the project duration of a network.	K2
CO5	Solve a game and find an optimum decision based on different approach.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	1			1	2		2
CO2	3	3	3	3	2	1			1	2		2
CO3	3	3	3	3	2	1			1	2		2
CO4	3	2	3	2	1	1			1	2		2
CO5	3	3	3	3	2	1			1	2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Industrial Engineering and Operations Research	Miller, D.M. and Schmidt, J.W.	John Wiley & Sons, Singapore.	1990
2	Operations Research	Paneerselvam,R.	Prentice Hall of India, New Delhi.	2008
3	“Operations Research – Principles and Applications”.	Srinivasan, G.	PHI Pvt.Ltd.	2010
4	Operations Research	Taha, H.A.	Pearson	2004

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Linear Programming	George Bernard Dantzig, Mukund Narain Thapa.	Springer series in operations research, 3 rd Edition	2003
2	“Optimization Theory and Applications”.	S.S.Rao	Second Edition, New Age International (P) Limited Publishers.	1995

SEMESTER S7

POWER ELECTRONICS

Course Code	PERUT747	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the basic principles of power electronics and the characteristics of power semiconductor devices.
2. To design various power converter circuits including rectifiers, DC-DC converters, inverters, and AC-AC converters.
3. To learn the application of power electronics in controlling electric drives and power systems.
4. To explore the emerging trends and technologies in power electronics and their applications in automation and robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Power Semiconductor Devices: Power Diodes, TRIACs, GTOs, Power BJTs, Power MOSFETs, and IGBTs. Thyristors, Static and Dynamic Characteristics. Gate characteristics. Firing methods. Gate drive circuits. Thyristor protection circuits. Series and parallel combinations. Commutation types (no analysis).	9
2	AC-DC Converters: Single-phase Uncontrolled and Controlled Rectifiers. Three-phase Controlled Rectifiers. Inductive, and Capacitive Loads. Power Factor Improvement in Rectifiers. DC-DC Converters: Step up and step down Chopper. Buck, Boost, Buck-Boost Converters. Isolated and non-isolated converters	9
3	Inverters: Single-phase and three-phase inverters. Voltage source inverters – series, parallel, and bridge inverters – current source inverters. 1800 and 1200 modes. PWM inverters.	9

	AC-AC Converters: Single-phase and three-phase AC-AC Converters. Cycloconverters (single phase)	
4	Applications of Power Electronics: D.C Motor control - phase control, Single phase SCR drive, Three phase SCR drive, Chopper controlled DC drives. A.C. Motor control: Controlled – slip system and slip power recovery system, stepper motor drive. UPS, Battery charging techniques. Recent Developments in Power Electronics for Automation and Robotics.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the operating principles of power semiconductor devices and their characteristics.	K2
CO2	Design rectifier and chopper circuits for various applications.	K3
CO3	Develop inverters and AC-AC converters for various applications.	K3
CO4	Apply power electronics concepts in the control of electric drives and power systems. Also, explore the recent trends.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2	1	1							2
CO2	3	3	3	2	2							3
CO3	3	3	3	3	3	1						3
CO4	3	3	3	3	3	2						3
CO5												

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Power Electronics: Circuits, Devices and Applications	Muhammad H. Rashid	Pearson	4th edition, 2013
2	Power Electronics: Converters, Applications, and Design	Ned Mohan, Tore M. Undeland, and William P. Robbins	Wiley	4th edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Modern Power Electronics and AC Drives	Bimal K. Bose	Pearson	1 st edition, 2001
2	Fundamentals of Power Electronics	Robert W. Erickson and Dragan Maksimovic	Springer	3 rd edition, 2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeeep.ra@adishankara.ac.in
2	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeeep.ra@adishankara.ac.in
3	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeeep.ra@adishankara.ac.in
4	https://onlinecourses.nptel.ac.in/noc24_ee88/preview?user_email=sreedeeep.ra@adishankara.ac.in

SEMESTER S7

DATABASE MANAGEMENT SYSTEMS

Course Code	PERUT748	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PERUT528 Algorithms and Data Structures/ Exposure to any high level programming languages	Course Type	Theory

Course Objectives:

1. To impart the basic understanding of the theory and applications of database management systems in real-world scenarios.
2. To enable students in designing, querying, and managing databases efficiently

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Basics: Overview of Database Management Systems - Characteristics of database approach vs file system approach – Advantages & Applications – Three Schema Architecture - Three tier Architecture. Entity Relationship model: Entity Types and Sets – Attributes and types, Keys - Relationships and types - Cardinality, Participation - Structural Constraints – Conventional and min-max Notations – Drawing and interpreting ER diagrams. Relational Model Concepts: Structure of Relational Databases - Domains, Attributes, Tuples, Relations – Types of Keys -Integrity Constraints - Synthesizing ER diagram to Relational Schema – Examples.	9
2	Introduction to Relational Algebra: Select, Project, Rename, Union, Intersection, Difference, Division - Cartesian product – Join and its types – Relational algebra query examples. Structured Query Language (SQL): Data Types – Data Definition Language (DDL): Create, Alter, Drop, Truncate, rename – Data Manipulation Language (DML): Select, Insert, Update, Delete - Nested queries (correlated and non-correlated) - Aggregation and Grouping –	9

	Views – Example SQL queries on single and multiple tables – Query Practise.	
3	Normalization: DB Design Anomalies - Functional Dependencies - Armstrong's Axioms – Closures - Equivalence of FDs, Minimal Cover - First Normal Form (1NF) - Second Normal Form (2NF) - Third Normal Form (3NF) - Boyce Codd Normal Form (BCNF) - Lossless Join Decomposition - Dependency Preserving Decomposition. Physical Data Organization: Physical and Logical records - Blocking Factor - Pinned and unpinned organization - Heap files – Sorted files – Indexing - Single-Level Ordered Indexes - Multilevel Indexes – Dynamic indexing – B Trees & B⁺ Trees.	9
4	Transaction Processing Concepts: Transaction States and Operations – ACID Properties - Serial Schedules - Concurrent and Serializable Schedules - Conflict Equivalence and Conflict Serializability – Recoverable, Cascade-less & Strict Schedules - Locking - Two-phase locking and its variations - System Log – Log based recovery - Deferred database modification – Check pointing.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts of databases.	K2
CO2	Model real word scenarios, given as informal descriptions, using Entity Relationship model.	K3
CO3	Discover solutions for efficiently representing and querying data using relational model.	K4
CO4	Examine the features of database storage structures and access techniques like indexing and hashing.	K3
CO5	Examine the different concurrency control and recovery mechanisms in database systems.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1	2	-	-	-	-	-	-	-	-	3
CO2	3	2	3	3	-	1	-	-	-	3	-	3
CO3	3	3	3	3	1	1	-	-	-	3	-	3
CO4	3	2	2	-	-	-	-	-	-	-	-	3
CO5	3	2	3	-	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Database Systems	Elmasri R. and S. Navathe	Pearson Education	7 th Edition
2	Database System Concepts	Sliberschatz A., H. F. Korth and S. Sudarshan	McGraw-Hill	7 th Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data base Management Systems	Raghurama Krishnan, Johannes Gehrke	McGraw Hill	3 rd Edition
2	An Introduction to Database systems	C.J. Date, A.Kannan, S.Swami Nadhan	Pearson Education	8 th Edition
3	SQL in 10 Minutes a Day, Sams Teach Yourself	Ben Forta	SAMS	5 th Edition
4	Database Management System Oracle SQLI And PL/SQL	Das Gupta, Pranab Kumar Krishna, P. Radha	PHI Learning	2 nd Edition

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.swayam2.ac.in/ini24_cs01/preview https://archive.nptel.ac.in/courses/106/105/106105175/
2	https://onlinecourses.swayam2.ac.in/ini24_cs01/preview https://archive.nptel.ac.in/courses/106/105/106105175/
3	https://onlinecourses.swayam2.ac.in/ini24_cs01/preview https://archive.nptel.ac.in/courses/106/105/106105175/
4	https://onlinecourses.swayam2.ac.in/ini24_cs01/preview https://archive.nptel.ac.in/courses/106/105/106105175/

SEMESTER S7

ROBOT NAVIGATION

Course Code	PERUT735	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	NONE	Course Type	Theory

Course Objectives:

1. Robot navigation deals with the ability of the robot to determine its own position in its frame of reference and then to plan a path towards a goal location
2. Students will learn the various aspects and techniques associated with robot navigation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Building blocks of navigation: Definitions - perception, localization, cognition, motion control Perception: Sensors for Perception - Schematic representation of the sensing unit, Obstacle Sensor (Bumper), the odometry sensor, Heading sensors - Gyroscopes, IMUs, Distance sensors - ToF, Phase shift, triangulation, ultrasonic rangefinders, Ground based beacons, GPS, Motion field and optical flow, Color tracking sensors, Feature Extraction - Feature Definition, Target Environment, Environment representation	9
2	Localization: General schematic for mobile robot localization, Challenge of Localization - Sensor noise and Sensor aliasing, Effector noise, Error model for odometric position estimation Localization-Based Navigation versus Programmed Solutions, Belief Representation, Map Representation - Continuous Representation, Decomposition Strategies, current challenges in map representation, Probabilistic Map-Based Localization - Markov localization, Kalman filter localization, Application to mobile robots: Kalman filter localization - schematic, Robot position prediction, Observation, Measurement prediction,	9

	Matching, Estimation, Landmark-based navigation, Globally unique localization, Positioning beacon systems, Route-based localization, Autonomous Map Building - stochastic map technique, Challenges - Cyclic environments, Dynamic environments	
3	Motion in Potential Field and Navigation Function Problem Statement, configuration space, free configuration space, Gradient Descent Method of Optimization, Gradient Descent without and with Constraints, Minkowski Sum, Potential Field, Navigation Function - in Static Deterministic Environment, in Static Uncertain Environment, Navigation Function and Potential Fields in Dynamic Environment - Estimation, Prediction, Optimization	9
4	Satellite Navigation Introduction to Satellite Navigation, Trilateration, Position Calculation - Multipath Signals, GNSS Accuracy Analysis, Dilution of Precision, Coordinate Systems, UTM Projection, Local Cartesian Coordinates, Velocity Calculation, Urban Navigation - Urban Canyon Navigation, Map Matching, Dead Reckoning – Inertial Sensors, Incorporating GNSS Data with INS - Modified Particle Filter, Estimating Velocity by Combining GNSS and INS, A-GPS, DGPS Systems, RTK Navigation, GNSS Threats	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Two Course Project applying the principles of nonlinear control system analysis:(10 marks each)

Project phases: Proposal, Implementation, Testing, Final Report, Presentations and Viva Voce:

Faculty may direct students for the selection of topics for projects from real time examples.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe sensors for perception in navigation	K2
CO2	Compare various methods for robot localization	K2
CO3	Explain the techniques of motion in Potential Field and Navigation Function	K2
CO4	Apply information from satellites for navigation	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	2									2
CO2	3	2	3									2
CO3	3	2	3									2
CO4	3	2	3									2
CO5	3	2	2									2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Autonomous Mobile Robots	Roland Siegwart and Illah R. Nourbakhsh	MIT Press	
2	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	
3	Probabilistic Robots	Thrun, Burgard	MIT Press	

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Planning Algorithms	Steven M. LaValle	Cambridge University Press	
2	Robot Motion Planning	Jean- Claude Latombe	Springer Science+Business Media	

SEMESTER S7

UNDER WATER ROBOTICS

Course Code	PERUT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To describe the principle of locomotion and different types of mobile robots.
2. To impart knowledge of kinematics of underwater vehicle.
3. To develop skills in motion path planning, control and stability of underwater robots.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Fundamentals of Mobile Robot: Introduction to mobile robots - principle of locomotion - types of mobile robots: ground robot, aerial robot, underwater robot and water-surface robot - principles of underwater vehicle construction	9
2	Kinematics of underwater vehicle: Equations for moving frame - rigid motion in a plane - representation of a rotated frame - holonomic and non-holonomic systems - kinematic modeling with respect to global coordinates.	9
3	Sensors for robot navigation: Types of sensors - magnetic and optical position sensor - gyroscope - accelerometer - magnetic compass inclinometer - tactile and proximity sensor - ultrasound range finder - laser scanner, infrared range finder - visual and motion sensing systems.	9
4	Motion path planning, control and stability: path planning algorithms - collision-free path planning - sensor-based obstacle avoidance - motion control methods: kinematic control, dynamic control, controllability and stability about a point and trajectory	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the principle of locomotion and described different types of mobile robots	K2
CO2	Summarize the degree of freedom to manoeuvrability of various robots	K2
CO3	Identify the use of various sensors deployed in autonomous robots	K2
CO4	Develop motion path planning and its control for an autonomous robot	K3
CO5	Explain the robust feedback control methods	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2		2								
CO2	3	2		2	2							
CO3	2	3	2	2								2
CO4	3	3	2	2	2							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Autonomous Underwater Vehicles	Sabiha Wadoo, Pushkin Kachroo,	CRC Press	2011
2	Visual Perception and Control of Underwater Robots	Yu Junzhi	CRC Press	2018
3	Underwater Robots	Gianluca Antonelli	Springer Cham	2018
4	Underwater Robots	Dan Faust	Rosen Publishing	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Under water Robotics Science Design & Fabrication	Steven W. Moore	Marine Advanced Technology Education	2010
2	Introduction to Autonomous Robots,	Nikolaus Correll	The MIT Press	2022
3	Navigation, Control and Sensing, Surface	Gerald Cook, Feitian Zhang,	Wiley-IEEE Press	2020

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://youtu.be/f5GMjGRsKi4?list=PLyqSpQzTE6M-YPOEUVrgd_2UnIRvWUcRr
2	https://youtu.be/U0gGoPKgays
3	https://youtu.be/lRm9GiGoZKg?list=PLLy_2iUCG87AjAXKbNMiKJZ2T9vvGpMB0
4	https://youtu.be/A7wHSr6GRnc

SEMESTER S7
SOFT ROBOTICS

Course Code	PERUT751	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles and characteristics of soft robotics.
2. To explore the design and fabrication methods used in creating soft robotic systems.
3. To study the control strategies and material properties that enable soft robotics functionality.
4. To investigate the applications and future trends of soft robotics in various domains

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Soft Robotics: Definition and Importance-Comparison with Traditional Rigid Robotics- Fundamental Principles of Soft Robotics: Compliance, Adaptability, and Safety Materials for Soft Robotics: Polymers, Gels, and Flexible Materials Historical Development and Milestones in Soft Robotics Current Trends and Future Directions in Soft Robotics	9
2	Design and Fabrication of Soft Robots :Design Principles for Soft Robots- Bio inspiration and Functional Requirements- Fabrication Techniques: Molding, Casting, 3D Printing, and Pneumatic Actuation- Materials Used in Soft Robotics: Silicone, Elastomers, and Hydrogels- Mechanical Properties and Characterization of Soft Materials- Case Studies: Soft Grippers, Soft Actuators, and Soft Sensors	9
3	Control and Actuation in Soft Robotics: Control Strategies for Soft Robotics: Kinematics and Dynamics- Actuation Mechanisms: Pneumatic, Hydraulic, and Electro active Polymers- Integration of Sensors and Feedback Systems- Modeling and Simulation of Soft Robotic Systems- Case Studies: Soft Robotic Manipulators and Crawlers	9

4	Applications and Future Trends in Soft Robotics: Applications of Soft Robotics-Industrial Automation, Healthcare, and Consumer Products- Soft Robotics in Medical Applications- Soft Endoscopy, Wearable Robots, and Rehabilitation -Research Trends in Soft Robotics- Bio hybrid Systems and Soft-Body Locomotion- Future Directions: Soft Robotics in Space Exploration and Extreme Environments- Challenges and Limitations: Durability, Control Precision, and Integration-Ethical and Societal Implications of Soft Robotics	9
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**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts and significance of soft robotics.	K2
CO2	Design and fabricate simple soft robotic systems.	K3
CO3	Apply appropriate control strategies for the movement and operation of soft robots.	K3
CO4	Evaluate the potential and challenges of soft robotics in real-world applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3	2		3						3
CO3	3		1			1						3
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Soft Robotics: Transferring Theory to Application"	Cecilia Laschi and Matteo Cianchetti	Springer	2016
2	"Fundamentals of Soft Matter Science"	Linda S. Hirst	CRC Press	2020
3	Soft Robotics: Modeling, Simulation, and Control of Deformable Robots" Ivan Petrov, Springer, 2021.	Ivan Petrov	Springer	2021
4	Soft Robotics: Technologies and Systems for Soft Machines and Robots".	Mohammad D. Hossain	Springer	2021

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Soft Robotics: Materials, Design, and Applications	Dong Sun and Shuozhi Xu	CRC Press	2020
2	Bio inspired Soft Robotics: Soft Matter in Engineering and Life Sciences	Eduard Arzt, De Gruyter,	CRC Press	2016
3	Fundamentals of Soft Robotics: Theory and Applications	Qingsong Xu,	Springer	2022
4	Soft Robotics: Challenges and Applications	Xianmin Zhang, Yujiang Shi, and Kwang Kim,	Springer	2021
5	Wearable Robotics: Challenges and Trends	Jose L. Pons	Springer	2019

SEMESTER S7

MEDICAL ROBOTICS

Course Code	PERUT753	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental principles and applications of medical robotics.
2. To explore the design and control of robotic systems used in medical applications.
3. To examine the role of robotics in surgery, diagnostics, and rehabilitation.
4. To Analyze current trends and future directions in medical robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Medical Robotics: Overview of Medical Robotics: Definition, Scope, and Importance-Historical Development and Milestones-Classification of Medical Robots: Surgical, Diagnostic, and Rehabilitation Robots Components and Architecture of Medical Robots-Safety, Regulatory, and Ethical Issues in Medical Robotics-Current Trends and Future Directions in Medical Robotics.	9
2	Introduction to Surgical Robotics: Definition and Importance-Components and Architecture of Surgical Robots- Types of Surgical Robotic Systems: Laparoscopic, Orthopaedic, and Cardiovascular Robots- Image-Guided Surgery and Integration with Robotic Systems	9
3	Introduction to systems hardware: Introduction to Rehabilitation and Assistive Robotics- Design and Control of Rehabilitation Robots-Types of Rehabilitation Robots: Upper Limb, Lower Limb, and Exoskeleton Robots-Assistive Robots: Robotic Wheelchairs, Prosthetics, and Daily Living Aids-Impact of Rehabilitation and Assistive Robots on Patient Care	9
4	Introduction to Diagnostic Robotics: Importance and Applications-Robotic Systems in Medical Imaging: MRI, CT, and Ultrasound Robots- AI	9

	in Medical Robotics: Definition, Importance, and Applications- Integration of Machine Learning and AI in Robotic Systems- Case Studies: Robotic Biopsy, AI in Diagnostic Imaging, Robotic Pathology- Challenges and Future directions in Diagnostic and AI-enhanced Medical Robotic	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the basic concepts and applications of medical robotics.	K2
CO2	Design and evaluate robotic systems for various medical applications	K3
CO3	Apply knowledge of robotics to enhance surgical diagnostic, and therapeutic processes.	K3
CO4	Assess the potential and limitations of current medical robotic technologies.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3											3
CO2	3	3	3	2		3						3
CO3	3		1			1						3
CO4	3											2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Medical Robotics: Minimally Invasive Surgery	Paula Gomes	Wood head Publishing	2012
2	Robot-Assisted Surgery: Principles and Applications	Hemant D. Tagare,	Springer	2010
3	Rehabilitation Robotics: Technology and Application	Sashi S. Kommu,	Intech Open	2007
4	Medical Image Analysis and Informatics: Computer-Aided Diagnosis and Therapy	Paulo Mazzoncini de Azevedo Marques,	CRC Press	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Surgical Robotics: Systems Applications and Visions	Jacob Rosen, Blake Hannaford, and Richard M. Satava	Springer	2011
2	Introduction to Robotics in Minimally Invasive Surgery	Thomas C. Roethlisberger and Gabor	CRC Press	2016.
3	Surgical Robotics: Systems Applications and Visions	Jacob Rosen, Blake Hannaford, and Richard M. Satava	Springer	2011
4	Artificial Intelligence in Medical Imaging: Opportunities, Applications and Risks	Lia Morra, Marco Ruella, and Luca Gatto	Springer	2021
5	Assistive Technology: Principles and Practice	Albert M. Cook and Janice M. Polgar,	Mosby	2014

SEMESTER S7

INDUSTRY 4.0

Course Code	PERUT754	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive understanding of Industry 4.0 and its significance in the modern industrial landscape
2. To introduce the fundamental concepts and technologies underpinning Industry 4.0, including IoT, Big Data, AI, and Cyber-Physical Systems
3. To explore the practical applications and implications of Industry 4.0 technologies in manufacturing and other industries
4. To prepare students to analyze and implement Industry 4.0 technologies in real-world scenarios, considering ethical, legal, and social aspects.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Industry 4.0 : Introduction to the Fourth Industrial Revolution, Evolution from Industry 1.0 to Industry 4.0, Key Components of Industry 4.0: IoT, Big Data, AI, Robotics, and Cyber-Physical Systems, Industry 4.0 Technologies and their Impact on Manufacturing and Supply Chain, Smart Factories and Smart Manufacturing	9
2	Internet of Things (IoT) and Cyber-Physical Systems: Introduction to IoT: Concepts and Architecture, Sensors, Actuators, and IoT Devices, Communication Protocols: MQTT, CoAP, and HTTP, Cyber-Physical Systems: Integration and Application, IoT Security and Challenges	9
3	Big Data and Artificial Intelligence in Industry 4.0: Big Data Analytics: Concepts and Tools, Data Collection, Storage, and Processing, Introduction to Artificial Intelligence and Machine Learning, AI Applications in Industry 4.0: Predictive Maintenance, Quality Control, and	9

	Autonomous Systems, Case Studies and Real-world Applications	
4	Robotics, Automation, and Future Trends: Basics of Robotics and Automation, Types of Industrial Robots and their Applications, Automation in Manufacturing: CNC, PLC, and SCADA Systems, Future Trends: Digital Twins, Augmented Reality, and Blockchain in Industry 4.0, Ethical, Legal, and Social Implications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the evolution and fundamental concepts of Industry 4.0 and its impact on modern manufacturing.	K2
CO2	Analyze the architecture, components, and security challenges of IoT and Cyber-Physical Systems.	K2
CO3	Understand the role of big data analytics and AI techniques to solve industrial problems and improve processes.	K2
CO4	Understand the role of robotics, automation, and emerging trends in Industry 4.0, considering ethical and social implications	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	1	2	1	1	1	1	1
CO2	3	3	2	2	3	2	2	1	1	1	1	1
CO3	3	3	3	3	3	2	2	1	1	1	1	1
CO4	3	3	3	3	3	2	3	2	1	1	1	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Industrial Internet of Things and Industry 4.0.	S. Misra, C. Roy, and A. Mukherjee	CRC Press.	2020
2	SMART FACTORY AND INDUSTRY 4.0. The current state of Application Technologies	Oliver Grunow	Study lab Publications	2016
3	Big Data in Practice: How 45 Successful Companies Used Big Data Analytics to Deliver Extraordinary Results.	Marr, B.	Wiley	2016
4	Robotics: Modelling, Planning and Control.	Siciliano, Bruno, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo	Springer	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Artificial Intelligence: A Modern Approach.	Russell, Stuart, and Peter Norvig	4th ed., Pearson	2021
2	Big Data Applications in Industry 4.0,	P. Kaliraj, T. Devi	SBN 9781032008110, CRC Press, Taylor & Francis Group	2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=WUYAjsxnwjU4&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=1
2	https://www.youtube.com/watch?v=d9QfVpCG00Y&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=7 https://www.youtube.com/watch?v=WdTHHXOI8kg&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=33
3	https://www.youtube.com/watch?v=mkc_nL6upZ0&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=43
4	https://www.youtube.com/watch?v=L3APegLIE8A&list=PLXDsvE7qtfNf_N99hJZbdTEM001mOii6_&index=1

SEMESTER S7
GENERATIVE AI

Course Code	PERUT756	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To learn the fundamentals of Neural Networks and their various types.
2. To explore Generative AI models like GANs, VAEs, and Transformers.
3. To analyze the limitations of traditional RNNs and LSTMs.
4. To discuss current trends and future directions in Generative AI research.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Foundations of AI and Neural Networks: History and evolution of AI/ML, Deep learning revolution, Transfer learning, History of Neural Natural Language Processing, Structure of Artificial Neural Networks, Steps in Training an Artificial Neural Network, Parameters and Hyperparameters, Backpropagation	9
2	Advanced Neural Network Architectures: Introduction to advanced architectures, Introduction to Generative AI Models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), Transformers, Attention Mechanism in detail Long Short-Term Memory Networks (LSTMs).	9
3	Data Pre-processing: Probability and Statistics, Data Pre-processing Techniques, Model Training Techniques	9
4	Generative AI Applications: Applications in Various Fields : Art and Creativity, Image and Video	9

	Generation, Text Generation, Music Composition, Healthcare Finance. Real-world use cases and challenges in deploying generative AI models	
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the evolution of AI and the significance of Deep Learning.	K2
CO2	Apply various Neural Network architectures for tasks like image recognition and sequence modelling.	K3
CO3	Analyze data pre-processing and training techniques for neural networks.	K4
CO4	Design practical solutions using advanced neural networks for diverse applications.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1		3	1	2		2	1		2	1	
CO2		2							1	1	2	
CO3	1		1	1	1		1	1	2	3	1	
CO4		1		2		1	1		3			

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Generative AI for everyone: Understanding the essentials and applications of this breakthrough technology".	Altaf Rehmani	Altaf Rehmani	February 12, 2024
2	Introduction to Generative AI	Numa Dhamani	Manning	March 5, 2024

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Generative Adversarial Networks Cookbook: Over 100 recipes to build generative models using Python, TensorFlow, and Keras	Josh Kalin	Packt Publishing	31 December 2018
2	Generative AI in Software Development: Beyond the Limitations of Traditional Coding	Jesse Sprinter	IBS	2024

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://elearn.nptel.ac.in/shop/iit-workshops/completed/leveraging-generative-ai-for-teaching-programming-course/s/?v=c86ee0d9d7ed
2	https://elearn.nptel.ac.in/shop/iit-workshops/completed/introduction-to-language-models/?v=c86ee0d9d7ed

SEMESTER S7

DIGITAL IMAGE PROCESSING

Course Code	PERUT757	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PERUT524 Signals and Systems	Course Type	Theory

Course Objectives:

1. To study the image fundamentals and mathematical transforms necessary for image transform
2. To study the image processing techniques like image enhancement, image reconstruction, image compression, image segmentation and image representation

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Digital Image Fundamentals: Image representation, basic relationship between pixels, elements of DIP system, elements of visual perception-simple image formation model. Brightness, contrast, hue, saturation, mach band effect. Colour image fundamentals -RGB, CMY, HIS models, 2D sampling, quantization.	9
2	Review of matrix theory: row and column ordering- Toeplitz, Circulant and block matrix. Image Compression: Need for compression, Basics of lossless compression – bit plane coding, run length encoding and predictive coding, Basics of lossy compression – uniform and non-uniform quantization techniques used in image compression, Concept of transform coding, JPEG Image compression standard.	9
3	Image Enhancement: Spatial domain methods: point processing- intensity transformations, histogram processing, image subtraction, image averaging. Spatial filtering smoothing filters, sharpening filters. Frequency domain methods: low pass filtering, high pass filtering,	9

	homomorphic filter. Simple MATLAB examples	
4	Image Restoration: Degradation model, Unconstraint restoration- Lagrange multiplier and constraint restoration Inverse filtering- removal of blur caused by uniform linear motion, Weiner filtering. Image segmentation: Classification of Image segmentation techniques, region approach, clustering techniques. Segmentation based on thresholding, edge based segmentation. Classification of edges, edge detection, Hough transform, active contour.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Distinguish/Analyse the various concepts and necessary for image processing	K4
CO2	Analyse basic image compression techniques	K4
CO3	Differentiate and interpret the various image enhancement techniques	K2
CO4	Apply different restoration models in image processing	K3
CO5	Analyse various image segmentation algorithms	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2		1							2
CO2	3	3	3		1							2
CO3	3	3	2		1							2
CO4	3	3	3		1							2
CO5	3	3	3		1							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Digital Image Processing	Gonzalez Rafael C	Pearson Education	4e/2009
2	Digital image processing	S Jayaraman, S Esakkirajan, T Veerakumar	Tata Mc Graw Hill	2e/2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of digital image processing	Jain Anil K	PHI	1998
2	Digital image processing	Kenneth R Castleman	Pearson Education	2/e,2003
3	Digital Image Processing	Pratt William K	John Wiley	4/e,2007

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/117/105/117105135/
2	https://onlinecourses.nptel.ac.in/noc22_ee116/preview
3	https://www.classcentral.com/course/swayam-digital-image-processing-14005

SEMESTER S7

COMPUTER NETWORKS

Course Code	PERUT758	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Study of this course provides the learners a clear understanding of how computer networks from local area networks to the massive and global Internet are built, how they allow computers to share information and communicate with one another. This course covers the physical aspects of computer networks, layers of OSI Reference model, and inter-networking. The course helps the learners to compare and analyze the existing network technologies and choose a suitable network design for a given system.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	(Introduction Physical Layer and Data Link Layer) Introduction - Uses of computer networks, Network hardware, Network software. Reference models- The OSI reference model, The TCP/IP reference model, Comparison of OSI and TCP/IP reference models. Physical Layer Modes of communication, Physical topologies, Signal encoding, Performance indicators- Bandwidth, Throughput, Latency, Queuing time, Bandwidth Delay product.	9
2	Data link layer - Data link layer design issues, Error detection and correction, Sliding window protocols. Medium Access Control (MAC) sublayer – Channel allocation problem, Multiple access protocols, Ethernet, Wireless LANs - 802.11, Bridges & switches - Bridges from 802.x to 802.y, Repeaters, Hubs, Bridges, Switches, Routers and Gateways.	9
3	(Network Layer) Network layer design issues. Routing algorithms - The Optimality Principle, Shortest path routing, Flooding, Distance Vector Routing, Link State	9

	Routing, Congestion control algorithms IP protocol, IP addresses, Internet Control Message Protocol (ICMP), Address Resolution Protocol (ARP), Reverse Address Resolution Protocol (RARP), BOOTP, Dynamic Host Configuration Protocol (DHCP), NAT, IPv6, ICMPv6.	
4	(Transport Layer and Application Layer) User Datagram Protocol (UDP). Transmission Control Protocol (TCP) Overview of TCP, TCP segment header, Connection establishment & release, Connection management modeling, Application Layer File Transfer Protocol (FTP), Domain Name System (DNS), Electronic mail, Multipurpose Internet Mail Extension (MIME), Simple Network Management Protocol (SNMP), World Wide Web (WWW), SMTP, HTTP	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the features of computer networks, protocols, and network design models (Cognitive Knowledge: Understand)	K2
CO2	Examine the fundamental characteristics of the physical layer and its role in network communication (Cognitive Knowledge: Apply)	K3
CO3	Explain the design issues of data link layer, link layer protocols, bridges and switches (Cognitive Knowledge: Understand)	K2
CO4	Select appropriate routing algorithms, congestion control techniques (Cognitive Knowledge: Apply)	K3
CO5	Illustrate functions and protocols of the network layer, transport layer, and application layer in inter-networking (Cognitive Knowledge: Understand)	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	-	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	-	-	-	-	-	-	-	3
CO3	3	3	2	-	-	-	-	-	-	-	-	3
CO4	3	3	3	2	-	-	-	-	-	-	-	3
CO5	3	3	3	-	-	2	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks	Andrew S. Tanenbaum	PHI (Prentice Hall India)	4/e
2	Data Communication and Networking	Behrouz A Forouzan	Tata McGraw Hill	4/e

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks – A Systems Approach	Larry L Peterson and Bruce S Dave	Morgan Kaufmann	5/e
2	Computer Networking and the Internet	Fred Halsall	Addison-Wesley	5/e
3	Computer Networking: A Top-Down Approach	James F. Kurose,	Keith W. Ross	6/e
4	An Engineering Approach to Computer Networks,	Keshav	Addison Wesley	1998

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://elearn.nptel.ac.in/shop/nptel/computer-networks-and-internet-protocol https://archive.nptel.ac.in/courses/106/106/106106091/ https://onlinecourses.swayam2.ac.in/cec21_cs19/preview
2	https://archive.nptel.ac.in/courses/106/105/106105081/ https://elearn.nptel.ac.in/shop/nptel/computer-networks-and-internet-protocol
3	https://onlinecourses.swayam2.ac.in/cec21_cs19/preview https://archive.nptel.ac.in/courses/106/106/106106091/ https://onlinecourses.nptel.ac.in/noc23_cs35/preview
4	https://archive.nptel.ac.in/courses/106/106/106106091/ https://elearn.nptel.ac.in/shop/nptel/computer-networks-and-internet-protocol

SEMESTER S7

VISUAL AND AUGMENTED REALITY FOR ROBOTICS

Course Code	PERUT755	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand the taxonomy, technology, and features of Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR)
2. Apply AR Tools and Techniques
3. Apply VR Hardware and Interaction

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction - Taxonomy, Technology and Features of Augmented Reality, Difference between AR, VR and MR, Challenges with AR, AR Systems and Functionality, Augmented Reality Methods, Visualization Techniques for Augmented Reality	9
2	Introduction to AR Tools - Camera Parameters and Calibration, Marker-Based Augmented Reality – AR Toolkit, Case Study: AR for Human Robot Communication - AR and Cobots.	9
3	Basic Features and Architecture - VR Input Hardware: Tracking Systems, Motion Capture Systems, Data Gloves, VR Output Hardware: Visual Displays	9
4	3D Manipulation Tasks, Manipulation Techniques and Input Devices, Interaction Techniques for 3D Manipulation, Applications of VR and AR: Medical Applications - Educational Applications - Public Safety and Military Applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Ex	Evaluate	Analyse	Total
5	15	10	10	40

Criteria for Evaluation (Evaluate and Analyse): 20 marks

Students may be given microprojects on the application of AR and VR tools

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the taxonomy, technology, and features of Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR)	K2
CO2	Apply AR Tools and Techniques	K3
CO3	Apply VR Hardware and Interaction	K3
CO4	Analyse AR and VR Applications	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	1	1	1	1	-	1	1	-	-	-	-	1
CO2	2	1	2	1	2	1	1	1	1	1	1	2
CO3	2	1	2	1	2	1	1	1	1	1	1	2
CO4	2	2	2	2	2	2	2	2	2	2	1	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Virtual & Augmented Reality for Dummies	Paul Mealy	John Wiley and Sons	2018
2	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile	Tony Parisi	O' Reilly	2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Understanding Augmented Reality: Concepts and Applications	Alan B. Craig	Morgan Kaufmann	2013
2	Developing Virtual Reality Applications: Foundations of Effective Design	Alan Craig, William Sherman and Jeffrey Will	Morgan Kaufmann	2009
3	Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR (Usability)	Steve Aukstakalnis	Addison-Wesley Publisher	1st Edition, 2016.
4	Virtual Reality Technology	Burdea, G. C. and P. Coffet	Wiley-IEEE Press	2nd Edition, 2017.

SEMESTER S7

BASICS OF MOBILE ROBOTICS

Course Code	OERUT721	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To give students the ability to understand the various types of mobile robots and sensors used for these robots
2. To give students the ability to apply different models and algorithms to kinematics, localization and navigation of mobile robots

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Types of Mobile Robots: Introduction, key issues for locomotion, Wheeled Mobile Robots, Wheeled locomotion: The design space, Wheeled locomotion: Case studies, Legged Mobile Robots - Leg configurations and stability, Examples of legged robot locomotion, aerial robots, underwater robots and surface water robots	9
2	Sensors for mobile robot navigation: Sensor classification, Characterizing sensor performance, Wheel /motor sensors, Heading sensors, Accelerometers, IMUs, Ground-based beacons, Active ranging, Motion/speed sensors, Robot Vision: Sensing, Pre-processing, Segmentation, Description, Recognition, Interpretation, Feature extraction	9
3	Mobile Robot Kinematics: Introduction, Kinematic Models and Constraints - Representing robot position, Forward kinematic models, Wheel kinematic constraints for Fixed standard wheel, Steered standard wheel and Castor wheel only, Robot kinematic constraints Example of Robot kinematic models and constraints: Differential-drive robot	9

	only Mobile Robot Maneuverability - Degree of mobility, Degree of steerability, Robot maneuverability, Mobile Robot Workspace - Degrees of freedom, Holonomic robots	
4	Robot Navigation: Robot Localization: Challenges in localization, An error model for odometric position estimation, Localization-Based Navigation versus Programmed Solutions, Belief Representation - Single-hypothesis belief, Multiple-hypothesis belief, Markov versus Kalman Localization (basics only) Path Planning: Local vs global path planning, Joint space versus Cartesian Space Trajectory Planning, Cubic Polynomial and LSPB methods of joint space trajectory planning, Graph search, Dijkstra's algorithm, A*, D* algorithms, Potential field based path planning, Bug algorithm	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the various types of mobile robots	K2
CO2	Explain the working and application of sensors used in mobile robots	K2
CO3	Derive kinematic constraints and models for mobile robots	K2
CO4	Apply various algorithms and techniques for localization and path planning of mobile robots	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	-	-	-	-	-	-	-	3
CO2	3	3	3	-	-	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Autonomous Mobile Robots	R Siegwart, IR Nourbakhsh, D Scaramuzza	The MIT Press	2nd edition, 2011
2	Introduction to Mobile Robot Control	Spyros G. Tzafestas	Elsevier Science Publishing Co Inc	1st edition, 2013
3	Sensors for mobile robot: Theory and Application	H.R. Everett	A K Peters/CRC Press	1st edition, 1995
4	Chapman & Hall_CRC Artificial Intelligence and Robotics Series - A First Course in Aerial Robots and Drones-	Yasmina Bestaoui Sebbane	CRC Press	1st edition, 2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Mobile Robotics: Mathematics, Models and methods	A Kelly	Cambridge University Press	2013
2	Computational Principles of Mobile Robotics	Gregory Dudek, Michael Jenkin	Cambridge University Press	2nd edition, 2010
3	Autonomous Mobile Robots and Multi-Robot Systems - Motion-Planning, Communication, and Swarming	Kagan, Shvalb, Ben-Gal	Wiley	1st edition, 2019

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_ge31/preview
2	https://nptel.ac.in/courses/112105249/
3	https://archive.nptel.ac.in/courses/112/106/112106298/
4	https://archive.nptel.ac.in/courses/112/106/112106298/

SEMESTER S7

PLC AND DISTRIBUTED CONTROL SYSTEMS

Course Code	OERUT722	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand PLC Fundamentals: Gain proficiency in the principles, operation, and programming of PLCs, including ladder logic programming and basic troubleshooting techniques.
2. Master DCS Architecture: Learn the architecture, components, and communication protocols of Distributed Control Systems (DCS), including their role in industrial automation and process control.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Process Control Loops: Feedback control, feed forward control, Cascade control, Ratio control, Split range control. Computerized control: Basic building blocks of Computer controlled systems Programmable Logic Controller: Architecture of PLC, Types of PLC, PLC modules, Input and output modules - Analog and discrete I/O modules, Examples of Digital and Analog Inputs/Outputs- PLC Configuration- Scan cycle, Capabilities of PLC, Selection criteria for PLC, PLC communication with PC and software, Power supply of PLCs, PLC Wiring- Installation of PLC and its modules.	9
2	PLC programming languages, Ladder programming, Functional Block Diagram, Sequential Flowchart SFC, Structured Text, Instruction List, Relay logic, Timers and counters, simple programs in PLC – ladder programming, math operations, sequencers, program control instructions, PTO/PWM generation. PLC Networking, Networking standards & IEEE standards, Protocols – Ethernet, PROFIBUS, CANopen	9

	Case studies – PLC based traffic light system, stepper motor & servo motor control using PLC, Analog sensor interfacing with PLC, Encoder interfacing with PLC, bottle filling system, gate control system	
3	SCADA: SCADA Fundamentals: Introduction, Types, Open system: Need and advantages, Building blocks of SCADA systems, Remote terminal unit (RTU), Human-Machine Interface (HMI) subsystem. SCADA communication systems, Master Station: Master station software components, Master station hardware components, Server systems in the master station, small, medium, and large master stations, software architecture of SCADA, Interfacing of SCADA with PLC, drive and other field devices. Distributed Control System: DCS - Architectures, Comparison, Local control unit, Process interfacing issues, Communication facilities. Various function Blocks.	9
4	LCU communication Facilities - Communication system requirements – Architectural Issues, Operator displays, Operator Interfaces – Engineering Interfaces. Introduction to HART and Field bus protocol. Interfacing Smart field devices (wired and wireless) with DCS controller. Introduction to Object Linking and Embedding (OLE) for Process Control. Case studies – Design of conveyor automation system using PLC, SCADA and Electrical drive: Design of inspection automation system using sensors, PLC and SCADA: Design of simple water management system using PLC, SCADA and Electrical Drive	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic process control loops for industrial control	K2
CO2	Describe the use of different types of inputs/outputs and networking used in PLC based systems.	K2
CO3	Design logic circuits to perform industrial control functions of medium complexity and realise using ladder logic	K3
CO4	Illustrate SCADA systems and its building blocks for industrial automation.	K2
CO5	Describe Distributed Control System and its applications	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	-	-	-	-	-	-	-	-	2
CO2	3	3	2	-	-	-	-	-	-	-	-	2
CO3	3	3	2	-	-	-	-	-	-	-	-	2
CO4	3	3	2	-	-	-	-	-	-	-	-	2
CO5	3	3	2	-	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Programmable Logic Controllers	Frank D. Petruzella	McGraw Hill	4th Edition, 2011
2	Distributed Control Systems: Their Evaluation and Design, Van Nostrand Reinhold Co., 1986	Michael P. Lukas	Van Nostrand Reinhold Co.,	1986
3	Programmable Logic Devices and Logic Controllers	Enrique Mandado, Jorge Macros, Serafin A Peres	Prentice Hall	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Programmable Logic Controllers	Gray Dunning	Delmar Cengage Learning	2005
2	Programmable logic controllers	W. Bolton	Elsevier Ltd.	2015
3	Distributed computer control for industrial Automation	D. Popovic and V.P.Bhatkar	Marcel Dekker, Inc., Newyork	1990
4	Practical SCADA for industry	David Bailey, Edwin Bright	Newnes Publishing	2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/108105088
2	https://nptel.ac.in/courses/108105088
3	https://nptel.ac.in/courses/108105088
4	https://nptel.ac.in/courses/108105088

SEMESTER S7

ROBOTIC WELDING TECHNOLOGY

Course Code	OERUT723	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce the fundamental principles and components of robotic welding systems.
2. To provide an in-depth understanding of various robotic welding processes and techniques.
3. To analyze the applications and performance of robotic welding systems in different industries.
4. To equip students with the skills to design, implement, and manage robotic welding systems, emphasizing economic, environmental, and safety considerations.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Robotic Welding: Overview of Welding Technology, Types of Welding Processes, Evolution of Welding Technology, Basic Concepts of Robotics, Historical Development of Robotic Welding, Components of Robotic Welding Systems, Robots and Controllers, Welding Equipment and Power Sources, Sensors and Vision Systems, Robotic Welding Cell Design, Layout and Configuration, Safety Considerations, Integration with Other Manufacturing Processes.	8
2	Robotic Welding Processes and Techniques: Types of Robotic Welding, Arc Welding (MIG, TIG, etc.), Resistance Welding, Laser Welding, Spot Welding, Programming Robotic Welders, Introduction to Robot Programming Languages, Path Planning and Trajectory Control, Offline Programming and Simulation, Welding Parameters and Quality Control, Selection of Welding Parameters, Monitoring and Controlling Weld Quality, Non-destructive Testing Methods, Advanced Welding Techniques, Adaptive	10

	Welding, Multi-Pass Welding, Welding of Advanced Materials	
3	Applications of Robotic Welding: Industrial Applications, Automotive Industry, Aerospace Industry, Construction and Heavy Machinery, Case Studies and Best Practices, Success Stories in Robotic Welding, Lessons Learned from Industry Leaders, Collaborative Robots in Welding, Applications and Limitations, Safety Standards and Compliance, Future Trends in Robotic Welding, Automation and Industry 4.0, Artificial Intelligence in Welding, Emerging Technologies	8
4	Implementation and Management of Robotic Welding Systems: Planning and Implementation, Feasibility Studies and ROI Analysis, Project Planning and Management, Selection of Robotic Welding Systems, Workforce Training and Development, Skills Required for Robotic Welding, Training Programs and Certifications, Managing Change in the Workplace, Maintenance and Troubleshooting, Preventive Maintenance Practices, Common Issues and Solutions, Upgrading and Retrofitting Systems, Economic and Environmental Considerations, Cost-Benefit Analysis, Environmental Impact and Sustainability, Regulatory Compliance	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the fundamental principles and components of robotic welding systems.	K1, K2
CO2	Apply knowledge of various robotic welding processes and techniques to select appropriate methods for different applications.	K3
CO3	Analyze and evaluate the performance of robotic welding systems in industrial applications, including quality control and troubleshooting.	K4
CO4	Design and implement robotic welding systems, considering economic, environmental, and safety factors.	K6

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	1	1	1	1	1	1	1	1
CO2	3	3	3	3	2	1	1	1	1	1	2	2
CO3	3	3	3	3	3	2	2	2	2	1	2	2
CO4	3	3	3	3	3	3	3	3	3	2	3	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotic Welding, Intelligence and Automation	Tzyh-Jong Tarn, Shan-Ben Chen, Sheng Liu, Demin Wang, Zhao Zhang	Springer	1st Edition, 2007
2	Robotic Welding, Intelligence and Automation: RWIA'2014	Tzyh-Jong Tarn, Shan-Ben Chen, Zhongkui Zhu, Naoyuki Kubota	Springer	1st Edition, 2014
3	Robotics and Automation in Welding, by Resistance and Friction: International Welding Engineering	Radovan Kovacevic	Springer	1st Edition, 2020

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Advanced Welding Technology	S. V. Nadkarni	Khanna Publishers	Revised Edition, 2010
2	Welding Engineering and Technology	R. S. Parmar	Khanna Publishers	1st Edition, 2012
3	Welding: Principles and Applications	Larry Jeffus	Cengage Learning	8th Edition, 2016

SEMESTER S7

ROBOTIC ASSEMBLY AND MATERIAL HANDLING SYSTEMS

Course Code	OERUT724	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	UCEST105, PCRUT303	Course Type	Theory

Course Objectives:

1. Understand the principles and components of robotic systems used in assembly and material handling.
2. Develop skills in programming and controlling robotic systems.
3. Analyze the performance of robotic systems in industrial applications.
4. Design and implement robotic solutions for material handling and assembly tasks

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Robotics and Industrial Automation - Introduction to Robotics- Definition and history of robotics - Types of robots and their applications - Overview of robotic systems in industrial automation. Robotic Components - Actuators, sensors, and controllers - End effectors and grippers - Robot kinematics and dynamics - Robotic Assembly Systems - Types of assembly robots - Applications in manufacturing - Advantages and limitations of robotic assembly	7
2	Robot Programming and Control - Basics of Robot Programming - Introduction to robot programming languages - Offline and online programming - Programming environments and simulation tools - Robot Control Systems - Open-loop and closed-loop control - PID control and advanced control strategies - Motion planning and trajectory generation - Sensor Integration and Feedback - Types of sensors used in robotics - Sensor fusion and data processing - Feedback control and real-time monitoring -	8

	Industrial Robot Programming - Programming industrial robots (e.g., ABB, FANUC) - Common programming tasks and challenges - Case studies and practical examples.	
3	Design and Analysis of Robotic Systems - Robotic System Design - Design principles and methodologies - Mechanical design and selection of components - Electrical and electronic considerations - Simulation and Modelling - Modelling robotic systems using CAD software - Kinematic and dynamic simulation - Validation and optimization of robotic designs - Performance Analysis - Metrics for evaluating robotic systems - Reliability, precision, and repeatability - Case studies on performance analysis - Safety and Standards - Safety considerations in robotic systems - International standards and regulations.	8
4	Material Handling Systems - Concepts of material handling - principles and considerations in material handling systems design - conventional material handling systems Types of material handling robots – industrial trucks, monorails, rail guided vehicles, conveyor systems, cranes and hoists, advanced material handling systems, automated guided vehicle systems, automated storage and retrieval systems (ASRS), bar code technology, radio frequency identification technology. Applications and Future Trends - Advanced Robotic Applications - Collaborative robots (cobots) - Flexible manufacturing systems - Robotics in warehousing and logistics - AI and Machine Learning in Robotics - Introduction to AI and machine learning concepts - Applications in robotic vision and decision-making - Case studies on AI-driven robotic systems - Trends in robotic automation - Developments in robotic hardware and software - Potential future directions in robotic assembly and material handling	13

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To understand the definition, history, and various types of robots and their applications in industrial automation.	K2
CO2	To identify and describe the key components of robotic systems, including actuators, sensors, controllers, end effectors, and grippers, and understand their roles in robotic kinematics and dynamics.	K2
CO3	To develop skills in robot programming, including the use of programming languages, offline and online programming techniques, motion planning, trajectory generation, and sensor integration.	K3
CO4	To design and analyze robotic systems using CAD software, understanding mechanical, electrical, and electronic considerations, and evaluating the performance of robotic systems with metrics such as reliability, precision, and repeatability.	K4
CO5	To explore and apply concepts of material handling, including the design principles of conventional and advanced material handling systems, and understand the role of technologies like AGVs, AS/RS, bar code technology, and RFID in industrial automation	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2			2							2
CO2	3	2			2							2
CO3	3	3	2	2	3				1			2
CO4	3	3	3	3	3				2			3
CO5	3	3	3		3				1			3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Robotics: Mechanics and Control.	John J. Craig	Pearson	3rd Edition Year: 2004
2	Robot Modelling and Control	Mark W. Spong, Seth Hutchinson, M. Vidyasagar	Wiley	1st Edition Year: 2005
3	Robotics: Fundamental Concepts and Analysis	Ashitava Ghosal	Oxford University Press	1st Edition Year: 2015
4	Material Handling Systems: Design, Analysis, and Simulation	Dimitris N. Kasapidis	CRC Press	1st Edition Year: 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robotic Engineering – An integrated Approach	Richaerd D Klafter, Thomas Achmielewski Mickael Negin	Prentice Hall India, New Delhi,	Year: 2001
2	Automation, Production Systems, and Computer Integrated Manufacturing	Mikell P. Groover	John Wiley & Sons, Inc.	2nd Edition Year: 2007
3	Introduction to Robotics in CIM Systems	James A Rehg	Prentice Hall of India	Year: 2002
4	Material Handling Automation	Jack M. Apple	John Wiley & Sons	1st Edition Year: 1999
5	Handbook of Material Handling Systems	René De Koster, G. S. B. J. Bozer, and C. J. M. van den Berg	Springer	1st Edition Year: 2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/112/105/112105249/
2	https://archive.nptel.ac.in/courses/112/105/112105249/
3	https://archive.nptel.ac.in/courses/112/105/112105249/
4	https://www.youtube.com/watch?v=Up1oSSJn6oM&t=3s https://onlinecourses.nptel.ac.in/noc20_me58/preview

SEMESTER 8

ROBOTICS AND AUTOMATION

SEMESTER S8

CO-OPERATIVE ROBOTICS

Course Code	PERUT861	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. The objective of this course is to give students a basic idea of cooperative robotics and design considerations of various algorithms.
2. Familiarise the various control problems, consensus algorithm and their applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to robotics: Components, Odometry, Kinematics, Control, Finite State Machines as Robot Controllers, State Transitions based on Robot–Robot Interactions, Early Micro-Macro Problems, Macroscopic Perspective, Expected Macroscopic Dynamics and Feedback	8
2	Discrete Consensus Achievement in Artificial Systems: Consensus Achievement, The Best-of-n Problem, Overview of Current Design Approaches, Bottom–Up Design Approaches, Top–Down Design Approaches. Consensus Algorithms in Cooperative Control: Fundamental Consensus Algorithms, Convergence Analysis of Consensus Algorithms, Synthesis and Extensions of Consensus Algorithms, Design of Coordination Strategies via Consensus Algorithms	10

3	Consensus-based Design Methodologies for Distributed Multivehicle Cooperative Control: Coupling in Cooperative Control Problems: Objective Coupling, Local Coupling, Full Coupling, Dynamic Coupling. Approach to Distributed Cooperative Control Problems with an Optimization Objective: Cooperation Constraints and Objectives, Coordination Variables and Coordination Functions, Centralized Cooperation Scheme, Consensus Building Approach to Distributed Cooperative Control Problems Without an Optimization Objective: Coordination Variable Constituted by a Group-level Reference State, Coordination Variable Constituted by Vehicle States	9
4	Applications to Multivehicle Cooperative Control: Rendezvous and Axial Alignment with Multiple Wheeled Mobile Robots, Distributed Formation Control of Multiple Wheeled Mobile Robots with a Virtual Leader, Deep Space Spacecraft Formation Flying, Cooperative Fire Monitoring with Multiple UAVs, Cooperative Surveillance with Multiple UAVs	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarise with various robotic components and controllers	K2
CO2	Describe various design approaches	K2
CO3	Understand various consensus algorithms and their applicability	K2
CO4	Analyse the use of coupling in cooperative control problems	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1										3
CO2	2	1										3
CO3	2	1										3
CO4	3	2	2									3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Swarm robotics: A formal approach	Hamann, Heiko	Springer	2018
2	Distributed consensus in multi vehicle cooperative control	Ren, Wei, and Randal W. Beard	Springer London	2008
3	Achieving consensus in robot swarms	Valentini, Gabriele	Studies in computational Intelligence, Springer	2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Path planning of cooperative mobile robots using discrete event models	Mahulea, Cristian, Marius Kloetzer, and Ramón González	John Wiley & Sons	2020
2	Cooperative Robots and Sensor	Koubâa, Anis, and	Springer	2014
3	Distributed autonomous robotic systems 2	Asama, Hajime, Toshio Fukuda, Tamio Arai, and Isao Endo	Springer Science & Business Media	2013

SEMESTER S8

PROBABILISTIC ROBOTICS

Course Code	PERUT862	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Teach students to design and implement probabilistic algorithms to enhance the autonomy and reliability of robotic systems.
2. Cover advanced topics such as sensor fusion, decision-making under uncertainty, and learning from data to prepare students for cutting-edge research and industry challenges in probabilistic robotics

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Probabilistic Robotics, Importance and applications of probabilistic methods in robotics. Historical context and development of probabilistic robotics. Fundamentals of Probability and Statistics Basic probability theory, random variables, and distributions. Statistical inference, Bayesian reasoning, and their significance in robotics. Probabilistic Models Types of probabilistic models used in robotics. Real-world examples and case studies demonstrating these models. Assignments and Assessments: Problem sets on probability and statistics. Readings and essays on the significance of probabilistic robotics	11
2	Bayesian Filters Understanding Bayes' rule and its application in filtering. Implementing Bayes filters for robotic localization.	10

	Kalman Filters Linear Gaussian models and the Kalman filter. Practical applications and limitations in robotics. Particle Filters Non-parametric filters and the particle filter algorithm. Handling non-linearities and non-Gaussian noise. Assignments and Assessments: Programming tasks to implement Bayesian, Kalman, and particle filters. Simulation projects for robotic localization	
3	Sensor Fusion Techniques Principles and methodologies for sensor fusion. Practical examples and case studies of sensor fusion in robotics. Decision Making Under Uncertainty Markov Decision Processes (MDPs) and Partially Observable MDPs (POMDPs). Algorithms for decision making in uncertain environments. Practical Applications Case studies and practical implementations. Hands-on projects with real or simulated robotic systems. Assignments and Assessments: Projects focused on sensor fusion and decision making. Case study analyses and practical demonstrations.	8
4	Learning Algorithms in Robotics Overview of supervised, unsupervised, and reinforcement learning. Implementing learning algorithms in robotic systems. Advanced Probabilistic Methods Exploration of advanced filtering techniques. Multi-robot systems and cooperative localization. Future Trends and Research Directions Emerging trends and future directions in probabilistic robotics. Assignments and Assessments: Final project integrating learned probabilistic algorithms. Review and presentation of contemporary research papers	8

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand and Explain the foundational principles of probability and statistics as they apply to robotic systems	K2
CO2	Design and Implement probabilistic algorithms, including Bayesian filters, Kalman filters, and particle filters, to enhance robotic localization and mapping capabilities.	K3
CO3	Analyze various sensor fusion techniques and decision-making algorithms to improve the autonomy and reliability of robotic systems under uncertain conditions.	K4
CO4	Develop and Synthesize advanced probabilistic methods and learning algorithms to address complex challenges in robotic research and industry applications.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	1			1							1
CO2	3	1			3							1
CO3	3	1		3	3							1
CO4	3	1		3	3							1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Probabilistic Robotics"	Sebastian Thrun, Wolfram Burgard, Dieter Fox	MIT Press	2005
2	"Introduction to Autonomous Mobile Robots"	Roland Siegwart, Illah R. Nourbakhsh, Davide Scaramuzza	MIT Press	2011
3	"Bayesian Reasoning and Machine Learning"	David Barber	Cambridge University Press	2012

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	"Artificial Intelligence: A Modern Approach"	Stuart Russell, Peter Norvig	Pearson	2021
2	"Machine Learning: A Probabilistic Perspective"	Kevin P. Murphy	MIT Press	2012

SEMESTER S8

COGNITIVE ROBOTICS

Course Code	PERUT863	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the science and technology behind cognitive thinking and to use it on autonomous robots
2. To familiarize the concepts of robot cognitive models, robot perceptions and 3D digital reconstruction
3. To develop knowledge about the robot programming packages utilized for mapping and localization

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to embodied cognitive science and cognitive robotics, How cognition works, Considerations for Modelling Cognitive Systems Cognition and intelligence, Intelligent architecture: Theories of intelligence, Kuramoto model, Child-Robot interaction. Levels of Abstraction in Modelling Cognitive Systems.	9
2	Overview, perspective of cognitive architecture, Desirable features, Designing a Cognitive Architecture. Introduction to the model of cognition, visual perception and recognition, Machine learning, soft computing tools, and robot cognition.	9
3	Necessity for 3D Reconstruction – Building Perception – Imaging Geometry – Global Representation – Transformation to Global Co-ordinate System. Map building: 2D world map, data structure for map building, Explanation of map building, procedure traverse boundary, robot simulation and robot	9

	map building programming.	
4	Robot Programming Methods: Program for Sensor Reading Display, Program for Avoid-Obstacles Behavior, Program for Wandering Within the Workspace, Program for Tele-operation. A Complete Program for autonomous mobile robots.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub-divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the philosophy of cognition and architecture of cognitive systems used in robotics	K2
CO2	Apply the cognitive intelligence and soft computing tool in the robot models.	K3
CO3	Apply the 3D digital reconstruction for the robot perception and map building.	K3
CO4	Describe various robot programming packages for display, tele-operation and other applications	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	3			2							3
CO3	3	3	3		2							3
CO4	3	3	3		3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot Cognition and Navigation - An Experiment with Mobile Robots	Patnaik, Srikanta	Springer-Verlag Berlin and Heidelberg,	2007
2	Artificial Cognitive Systems: A Primer	David Vernon	The MIT Press	1st Edition, 2014
3	Principles of Robot Motion Theory, Algorithms, and Implementation	Howie Choset, Kevin Lynch, Seth Hutchinson, George Kantor, Wolfram Burgard, Lydia Kavraki, and Sebastian Thrun	MIT Press, Cambridge	2005

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cognitive Robotics	Hooman Somani,	CRC Press	2015
2	Cognitive Robotics: Intelligent Robotic Systems	Jared Kroff	Wilford Press	2016
3	Control Systems: Classical, Modern, and AI-Based Approaches	Jitendra R. Raol, Ramakalyan Ayyagari	CRC Press	

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc23_me84/preview
2	https://www.youtube.com/watch?v=dBysgv3Q_Yc
3	https://onlinecourses.nptel.ac.in/noc23_me84/preview
4	https://onlinecourses.nptel.ac.in/noc23_me84/preview

SEMESTER S8

MICROROBOTICS

Course Code	PERUT864	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understanding Core Concepts and able to explain the science and technology behind micro-robots
2. Explain the design methodology, Microfabrication and Technology
3. Explain design principles of actuators
4. Explain the design selection criteria for micro machining applications

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Micro Robot: Science and Technology behind micro-robot – definition of microrobot. Fundamental concepts of linear elasticity, stress, strain, torsion, yield criteria. Fundamental concepts of kinematics – kinetics, dynamics, linear and angular momentum, equation of motion, multi body systems	9
2	Physics for Micro Robots: Scaling effect – scaling laws – scaling effect on surface and volume, electrical properties, fluids, physical forces. Concept of torsion – dummy suffix notation – matrix notation – transformations. Flexures: Definition of flexure – design methodology – bending mode, torsion mode. Elemental flexure – cantilever beam. Range motion, Notch Hinge, Cross pivot – cross strip flexure–monolithic cross pivot. Flexure systems – Linear guidance systems.	9

3	Actuators for Micro Robot: Introduction to Actuators – Design principles of actuators – Amplification of motion – Bimorph Assembly – Motion Amplification in plane – electrostatic actuator – Thermal based Actuator – Shape memory alloys – piezoelectric actuators – piezoelectric motors – magnetic shape memory alloys.	9
4	Sensors for Micro Robots: Sensors in micro-robots, sensors terminology, sensing technology for displacements – Types of sensors –tracking methods - Implementation, application and future prospects of Micro Robots:	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe micro system technology and micro robots	K2
CO2	Explain the physics of micro robots and describe flexures used in micro robots	K2
CO3	Describe various actuators used in micro robots	K2
CO4	Discuss about the various sensors used in micro robots	K2
CO5	Apply appropriate design criteria suitable for micromachining and micro assembly	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	0	0	0	1	1	1	1	2	1	0	2
CO2	2	1	2	0	1	1	1	1	1	0	0	1
CO3	2	0	0	0	1	1	1	1	2	1	0	2
CO4	2	1	2	0	1	1	1	1	1	0	0	1
CO5	2	0	0	0	1	1	1	1	2	1	0	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Microrobotics Methods and Applications	Yves Bellouard	CRC Press, Massachusetts	2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	“Microrobotics for Micromanipulation”	Nicolas Chaillet, Stephane Regnier	Wiley	2013
2	SergejFatikow, Ulrich Rembold,	Microsystem Technology and Microrobotics	Springer	2013
3	Vikas Choudhry, Krzystof	MEMS: Fundamental Technology and Applications	CRC Press	2013
4	NadimMaluf and Kirt Williams	An Introduction to Microelectromechanical systems Engineering	Artech House, MA	2002

SEMESTER S8

CLOUD ROBOTICS

Course Code	PERUT866	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Provide an overview of cloud robotics
2. Understand the concept of programmable robots for automated tasks, principles of cloud computing technologies and robotics-related software paradigm such as the Robotics Operating System (ROS)
3. Provide knowledge on how cloud computing techniques can automate manufacturing tasks using algorithms designed based on machine learning and big data analytics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Challenges in cloud robotics: Technology for cloud computing. Constraints and features of programmable robots. Development tools and software. Robot operating systems and software services. Current trends and advancements in cloud robotics computing.	9
2	Programmable robots: Microprocessors such as Arduino/Raspberry Pi with consumer robotic appliances. Robotics design. Navigation planning. Simultaneous Localisation and Mapping algorithm. Robot operating systems and its architecture	9
3	Cloud computing and networking: Cloud computing systems. Client-server programming and programming API in the cloud. Software programming and API services to integrate ROS and the Cloud.	9
4	Cloud robotics and automation architectures: robotics and human automation by cloud computing, current trends of networked robotics using	

	Internet and the Cloud. System components and architectures of the Internet of Things. Optimisation and Data analytics algorithms	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Identify the basic problems, limitations, strengths and current trends of programmable robotics	K2
CO2	Explain the current cloud computing technologies and computing mechanisms for robotics such as ROS	K2
CO3	Analyse and critique the performance of robotics algorithms and data analytics algorithms for cloud robotics	K4
CO4	Develop the attitude to use software programming and cloud computing solutions to create cloud robotics prototype	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2										3
CO2	3	3			2							3
CO3	3	3	3		3							3
CO4	3	3	3		3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud Robotics – Distributed Robotics using Cloud Computing	Joao Pedro, Carvalho Rosa		Coimbra, 2016
2	A Survey of Research on Cloud Robotics and Automation	Ben Kehoe, Sachin Patil, Pieter Abbeel, Ken Goldberg	IEEE Transactions on Automation Science and Engineering (T-ASE)	Special Issue on Cloud Robotics and Automation. Vol. 12, no. 2, 2015
3	Programming Robots with ROS: A Practical Introduction to the Robot Operating System	Morgan Quigley, Brian Gerkey and William Smart	O'Reilly Media	1st Edition, 2015.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Gentle Introduction to ROS	Jason M. O’Kane,	CreateSpace Independent Publisher;	1st Edition, 2013.
2	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej M. Goscinski	Wiley	2011
3	Enterprise Cloud Computing - Technology, Architecture, Applications,	Gautam Shroff,	Cambridge University Press,	2010

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
2	https://www.youtube.com/watch?v=Bcz3a-klKGk
3	https://onlinecourses.nptel.ac.in/noc21_cs14/preview
4	https://onlinecourses.nptel.ac.in/noc21_cs14/preview

SEMESTER S8

ENGINEERING OPTIMIZATION

Course Code	PERUT867	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To introduce various optimization techniques for solving optimization problems in real world situations.
2. To explain the concept of Dynamic programming and its applications to project implementation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction and Classical Optimization Techniques: Statement of an optimization problem-design vector-design constraints-constraint surface-objective function-classification of optimization problems Classical Optimization Techniques: Single variable Optimization-multi variable optimization without constraints-necessary and sufficient condition for minimum/maximum-multivariable Optimization with equality constraints, Solution by the method of Lagrange multipliers-Multivariable Optimization with inequality constraints –Kuhn-Tucker conditions.	9
2	Unconstrained Nonlinear Programming: One dimensional minimization methods-Classification-Fibonacci method and Quadratic Interpolation method. Unconstrained Optimization Techniques: Univariate method-Powell's method and Steepest Descent method.	9
3	Constrained Nonlinear Programming: Characteristics of a Constrained	9

	Problem- classification- Basic approach of Penalty Function method-Basic approaches of Interior and Exterior penalty function methods- Introduction to convex programming problem.	
4	Dynamic Programming: Dynamic Programming multistage decision processes-types-concept of sub optimization and the principle of optimality-computational procedure in dynamic programming-examples illustrating the calculus method of solution-examples illustrating the tabular method of solution.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	To introduce classical optimization techniques for solving optimization problems.	K2
CO2	To solve non linear programming problems with unconstrained optimization techniques.	K2
CO3	To solve constrained non linear programming problems using different optimization techniques.	K2
CO4	To introduce convex programming problems	K2
CO5	To solve dynamic programming using different optimization techniques.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	2	1			1	2		2
CO2	3	3	3	3	2	1			1	2		2
CO3	3	3	3	3	2	1			1	2		2
CO4	3	2	3	2	1	1			1	2		2
CO5	3	3	3	3	2	1			1	2		2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Engineering Optimization: Theory and Practice	Singiresu S. Rao	John Wiley and Sons	4 th Edition, 2009
2	Introductory Operations Research	H .S. Kasene and K.D.Kumar	Springer(India),Pvt.Ltd	2004
3	An Introduction to Optimization	Edwin K .P. Chong & Stanislaw H . Zak	Wiley	2 nd Edition, 2013

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	H.A.Taha	Operations Research: An introduction	Pearson/Prentice Hall	8 th Edition, 2007
2	Kalyanmoy Deb	Optimization for Engineering Design- Algoritms and Examples	PHI Learning Pvt. Ltd, New Delhi	2005

SEMESTER S8

ADAPTIVE CONTROL

Course Code	PERUT868	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None/ (Course code)	Course Type	Theory

Course Objectives:

1. Understand basic theories of adaptive control
2. Be able to analyse an adaptive control system
3. Differentiate between MRAS and STR

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Adaptive Control – Introduction, block diagram, adaptive systems – 4 approaches with block diagrams, applications of adaptive control, importance of adaptive control, Robust Adaptive control – overview, robust control laws	9
2	Deterministic Self tuning regulators (STR): Pole placement design, Indirect self tuning regulators, Continuous time self tuners, Direct self tuning regulators, disturbances with known characteristics	9
3	Stochastic and Predictive Self tuning regulators: Design of minimum variance and moving average controllers Stochastic self tuning regulators, Unification of direct self tuning regulators. Linear quadratic STR, adaptive predictive control	9
4	Model reference adaptive control (MRAS): The MIT Rule, Determination of adaptation gain, Lyapunov theory, Design of MRAS using Lyapunov theory, BIBO stability, Output feedback, Relations between MRAS and STR	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Familiarize with different types of adaptive control and its importance	K1,K2
CO2	Analyze pole placement design and STR	K2,K3
CO3	Differentiate between Deterministic Self tuning regulators and Stochastic Self tuning regulators	K2,K3
CO4	Discuss the Model Reference Adaptive Control method	K2,K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2						1	1	1			1
CO2	2		1				2	1	1			1
CO3	2		1				1	1	1			1
CO4	2						1	1	1			1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Adaptive control	Karl.J.Astrom and Bjorn Wittenmark	Dover publications	2 nd edition, 2008
2	Adaptive Control: Stability, Convergence and Robustness.	S. Sastry and M. Bodson	Prentice Hall	1989
3	Robust Adaptive Control	P. A. Ioannou and J. Sun	PTR Prentice-Hall	1996

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Stable Adaptive Systems	K. S. Narendra and A. M. Annaswamy	Prentice-Hall	1989
2	Adaptive Control Verlag	I.D. Landau, R. Lozano, and M. M'Saad	Springer	1998

SEMESTER S8

REINFORCEMENT LEARNING FOR ROBOTICS

Course Code	PERUT 865	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs.30 Min.
Prerequisites (if any)	NONE	Course Type	Theory

Course Objectives:

1. Students will be able to understand and apply reinforcement learning in the field of robotics.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Reinforcement Learning (RL): Introduction, Example applications of RL, Elements of Reinforcement Learning Finite Markov Decision Processes: The Agent–Environment Interface, Examples - Pick-and-Place Robot, Recycling Robot; Goals and Rewards, Returns and Episodes, Example – Pole Balancing, Unified Notation for Episodic and Continuing Tasks, Policies and Value Functions, Example – Gridworld, Optimal Policies and Optimal Value Functions, Bellman optimality equation, Example - Bellman Optimality Equations for the Recycling Robot	9
2	Dynamic Programming: Introduction, Policy Evaluation (Prediction), Iterative Policy Evaluation, Example – 4x4 gridworld, Policy Improvement, Policy Iteration, Value Iteration Monte Carlo Methods: Monte Carlo Prediction, Monte Carlo Estimation of Action Values, Monte	9

	Carlo Control, Monte Carlo Control without Exploring Starts, Off-policy Prediction via Importance Sampling, Incremental Implementation, Off-policy Monte Carlo Control	
3	Temporal-Difference Learning: TD Prediction, Advantages of TD Prediction Methods, Optimality of TD(0), Sarsa: On-policy TD Control, Q-learning: Off-policy TD Control, Expected Sarsa n-step Bootstrapping: n-step TD Prediction, n-step Sarsa, n-step Off-policy Learning, Off-policy Learning Without Importance Sampling: The n-step Tree Backup Algorithm	9
4	On-policy Prediction with Approximation: Value-function Approximation, The Prediction Objective, Stochastic-gradient and Semi-gradient Methods, Linear Methods Eligibility Traces: The λ -return, TD(λ), n-step Truncated λ -return Methods, Sarsa(λ) Policy Gradient Methods: Policy Approximation and its Advantages, The Policy Gradient Theorem, REINFORCE: Monte Carlo Policy Gradient, REINFORCE with Baseline, Actor–Critic Methods	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

<i>Attendance</i>	<i>Internal Ex</i>	<i>Evaluate</i>	<i>Analyse</i>	<i>Total</i>
5	15	10	10	40

Criteria for Evaluation(Evaluate and Analyse): 20 marks

Two Course Project applying the principles of nonlinear control system analysis:(10 marks each)

Project phases: Proposal, Implementation, Testing, Final Report, Presentations and Viva Voce:

Faculty may direct students for the selection of topics for projects from real time examples.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	2 questions will be given from each module, out of which 1 question should be answered. Each question can have a maximum of 3 sub divisions. Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Formulate Reinforcement Learning problems for various applications	K3
CO2	Apply dynamic programming and Monte Carlo methods to Reinforcement Learning	K3
CO3	Describe temporal difference Reinforcement Learning algorithms	K2
CO4	Explain On-policy Prediction and Policy gradient methods with approximation	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3
CO5	3	3	3	-	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Reinforcement Learning: An Introduction	Sutton, Barto, Bach	MIT Press	2nd edition, 2018
2	Reinforcement Learning in Robotics: A Survey	Kober, Peters	Springer Tracts in Advanced Robotics	2014

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Survey on Policy Search for Robotics	Deisenroth, Neumann, Peters	Now Publishers Inc	2013
2	Algorithms for Reinforcement Learning	Csaba Szepesvári	Morgan & Claypool Publishers	2009

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/106/106/106106143/
2	https://archive.nptel.ac.in/courses/106/106/106106143/
3	https://archive.nptel.ac.in/courses/106/106/106106143/
4	https://archive.nptel.ac.in/courses/106/106/106106143/

SEMESTER S8

SAFETY AND RELIABILITY IN ROBOTIC SYSTEMS

Course Code	OERUT831	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	4/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. Understand basic safety guidelines for robotic applications
2. Identify key elements of the robot safety standard
3. Apply provisions of the standard in daily operations
4. Illustrate the basic concepts and techniques of modern reliability engineering tools

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	ROBOT SAFETY: Safety Basics and Features, Need for Safety and the Role of Engineers Robot Safety: Problems and Hazards, Roles of Robot Manufacturers and Users in Robot Safety, Safety Considerations in Robot Design, Installation and Operations, Robot Safeguard Approaches, Robot-Related Safety Problems Causing Weak Points in Planning, Design, and Operation, Workplace Safety	9
2	ROBOT SYSTEM RELIABILITY: Reliability Basics and Configurations, Classifications of Robot Failures and their Causes and Corrective Measures, Robot Effectiveness Dictating Factors, Robot-Related Reliability Measures, Robot Reliability Analysis, Reliability Analysis of Hydraulic Robots	9
3	ROBOETHICS: Roboethics and Levels of Robomortality, Ethics Fundamental Elements and Theories, Top-Down and Bottom Up Roboethics Approach, Ethics in Human-Robot Symbiosis, Robot Rights, Socialized Roboethics, Ethical Issues of Socialized Robots	9

4	ROBOT TESTING: Robot Performance Testing Methods: Test-Robot Program Method, Ford Method, IPA- Stuttgart Method, National Bureau of Standards Method, Robot Reliability Testing, Robot Testing and Start-Up Safety-Related Factors, Testing Equipment and Procedures-Test Reports	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand basic safety guidelines for robotic applications	K1,K2
CO2	Understand robot system reliability	K1,K2
CO3	Understand roboethics	K1,K2
CO4	Understand the robo performance testing in daily operations	K1,K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	-	-	2	-	2	-	-	-	2
CO2	2	-	1	-	-	2	-	2	-	-	-	2
CO3	2	-	1	-	-	2	-	2	-	-	-	2
CO4	2	-	1	-	-	2	-	2	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Robot System Reliability and Safety: A Modern Approach	B. S. Dhillon	CRC Press, Taylor & Francis Group	2015
2	Roboethics: A Navigating Overview	Spyros G, Taffetas	1st Edition, Springer	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Human-Robot Interaction: Safety, Standardization, and Bench marking	Paolo Barattini, Federico Vicentini	CRC Press	2019
2	Robot Ethics: The Ethical and Social Implications of Robotics	Abney, Keith, Bekey, George A, Lin, Patrick	The MIT Press	2012
3	ISO/TC 299, Robotics	ISO	ISO	2018
4	Robot Framework Test Automation	SumitBisht	PACKT Publishing	2013

SEMESTER S8

INTRODUCTION TO INDUSTRY 4.0

Course Code	OERUT832	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To provide a comprehensive Understanding of the drivers and enablers of Industry 4.0
2. To Appreciate the smartness in Smart Factories, Smart cities, smart products and smart services
3. To explore the various systems used in a manufacturing plant and their role in an Industry 4.0 world
4. To prepare the students to understand the role of cyber physical systems in Industry 4.0.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Industry 4.0, Industry 4.0: The Fourth Industrial Revolution, Drivers, Enablers, Compelling Forces and Challenges for Industry 4.0. The opportunities in Industry 4.0, Industrial Internet, Smart Factory, Smart Buildings, Smart Manufacturing, Smart Farming	9
2	LEAN Production Systems, Smart and Connected Business Perspective, Smart Factories, Next Generation Sensors, Collaborative Platform and Product Lifecycle Management, Big Data and Advanced Analysis. Internet of Things (IoT) & Industrial Internet of Things (IIoT) & Internet of Services, Smart Manufacturing, Smart Devices and Products, Smart Logistics, Smart Cities, Predictive Analytics	9
3	Introduction: Sensing & actuation, Communication, Communication protocols, Integration of Robotic Automation and Collaborative Robots, Artificial Intelligence, Human-Machine Interaction, Application Domains:	9

	Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Case Studies	
4	Introduction to Cyber Physical Systems (CPS), Architecture of CPS, Data science and technology for CPS, Mobile Computing, Prototypes of CPS, , Networking systems for CPS applications, Wearable cyber physical systems and applications	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the evolution and fundamental concepts of Industry 4.0 and its impact on modern manufacturing.	K2
CO2	Understand the role of IoT and IIoT in Smart Factories, Smart cities, smart products and smart services.	K2
CO3	Understand the role of robotic systems and AI techniques to solve industrial problems and improve processes	K2
CO4	To understand the role of cyber physical systems in IoT	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	3	1	2	1	1	1	1	1
CO2	3	3	2	2	3	2	2	1	1	1	1	1
CO3	3	3	3	3	3	2	2	1	1	1	1	1
CO4	3	3	3	3	3	2	3	2	1	1	1	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Industrial Internet of Things and Industry 4.0.	S. Misra, C. Roy, and A. Mukherjee	CRC Press.	2020
2	SMART FACTORY AND INDUSTRY 4.0. The current state of Application Technologies	Oliver Grunow	Study lab Publications	2016
3	Principles of Cyber-Physical Systems	Rajeev Alur	MIT Press	2015
4	Robotics: Modelling, Planning and Control.	Siciliano, Bruno, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo	Springer	2009

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Artificial Intelligence: A Modern Approach.	Russell, Stuart, and Peter Norvig	4th ed., Pearson	2021
2	Big Data Applications in Industry 4.0,	P. Kaliraj, T. Devi	SBN 9781032008110, CRC Press, Taylor & Francis Group	2022

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=WUYAjsxwU4&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=1
2	https://www.youtube.com/watch?v=mkn_L6upZ0&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=43
3	https://www.youtube.com/watch?v=L3APegLIE8A&list=PLXDsvE7qtfNf_N99hJZbdTEM001mOii6_&index=1
4	https://www.youtube.com/watch?v=d9QfVpCG00Y&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=7 https://www.youtube.com/watch?v=WdTHHXOI8kg&list=PLaxu2gn-9WXMf_ln5pMvxjf043jzof4-i&index=33

SEMESTER S8

INTRODUCTION TO VIRTUAL INSTRUMENTATION

Course Code	OERUT833	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To impart the concepts of virtual instrumentation.
2. To apply concepts of LabVIEW in developing graphical programs.
3. To develop skills in data acquisition, instrumentation and control

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Virtual instrumentation: Virtual instrument and traditional instrument, hardware and software in VI, VI for test, control and design, VI in engineering process, virtual instruments beyond personal computer, graphical system design using LabVIEW. Introduction to LabVIEW: Advantages, software environment, creating and saving VI, front panel and block diagram toll bar, palettes, controls and indicators, block diagram, data types, data flow program.	9
2	Modular programming: Build a VI front panel and block diagram, building a connector pane, displaying sub VIs and express VIs, creating sub VIs, Repetition and loops: For loops, while loops, structure tunnels, terminal inside or outside loops, shift registers, feedback nodes, control timing, communication among multiple loops, local and global variables.	9
3	Arrays: Creating one dimensional, two dimensional, multi-dimensional arrays, array initialization, deleting, inserting, replacing elements within an array, array function, auto indexing. Structures: Case, sequence, customizing, timed structures, formula nodes, event structures.	9

4	Data acquisition: Signals, signal conditioning, DAQ hardware configuration, DAQ hardware, analog inputs, outputs, counters. Motion control: Components, software for configuration, prototyping and development, motion controller, move types, motor amplifiers and drives, feedback devices and motion I/O.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Comprehend the aspects of VI	K2
CO2	Apply the concepts of VI for the given logic.	K3
CO3	Describe the purpose and typical uses of arrays and structures within virtual instrumentation.	K2
CO4	Analyze the software and hardware components of VI.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3				2							3
CO2	3	3	3	2	2	2						3
CO3	3		1			1						3
CO4	3				1							2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	LabVIEW Graphical Programming	Richard Jennings and Fabiola De la Cueva	McGraw Hill	5 th , 2019
2	Virtual Instrumentation Using Labview	Jerome J	Prentice Hall India Learning Private Limited	2010
3	Virtual Instrumentation Using Labview	Sanjay Gupta & Joseph John	Tata McGraw Hill India	2005
4	LabVIEW for Everyone: Graphical Programming Made Easy and Fun	Wells Lisa K and Travis Jeffrey	Prentice Hall	3 rd , 2006

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	LabVIEW for Engineers	Ronald Larsen	Pearson	1 st , 2010
2	LabVIEW for Data Acquisition	Bruce Mihura	Prentice Hall	1 st , 2001
3	Learning With Labview 7 Express	Robert H. Bishop	Pearson College Div	1 st , 2004

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://www.youtube.com/watch?v=pP0ETel1myI&list=PLckZZXwYAE8qSUyc5hZsLHXJrArYl00r2
2	https://www.youtube.com/watch?v=90yKPBLmDJ4&list=PL71M4jZFYgQV4otlNftbCPKWBrl2alZab
3	https://youtu.be/5SRXnZydV3o?si=2r9ypmcR4wqY0xnt
4	https://www.youtube.com/watch?v=GBhJk5Tnshc

SEMESTER S8

MACHINE VISION

Course Code	OERUT834	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To give students the ability to understand the concepts of image formation and processing
2. To give students the ability to apply machine learning to various problems related to vision

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Computer Vision – Definition, Difference between computer vision and machine vision, real world applications Image Formation: Geometric primitives and transformations, 2D transformations, 3D transformations – Translation, Rotation + Translation, 3D to 2D projections - Orthography and para-perspective, Perspective, Camera intrinsics, Camera matrix, Photometric image formation - Lighting, The digital camera - Image sensing pipeline, factors affecting the performance of a digital image sensor	9
2	Image Processing: Point operators - Pixel transforms, Color transforms, Histogram equalization; Linear filtering (Basic explanation only), Non Linear Filtering – Median Filtering; Binary Image Processing – Morphology, Distance Transforms, Connected Components	9
3	Machine Learning: Traditional, machine learning, and deep learning pipelines; Supervised learning - Nearest neighbors, Bayesian classification; Unsupervised learning – Clustering, K-means and Gaussians mixture models; Deep neural networks - perceptron unit, Weights and layers, Activation functions, Loss functions, Backpropagation; Convolutional	9

	neural networks	
4	Recognition: Instance recognition, Image classification - Feature-based methods, Deep networks, Object detection - Face detection, General object detection - Precision vs. recall, Semantic segmentation – Basic understanding only Feature Detection and Matching: Points and patches, Edge Detection, Contour Detection, Lines and vanishing points, Segmentation - Region splitting, Region merging, Watershed	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the mathematics underlying image formation	K2
CO2	Apply different techniques to the processing of images	K3
CO3	Describe various machine learning algorithms and models	K2
CO4	Perform recognition and feature detection in images using different techniques	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	-	3	-	-	-	-	-	-	3
CO2	3	3	3	-	3	-	-	-	-	-	-	3
CO3	3	3	3	-	3	-	-	-	-	-	-	3
CO4	3	3	3	-	3	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Vision: Algorithms and Applications	Richard Szeliski	Springer	2010
2	Computer Vision a Modern Approach	Forsyth and Ponce	Pearson Education India	2nd edition, 2015

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Practical Introduction to Computer Vision with OpenCV	Kenneth Dawson-Howe	Wiley	1st edition, 2014
2	Robotics, Vision and Control: Fundamental Algorithms in MATLAB	Peter Corke	Springer	1st edition, 2011
3	Computer Vision: Principles, Algorithms, Applications, Learning	E. R. Davies	Academic Press Inc	5th edition, 2017

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://nptel.ac.in/courses/106105216/
2	https://nptel.ac.in/courses/106105216/
3	https://onlinecourses.nptel.ac.in/noc21_cs93/preview
4	https://nptel.ac.in/courses/106105216/